Wave-Constrained Computation

Formal Equivalence, Physical Resource Bounds, and the P versus NP Problem

Consolidated Technical Volume

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Original studies released in 2025

Abstract

This technical volume binds the three Wave-Constrained Computation (WCC) manuscripts and an independent fixed-mass nodal-coercivity supplement into one navigable PDF. The earlier revised documents are reproduced in full as source-preserving parts. This introduction distinguishes exact uniform discrete WCC results, charged physical wave realizations, restricted lower bounds, and unresolved universal separation claims. Neither classical P versus NP nor physical WCC separation is proved by this collection.

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1 Purpose and publication provenance

This is a consolidated edition of earlier research, not four additional newly authored papers. The historical titles, mathematical qualifications, original citations, and internal references are preserved in the bound originals. A later corrected theorem takes precedence over a broader historical informal formulation.

Part	Original work	Historical publication
I	Curvature-bounded wave computation, May 7, 2025	Zenodo 17743607
II	Discrete WCC and Turing equivalence, November 26, 2025	Zenodo 17732642
III	Physical complexity reassessment, December 1, 2025	Zenodo 17783074
Appendix	Fixed-mass nodal coercivity, October 2026	Independent auxiliary note

These original-release dates are not reset by the 2026 LaTeX revisions or this compilation. The DOI links identify historical records, not a newly assigned consolidated-volume DOI.

2 Results and scope

2.1 Uniform discrete computation

With finite alphabets and local update rules, polynomial-time constructible lattices, and charged uniform encoders and decoders, the discrete model has the formally established class equivalences

$$\mathbf{P}_{\mathrm{WCC}}^{\mathrm{disc}} = \mathbf{P}_{\mathrm{TM}}, \qquad \mathbf{NP}_{\mathrm{WCC}}^{\mathrm{disc}} = \mathbf{NP}_{\mathrm{TM}}.$$

The original two-phase machine W_M has its complete canonical window and input-code writers formally verified as polynomial-time transducers in the amended discrete model (II-10, Phase XVIII). This does not establish a physically resource-preserving continuum realization.

2.2 Physical realization and mathematical lower bounds

The continuum WCT resource model additionally requires uniform field preparation, precision, stable readout, and full accounting of energy and computational cost. The discrete equivalences do not prove that a concrete nonlinear PDE realizes them with polynomial overhead. Finite-domain coercivity, phase readout, and separator estimates have explicit assumptions that need not be input-size uniform.

The repository includes uniform Cook–Levin reductions and a characterization of the target dense-SAT lower bound. Restricted one-tape crossing-sequence arguments yield quadratic-type lower bounds. They do not show that every deterministic SAT algorithm requires superpolynomial time. In particular,

$$\mathrm{SAT}_{\mathrm{dense}} \notin \mathbf{P}_{\mathrm{TM}}$$

remains an open conjectural target. The uniform separator-price hypothesis is refuted for frozen growing-torus trajectories; the encoding-independent expander and universal complexity lower-bound claims remain open.

3 Maintained formalization ledger and verification

The 91-claim inventory recorded 78 Exact, 2 Restricted, 4 Refuted and 7 Open on 7 November 2026. After original II-10 was completed on 12 November, the totals became 79 Exact, 1 Restricted, 4 Refuted and 7 Open. The I-31 counterexample subsequently established the current 79 Exact, 1 Restricted, 5 Refuted and 6 Open. The Exact category includes proved conditional implications whose premises are not established; it does not mean classical P is unequal to NP. See the continuously maintained manuscript formalization inventory and Lean source tree.

Topic	Established limit
Uniform discrete WCC	Proved classical P and NP class equivalences within the formal discrete model.
Original two-phase W_M	Original finite discrete encoders, writers, and simulation support are Exact at the amended model scope; physical realization remains separate.
Dense SAT	Uniform Cook–Levin and restricted quadratic single-tape lower bounds; universal superpolynomial target open.
Physical WCC	Charged, robust, uniform physical-to-discrete resource bridge missing.
Curvature interfaces	Uniform interface price refuted in the frozen growing-torus class (I-31); independent expander hardness open.
Phase readout	Logical-energy SAT certification does not imply efficient physical convergence or preparation.
Auxiliary nodal estimate	Fixed-domain analytical result; its finite two-site algebra is formalized, not the full continuum theorem.

Build caveat. Hosted GitHub Actions jobs have failed before execution. At commit `c92158f`, the author reported a local full Lean build and allowed-axiom audit; this must not be described as hosted CI or as a check of every later edit. A PDF compilation is not evidence of a Lean proof. The final publication still requires inspection of the generated PDF and disclosure controls.

3.1 Corrections to the historical separation program

Part I's original claims I-25 through I-28 inferred a decision-language separation from the non-constructibility of individual target states or failure of a designated constructor. Those inferences are refuted and are explicitly withdrawn in the revised Part I text. The requisite missing lower bound instead quantifies over every correct uniform *decision algorithm* for one fixed language. Part II's II-10 concerns the original two-phase W_M; its complete canonical window and input-code transducers are proved polynomial time within the specified finite discrete model. The independent class inclusion uses a different, fully charged one-cell construction. Neither result is a physical continuum realization. The Part III discussion distinguishes proven finite and algebraic estimates from continuum PDE and physical-measurement conditions that are not all kernel-formalized.

3.2 Publication and verification gates

The separate historical standalone PDFs are prior compilations and may predate the latest revised LaTeX. The release must rebuild all four source documents and bind them into this single volume. The exact-head Lean build and axiom audit are separate from PDF compilation. The exact-head

local Lean build and axiom audit at `c92158f` are author-reported rather than hosted-CI results; later changes require separate provenance. Until the freshly regenerated PDF is checked against its sources and publication rights are cleared, this is a research manuscript rather than an approved public-release artifact.

4 Unresolved original-claim register

The following items are *not* completed by the supporting Lean results. They are the original amended-manuscript obligations, not probabilities of a future proof.

Claim	Status	Missing requirement
I-2	Open	Verifiable wavefield with independently proved nonconstructibility.
I-3	Open	Formula-dependent physical dynamics and robust readout, beyond logical-energy certificates.
I-11	Restricted	A charged, robust continuum physical-to-discrete resource bridge.
I-16	Open	An unrestricted superpolynomial SAT time lower bound against every correct decider.
I-32	Open	An encoding-independent exponential interface lower bound for every decider.
I-37	Open	Sound and complete arbitrary-formula toroidal physical locking.
III-22	Open	Interpretive arrow-of-time remark, not a formal theorem.

The four refuted historical Part I state-to-language claims I-25–I-28 remain in the audit and have been withdrawn from theorem status in the revised text. I-31 is also refuted for the frozen growing-torus trajectory class, not for all fixed-domain or hard-language variants. The current inventory is 79 Exact, 1 Restricted, 5 Refuted and 6 Open. No universal classical or physical separation follows.

Addendum, 12 November 2026. II-10 is no longer unresolved. The original complete polynomial-time window and initial-configuration transducers are verified in Lean for every polynomial radius (`TwoPhasePolynomialWriter.originalWindowCode_fp`, `TwoPhasePolynomialWriter.originalInputCode_fp`), so the author's two-phase W_M is a uniform discrete WCC machine without hypotheses. The completion of II-10 raised Exact to 79 and reduced Restricted to 1. The later I-31 counterexample below refutes the frozen-model hypothesis, leaving the current 79 Exact, 1 Restricted, 5 Refuted and 6 Open. The unresolved table above excludes both claims. No universal classical or physical separation follows. A second addendum of the same date refutes I-31 for the frozen trajectory class: exact plane waves of fixed amplitude keep the curvature functional bounded while their decision interface grows with the torus side (`SeparatorFrozen.no_uniform_separator_price`). The inventory is then 79 Exact, 1 Restricted, 5 Refuted and 6 Open.

5 Reproduction and source maintenance

The authoritative historical LaTeX sources remain under the four original manuscript directories in the repository. The single-release build first compiles them into temporary PDFs, binds them behind

this overview, and installs only one primary consolidated PDF. Run `bash scripts/build_pdfs.sh` to regenerate it. For the Lean formalization, use the pinned toolchain and run `lake build PVNP` followed by the standard axiom audit from within the `lean` directory.

The reproduced parts retain their own title pages, internal pagination, bibliographies, and reference numbering, to preserve provenance and avoid silently rewriting earlier published work. They constitute one *bound research volume*, not yet an editorially merged single-paper text.

Part I: Curvature-Bounded Wave Computation

Original research May 7, 2025. Revised source reproduced in full.

P vs NP in Curvature-Bounded Wave Computation
A Model-Relative Separation Program and Corrected Proof Obligations

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Revised October 27, 2026
Original Release: May 7, 2025

Abstract

We study curvature-regulated wave computation (WCC) as a possible physical resource model for decision and verification problems. We distinguish the *uniform discrete* finite-state WCC abstraction from WCT implementations additionally constrained by curvature, field regularity, precision, and physical energy rails. Under polynomial-time uniform encodings and readouts, the discrete WCC model simulates and is simulated by classical Turing machines with polynomial overhead, as developed separately in *Discrete Wave-Constrained Computation and Classical Complexity*. This equivalence alone proves no equivalence of physical curvature costs.

We correct the Boolean phase encoding of 3-SAT: a signed literal score and a product-of-unsatisfied-literals clause penalty give an exact zero-penalty characterization of SAT assignments. This is a logical encoding theorem, not a proof that a satisfying assignment is a stationary point of the full WCT functional. We also exhibit a smooth finite-energy four-dimensional field, refuting an earlier inference from failure of a Sobolev embedding to universal infinite curvature energy.

The proposed locked-expander curvature lower bound $C_\Theta(F_n) \geq 2^{\Omega(n)}$ is stated as a *conditional research hypothesis*: expansion alone

does not establish an exponential area lower bound for every physical encoding, and a cost of enumerating clause-signature sectors does not bound all decision algorithms. A conditional separation of physical WCC classes would require a genuine language-level lower bound against every admissible deterministic decider, along with a physically admissible polynomial-resource verifier. The classical **P** versus **NP** problem is not resolved.

This revision records explicit mathematical counterexamples and precise remaining obligations under tags `WIN-01`–`WIN-08`.

Current Lean formalization status (7 November 2026)

The amended Part I manuscript has 41 inventory claims: 30 Exact under their stated hypotheses, one Restricted, four Refuted, and six Open. Exact includes *proved conditional implications* and does not certify their antecedents or a solution to classical **P** versus **NP**. The four false state-to-language inferences (I-25 through I-28) have been explicitly withdrawn in the revised sections below. Their Lean counterexamples remain in the historical audit. The Restricted item I-11 concerns the missing resource-preserving *physical* WCC bridge; the Open claims I-2, I-3, I-16, I-31, I-32 and I-37 require additional independent theorems.

Exact discrete model and scope. The repository formalizes $P^{\mathrm{disc}}_{\mathrm{WCC}} = P_{\mathrm{TM}}$ and $NP^{\mathrm{disc}}_{\mathrm{WCC}} = NP_{\mathrm{TM}}$ for its explicitly defined finite-state models, including the uniform version that charges lattice generation, input encoding and readout (`PartIIUniform.P_disc_eq_P_TM`, `PartIIUniform.NP_disc_eq_NP_TM`). The older fixed-radius, origin-marked one-dimensional models and higher-dimensional sweep simulations have separate verified equivalences in `PVNP.Discrete`. The original two-phase machine W_M still needs proofs that its *complete original* lattice and input-state writers run in polynomial single-tape time (Part II II-10). Generic class equivalence does not complete that particular construction. *Addendum, 12 November 2026:* both writer proofs now exist in Lean (`TwoPhasePolynomialWriter.originalWindowCode_fp`, `TwoPhasePolynomialWriter.originalInputCode_fp`), and II-10 is Exact.

What the SAT and curvature results establish. The ψ-SAT logical penalty characterizes Boolean satisfiability under the stipulated field representation. A logical-energy value strictly below 1/8 supplies a formula-dependent SAT certificate; it does not construct the phase field or prove convergence of a physical PDE. The classical discrete Cook–Levin reduction and the restricted crossing-sequence single-tape SAT lower bounds are supporting formal results, but do not yield $\mathrm{SAT}_{\mathrm{dense}} \notin P_{\mathrm{TM}}$. The uniform separator-price, physical expander hardness and construction lower-bound premises remain Open. Exact finite symbol counts and stable sign encoding likewise do not yield decision lower bounds when decoder cost is unconstrained.

Physical and numerical boundary. The repository has scoped analytic and Lean results for frozen cubic–quintic field equations, including a separate H^2 periodic-torus well-posedness formalization in Phase VI, stated under its fixed equation and parameter hypotheses. The particular preparation, precision, detector and evolution cost required for a universal physical SAT lower bound have not been established. Finite-grid numerical overlap observations are empirical and cannot be generalized to all polynomial-time deciders.

Verification provenance. The old October and early-November partial-status assertions are preserved in the version history and claim ledger, but are not current statements. Some Phase XII and XIII sources were merged while hosted Actions jobs failed *before executing any steps.* A fresh exact-head full Lean kernel build and standard-axiom audit are required before newly added declarations are described as independently verified. A successful PDF build is separate from this mathematical check.

1 Introduction

The **P** vs. **NP** problem asks whether every computational problem whose solution can be verified in polynomial time can also be solved in polynomial time. Despite decades of effort, no proof resolving $\mathbf{P} \stackrel{?}{=} \mathbf{NP}$ has been universally accepted. The present work does not claim to resolve this classical question; instead, it develops a curvature-regulated model of computation and isolates the hypotheses necessary for a model-relative lower bound and an internal separation program.

We work within Wave Confinement Theory (WCT), where computation is defined as the physical or symbolic evolution of confined oscillatory systems under curvature feedback. In this setting, stable coherent wave states play the role of solutions, and the feasibility of reaching them under bounded evolution defines a notion of computational difficulty.

We introduce a discrete curvature-regulated computation model (WCC) and define complexity classes $\mathbf{P}_{\mathrm{WCC}}$ and $\mathbf{NP}_{\mathrm{WCC}}$ by imposing polynomial bounds on evolution time and curvature/entropy budgets. Within this model we pursue two complementary tracks:

- **Analytic curvature lower bound.** We define a semantic curvature complexity measure $\mathbf{C}_\Theta(f)$ based on the WCT curvature operator and show that, for a locked expander family of 3-SAT instances inside the WCT curvature machine, an exponential cost has been proposed, $\mathbf{C}_\Theta(F_n) \geq 2^{\Omega(n)}$. The interface-energy and universal-encoding hypotheses required for this bound are not established here. It remains an open mathematical objective, not a proved lower bound.

- **Symbolic complexity and separation program.** We construct a verifier for wave-coherence predicates in NC^1, encode SAT instances as symbolic wave-interference constraints (a Cook–Levin style reduction into WCC), and formalize WCC-level complexity classes $\mathbf{P}_{\mathrm{WCC}}$ and $\mathbf{NP}_{\mathrm{WCC}}$. We then formulate a *model-internal separation program*: under explicit geometric and dynamical hypotheses (existence of curvature-locked states that are verifiable but not reachable within polynomially many curvature-bounded WCC updates), one would obtain an internal separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$. In this manuscript these hypotheses are stated as conjectures rather than claimed theorems.

Finally, we outline a conditional bridge to classical complexity. If WCC computations and deterministic Turing computations simulate each other with only polynomial overhead, then the WCC classes would satisfy $\mathbf{P}_{\mathrm{WCC}} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCC}} = \mathbf{NP}$. Under this *TM–WCC compatibility conjecture*, any internal separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$, combined with the curvature lower bound described above, would imply the classical separation $\mathbf{P} \neq \mathbf{NP}$. The specified frozen one-dimensional radius-one discrete model has a kernel-checked polynomial-time TM/WCC equivalence. Its physical curvature-preserving realization and any internal physical separation are unproved. We focus on

the exact logical encoding and explicit hypotheses needed for a future model-relative analytic curvature lower bound.

Author's Note: On the Geometry of Computation and the Dimensional Lock

This work proposes a new approach to the **P vs. NP** problem, one that is not purely combinatorial or symbolic, but instead *geometric*, *topological*, and *physical*. Computation here is modeled not as an abstract string-manipulation process, but as the evolution of a wavefunction constrained by curvature, entropy, and dimension.

The central mechanism is what I term a **dimensional lock**: in the curvature-regulated model, attempting to force a deterministic, uniformly controlled evolution (the analogue of **P**) to realize certain non-uniform, combinatorially large configurations (the analogue of **NP**) appears to require crossing a geometric barrier that is forbidden by the curvature and dimensionality constraints. At an intuitive level, this structural obstruction prevents uniform computation from accessing the full non-uniform solution space and motivates an internal separation

$$\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$$

within the WCC model.

In the body of the paper, this idea is made precise in two steps. First, a semantic curvature complexity measure is introduced and an exponential curvature lower bound is proposed as a conditional goal for locked-expander 3-SAT instances inside the WCT curvature machine. Second, a discrete symbolic abstraction of WCC is developed, together with a verifier and a set of explicit geometric and dynamical hypotheses which, if established, would yield a formal internal separation $\mathbf{P}_{\mathrm{WCC}} \neq \mathbf{NP}_{\mathrm{WCC}}$.

This perspective is simple but nontraditional. It draws from physics and information theory, disciplines not typically employed in classical complexity theory. For that reason, it may appear unfamiliar or even unorthodox to some readers. The claim here is not that a physical metaphor proves a classical complexity result, but that a curvature- and dimension-constrained *symbolic* model admits (i) a precise conditional curvature-bound program for SAT, and (ii) a carefully stated internal separation program.

The extrapolation to the classical statement $\mathbf{P} \neq \mathbf{NP}$ is treated explicitly as conditional. If curvature-regulated WCC machines and standard Turing machines are polynomially equivalent in computational power, then the dimensional lock and curvature bounds described here would provide a route to

the classical separation. That simulation equivalence is isolated as a distinct conjecture rather than assumed silently; this paper does not claim to resolve the classical **P** vs. **NP** problem.

2 Curvature-Confinement Formalism and a Model-Internal Separation

We propose that a confined wavefield $\psi(x,t)$ under curvature feedback evolves toward symbolic configurations that cannot be constructed in polynomial time but can be verified via coherence constraints. Within the rail-constrained WCC abstraction, this creates a natural asymmetry between generation and verification, aligning with a prospective distinction between $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$, where the superscript phys denotes photonic WCT/WCC machines subject to finite-curvature, finite-band rails (Section 10).

2.1 Definitions

Curvature-Locked State: A wavefield ψ^* is curvature-locked if it satisfies

$$W_\psi = \frac{-\nabla^2 \psi}{\psi + \epsilon e^{-\alpha|\psi|^2}}, \quad \text{with } \frac{dS[\psi]}{dt} \to 0 \text{ and } \nabla W_\psi \to 0 \tag{1}$$

under the WCT/WCC evolution rules.

Symbolic Configuration Set $\mathcal{S}$: A set of discrete symbolic regions R_i where

$$\psi(x) \approx \psi_i \quad \forall x \in R_i, \quad R_i \subseteq \mathbb{R}^n$$

implements a finite symbolic configuration (assignment, tape content, etc.) inside the WCC model.

Verification Operator $\mathcal{V}$: A polynomial-time procedure on the WCC encoding which confirms that

$$\mathcal{V}(\psi^*, \mathcal{S}) = \text{True} \iff \psi^* \text{ exhibits symbolic coherence consistent with } \mathcal{S},$$

with an implementation in NC^1 on a classical circuit model.

Non-Constructibility Condition (WCC): A curvature-locked state ψ^* is *non-constructible in* $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}}$ if no deterministic polynomial-time WCC machine $\mathcal{T}_{\mathrm{WCC}}^{\mathrm{phys}}$ (obeying WCT rails) can output ψ^* from a trivial input configuration, while a polynomial-time WCC verification $\mathcal{V}$ remains available.

2.2 Model-Internal Complexity Schema

Fix a state-construction model. Write P_{st} for states a polynomial-time machine produces and NP_{st} for states the verification predicate accepts. In Lean these sets are `ConstructionModel.Pst` and `ConstructionModel.NPst`.

Proposition 2.1 (Curvature-Confinement Complexity Schema in Physical WCC). Let C be a real-valued functional and ψ^* a verifiable state. Suppose that for each polynomial-time constructor M there is a number $B_M < C(\psi^*)$ such that $C(\phi) \leq B_M$ for every state ϕ that M produces. Then $\psi^* \in NP_{\mathrm{st}} \setminus P_{\mathrm{st}}$.

Proof. If M produced ψ^*, its bound would give $C(\psi^*) \leq B_M < C(\psi^*)$. Verifiability gives membership in NP_{st}. This proves the state-set conclusion of `PartISeparation.complexity_schema`. Deriving the cost premise from the physical dynamics remains open. □

2.3 ψ-SAT Encoding Scheme (WCC Rail)

Let each Boolean clause C_i in a SAT instance be mapped to a spatial clause kernel:

$$\psi(x,0) = \sum_i \chi_i(x) e^{i\phi_i}, \quad \chi_i(x) \text{ localized curvature excitation}, \tag{2}$$

with a prescribed WCC encoding from symbolic clauses to localized modes.

The proposed physical lock tests whether local phase gradients decay and intensity reaches a local maximum. Its equivalence to SAT remains open (I-3). The formula-dependent logical penalty in Section 5.3 has a proved witness criterion.

3 Complexity Class Definitions

Definition 3.1 (Class P). **P** is the class of decision problems solvable by a deterministic Turing machine in polynomial time, i.e., there exists a machine M and constant k such that for all inputs x, $M(x)$ halts in $\mathcal{O}(|x|^k)$ steps.

Definition 3.2 (Class NP). **NP** is the class of decision problems for which a proposed solution (certificate) can be verified in polynomial time by a deterministic Turing machine. Formally, $L \in \mathbf{NP}$ if there exists a polynomial-time verifier $V(x,w)$ and polynomial q such that

$$x \in L \iff \exists w : |w| \leq q(|x|) \ \wedge \ V(x,w) = 1.$$

In the WCC setting we distinguish two model-relative classes:

- $\mathbf{P}^{\text{phys}}_{\text{WCC}}$ and $\mathbf{NP}^{\text{phys}}_{\text{WCC}}$: rail-constrained photonic WCC machines under WCT curvature/energy caps (finite universe band).
- $\mathbf{P}^{\infty}_{\text{WCC}}$ and $\mathbf{NP}^{\infty}_{\text{WCC}}$: ideal discrete WCC machines without global physical caps, suitable for asymptotic Turing-equivalence considerations.

The main model-internal separation conjectures below concern $\mathbf{P}^{\text{phys}}_{\text{WCC}}$ and $\mathbf{NP}^{\text{phys}}_{\text{WCC}}$. Implications for classical $\mathbf{P}$ vs. $\mathbf{NP}$ via the ideal classes $\mathbf{P}^{\infty}_{\text{WCC}}, \mathbf{NP}^{\infty}_{\text{WCC}}$ are discussed only as conditional corollaries.

4 Main Model-Relative Separation Statement ($\mathbf{P}^{\text{phys}}_{\text{WCC}} \neq \mathbf{NP}^{\text{phys}}_{\text{WCC}}$)

Conjecture 4.1 (Symbolic Separation Program in Physical WCC). Under one fixed preparation, verification and resource model, there exists a coherent target state in $NP_{\text{st}} \setminus P_{\text{st}}$.

The proposed encoding associates SAT constraints with curvature-locked fields. This conjecture requires a verifiable target and a bound against every permitted polynomial-time constructor of that target. Decision hardness requires one fixed input language and a worst-case bound for every correct decider, including those that bypass the target. Section 10 discusses the proposed geometric obstruction.

4.1 ψ-SAT Encoding and Symbolic Map

We formally define the symbolic encoding from a Boolean SAT instance into curvature-locked wave states. Let a Boolean clause $C_i \in \text{SAT}$ be

$$C_i = (x_1 \vee \neg x_2 \vee x_3).$$

We map each clause C_i directly into a wave symbol region $R_i \subseteq \mathbb{R}^n$:

$$C_i \mapsto \psi_i(x) = A_i e^{i\phi_i(x)}, \quad x \in R_i,$$

where A_i is a localized amplitude distribution enforcing clause locality and $\phi_i(x)$ encodes logical states (True/False) in terms of constructive or destructive interference.

We construct the global wavefield as a coherent superposition of all clause wavefunctions:

$$\psi(x) = \sum_i \psi_i(x).$$

The phase-gradient and local-intensity test proposed here is an open physical readout criterion. The uniform Cook–Levin reduction applies to the discrete symbolic language.

4.2 Model-Internal Symbolic Separation

Theorem 4.2 (Conditional Symbolic Separation in Physical WCC). Fix the preceding state-construction model. If its verification predicate accepts ψ^*

and every polynomial-time constructor fails to produce ψ^*, then $\psi^* \in NP_{\mathrm{st}} \setminus P_{\mathrm{st}}$ and $P_{\mathrm{st}} \neq NP_{\mathrm{st}}$.

Proof. Verification puts ψ^* in NP_{st}. Membership in P_{st} would supply a polynomial-time constructor producing ψ^*, contrary to the second hypothesis. The two sets therefore differ on ψ^*. □

4.3 Symbolic Turing Abstraction and Conditional Classical Corollary

In an *ideal* discrete WCC model (ignoring global WCT caps), we encode each $\psi[i,j] = a + ib \in \mathbb{C}$ as a binary string on a Turing tape.

Encoding: Each cell amplitude $\psi[i,j]$ is represented by a finite-precision binary code on one or more Turing tapes.

Evolution: Polynomial-time ideal WCC updates are simulated by a deterministic Turing machine using local symbolic gates that implement the curvature-regulated transition rules, defining classes $\mathbf{P}^{\infty}_{\mathrm{WCC}}, \mathbf{NP}^{\infty}_{\mathrm{WCC}}$.

Verifier: A polynomial-time predicate on uniformly encoded pairs (x, w), with a polynomial bound on $|w|$, defines an NP language by existential quantification over witnesses. Physical overlap and entropy tests additionally require precision and margin estimates.

4.4 Formal Completion (Cook–Levin Consistency and Conjectural Bridge)

The defined finite discrete model admits the two polynomial-overhead simulations proved in Part II. Extending the classification to a continuum PDE requires a uniform resource-preserving simulation of preparation, evolution and readout.

Conditional Classical Corollary. If, in addition, (i) every polynomial-time Turing machine can be simulated by a polynomial-time *ideal* WCC machine (TM $\Rightarrow$ WCC$^{\infty}$ with polynomial overhead) and (ii) every polynomial-time ideal WCC machine can be simulated by a polynomial-time Turing machine (WCC$^{\infty}$ $\Rightarrow$ TM with polynomial overhead), then

$$\mathbf{P}^{\infty}_{\mathrm{WCC}} = \mathbf{P}, \quad \mathbf{NP}^{\infty}_{\mathrm{WCC}} = \mathbf{NP}.$$

Any eventual internal separation $\mathbf{P}^{\infty}_{\mathrm{WCC}} \neq \mathbf{NP}^{\infty}_{\mathrm{WCC}}$, established by strengthening the geometric barrier program to the ideal model, would then imply the classical separation $\mathbf{P} \neq \mathbf{NP}$.

For the defined discrete language classes the simulations are proved. The decision lower bound and the continuum correspondence remain open.

5 Internal Complexity Structure of WCC

We present the internal complexity hierarchy of the Wave Curvature Computation (WCC) model. All results in this section are strictly model-internal and do not concern classical Turing complexity.

5.1 Preliminaries

Let $\Lambda \subset \mathbb{Z}^d$ be a finite lattice with $d = 2$ or $d = 3$. A WCC configuration is a field

$$\psi : \Lambda \to \mathbb{C}.$$

The curvature functional is

$$E[\psi] = \sum_{i \in \Lambda} \left(|\nabla \psi(i)|^2 + |\Theta[\psi(i)]|^2 + V_F(\psi(i)) \right), \tag{3}$$

where

$$\Theta[\psi] = -\frac{\Delta \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}}$$

and V_F encodes logical constraints.

A WCC evolution is any sequence $\psi_0, \psi_1, \psi_2, \ldots$ obeying r-local update rules and satisfying

$$E[\psi_t] < \infty \quad \forall t.$$

5.2 Definition of PWCC and NPWCC

Definition 5.1 (PWCC). The class **PWCC** consists of all decision problems solvable by a WCC evolution in $\mathrm{poly}(n)$ update steps, where n is the instance size, with the additional constraints:

$$E[\psi_t] < \infty, \qquad \dim_{\text{eff}}(\psi_t) \leq 3 \quad \text{for all } t.$$

Here $\dim_{\text{eff}}$ denotes the effective geometric embedding dimension (Definition 5.4).

Definition 5.2 (NPWCC). The class **NPWCC** consists of all decision problems F for which there exists a configuration ψ satisfying all constraints (i.e. $E_{\text{logic}}[\psi] = 0$), and for which such ψ can be verified in $\mathrm{poly}(n)$ time by a WCC evolution.

5.3 ψ-SAT Encoding

WIN-01 and WIN-02 (corrected). Let $F = \bigwedge_{j=1}^{m} C_j$ be a 3-CNF formula on Boolean variables $x_1, \ldots, x_n$. On each variable region, use a nonzero field with encoded phase $\theta_i \in \{0, \pi\}$ and set

$$x_i = 1 \iff \theta_i = 0, \qquad x_i = 0 \iff \theta_i = \pi.$$

For a literal ℓ, write $v(\ell)$ for its variable index and set $\eta_\ell = +1$ for $x_{v(\ell)}$ and $\eta_\ell = -1$ for $\neg x_{v(\ell)}$. Define its truth score by

$$s(\ell; \psi) := \frac{1 + \eta_\ell \cos \theta_{v(\ell)}}{2}.$$

For Boolean phases the score is exactly 1 when the literal is true, and exactly 0 when it is false. For a clause $C_j = (\ell_{j1} \vee \ell_{j2} \vee \ell_{j3})$ define

$$V_j(\psi) := \prod_{k=1}^{3} \big(1 - s(\ell_{jk}; \psi)\big), \qquad E_{\text{logic}}[\psi] := \sum_{j=1}^{m} V_j(\psi).$$

The product, rather than a sum of unsatisfied literals, encodes the Boolean *OR* relation. The energy is nonnegative on all nonzero phase encodings; no division by a vanishing field amplitude is used.

5.4 ψ-SAT Logical Completeness (Corrected)

Theorem 5.3 (Exact logical zero-penalty characterization). For every 3-CNF formula F, the following are equivalent:

1. F has a satisfying Boolean assignment.
2. There exists a binary-phase configuration ψ with $E_{\text{logic}}[\psi] = 0$.

The same existential equivalence holds if arbitrary phases are permitted and all variable amplitudes remain nonzero. Neither equivalence asserts stationarity of the *full* WCT energy functional.

Proof. If F is satisfiable, give each variable phase 0 for true and π for false. In every clause some literal has score 1, and therefore every V_j vanishes. Conversely, all V_j are nonnegative, so $E_{\text{logic}} = 0$ implies that every clause has at least one literal with score 1. Since $s(\ell) = 1$ forces the corresponding phase to be exactly 0 or π with the appropriate sign, these literal truth requirements are mutually consistent across clauses. Complete any remaining unspecified Boolean variables arbitrarily. The resulting assignment satisfies F. □

Quantitative witness readout. Assign $x_i = 1$ when $\cos\theta_i \geq 0$. In any clause falsified by this assignment each of the three factors $1-s$ is at least $1/2$. Since all clause penalties are nonnegative, $E_{\text{logic}} < 1/8$ guarantees that this one assignment satisfies every clause. Along any trajectory with $E_{\text{logic}} \to 0$ this holds at every sufficiently late observation. The strict threshold for this decoder is sharp: for the single clause $(\neg x \vee \neg x \vee \neg x)$, the phase $\pi/2$ gives energy $1/8$ and the tie rule chooses $x = 1$. Convergence of the actual dynamics remains an independent hypothesis.

Remark (WIN-03: stationarity is a separate obligation). The implication $E_{\text{logic}}[\psi] = 0 \Rightarrow \delta(E_{\text{geom}} + \lambda E_{\text{logic}})[\psi] = 0$ is invalid without conditions on the geometric first variation, boundary data, and admissible function space. A zero of a nonnegative logical penalty need not be stationary for an independently varying geometric energy. Existence of a full PDE equilibrium realizing an assignment is therefore an open problem here.

5.5 Effective Dimension

Definition 5.4 (Effective Dimension). For a specified admissible representation class, let

$$\dim_{\text{eff}}(\psi) := \min\{k \in \mathbb{N} : \psi \text{ has the prescribed regular encoding in } \mathbb{Z}^k\},$$

when the set is nonempty. Any dimension-dependent obstruction must also specify the encoding and ambient function spaces.

5.6 The Dimensional-Lock Claim Requires Additional Hypotheses

Proposition 5.5 (WIN-04: failure of Sobolev embedding is not infinite energy). For $d = 4$, there exist nonzero smooth localized fields with finite $\|\nabla\psi\|_{L^2}^2$ and finite corrected regularized curvature energy $\|\Theta_\varepsilon[\psi]\|_{L^2}^2$, where

$$\Theta_\varepsilon[\psi] = -\frac{\Delta\psi\,\overline{\psi}}{|\psi|^2 + \varepsilon e^{-\alpha|\psi|^2}}, \qquad \varepsilon > 0, \quad \alpha \geq 0.$$

Consequently the failure of an $H^1 \hookrightarrow L^\infty$ embedding for $d > 3$ does not prove that every higher-dimensional configuration has infinite WCT curvature energy.

Proof. Take $\psi(x) = e^{-|x|^2}$ on $\mathbb{R}^4$. Then $\nabla\psi = -2xe^{-|x|^2}$ and $\Delta\psi = (4|x|^2 - 8)e^{-|x|^2}$, both square-integrable. For this real positive field, the denominator is positive and $|\overline{\psi}|/(|\psi|^2 + \varepsilon e^{-\alpha|\psi|^2})$ is globally bounded: near infinity its numerator tends to zero while the denominator tends to ε, and on compact intervals of amplitudes it is continuous. Thus $\Theta_\varepsilon[\psi] \in L^2(\mathbb{R}^4)$. □

Remark. This does not prove stability or existence of physically confined four-dimensional solitons. It invalidates the claimed *universal infinite-energy deduction*; a dimensional obstruction requires a separate theorem specifying its PDE, admissible states, and rails.

5.7 Internal Separation: Conditional Language-Level Statement

Proposition 5.6 (WIN-05: conditional physical-WCC separation). Fix the same physical encoding, resource bounds, uniformity conditions, and language definition for both $\mathbf{P}^{\text{phys}}_{\text{WCC}}$ and $\mathbf{NP}^{\text{phys}}_{\text{WCC}}$. If some language L has a polynomial-resource physical WCC verifier with polynomial-size witnesses, but every physically admissible *deterministic decider* for L exceeds at least one polynomial resource rail infinitely often, then

$$\mathbf{P}^{\text{phys}}_{\text{WCC}} \neq \mathbf{NP}^{\text{phys}}_{\text{WCC}}.$$

This proposition is conditional; the requisite universal lower bound is not proved here.

Proof. The verifier hypothesis gives $L \in \mathbf{NP}^{\text{phys}}_{\text{WCC}}$. The lower-bound hypothesis gives $L \notin \mathbf{P}^{\text{phys}}_{\text{WCC}}$. Both refer to the same language and physical resource accounting, so the classes differ. □

6 Curvature collapse and loss of grammatical structure

This subsection separates a high-curvature condition from a distinct, testable mixing hypothesis on observable transitions. Exceeding a curvature rail alone does not imply the loss of a finite symbolic partition, language triviality, or inability to infer the underlying dynamics. The conditional obstruction

below applies only when a specified observational transition relation satisfies an explicit common-successor property.

Recall that for a configuration ψ we write $C_\Theta(\psi)$ for the global curvature budget (for example $C_\Theta \sim E_{\mathrm{grad}} + E_{\mathrm{fb}}$) and $\overline{C}_\Theta(\psi)$ for the average curvature per site. The WCC rails partition the evolution into regimes (PWCC, NPWCC, OUT_OF_WCC) via bounds on $(\mathrm{steps}, \mathrm{cells}, \overline{C}_\Theta)$.

Definition 6.1 (Curvature-regular and curvature-collapsed regimes). Let ψ be a WCT configuration on a finite lattice Λ and let $\overline{C}_\Theta(\psi)$ be its average curvature budget per site.

- We say that ψ is *curvature-regular* if

$$\overline{C}_\Theta(\psi) \leq C_{\mathrm{reg}}$$

  for some fixed model-relative threshold $C_{\mathrm{reg}} > 0$ compatible with the PWCC rail. In this regime the WCC update rule induces a finite alphabet S of metastable states and a well defined local transition map $U \colon S^{N_r} \to S$.

- We say that ψ is *curvature-collapsed* if

$$\overline{C}_\Theta(\psi) \geq C_{\mathrm{coll}}$$

  for some threshold $C_{\mathrm{coll}} \gg C_{\mathrm{reg}}$ above the quantum classical crossover bound and outside all WCC rails. The threshold defines a high-curvature regime but does not itself assert symbolic mixing, non-identifiability, or any language-class property.

In the curvature-regular regime the induced symbolic dynamics admits a natural language model in the sense of Chomsky's hierarchy.

Definition 6.2 (Induced grammar of a curvature-regular configuration). Let ψ be curvature-regular in the sense of Definition 6.1 with associated finite state set S and local update rule U. Fix an encoding of space-time cylinders into words over a finite alphabet Σ.

The *induced grammar* $\mathcal{G}_\psi$ is any grammar (for example, finite state, context free, or context sensitive) that generates exactly the set of words corresponding to valid WCC space-time trajectories compatible with (S, U) and the WCC rails. The associated language $L(\psi)$ is the set of all such words.

Thus in the curvature-regular regime the pair (ψ, U) determines a nontrivial language $L(\psi)$ lying somewhere in Chomsky's hierarchy (for example regular, context free, or recursively enumerable).

To test an additional obstruction to symbolic computation beyond curvature thresholds, we consider a separately specified transition relation on observations.

Proposition 6.3 (Common-successor obstruction to invariant symbolic partitions). Let X be a nonempty set of observable, coarse-grained configurations in a specified regime (which may, but need not, be curvature-collapsed). Let $R \subseteq X \times X$ be the relation of *possible one-step observable transitions*. Assume the explicit common-successor property

$$\forall x, y \in X \; \exists z \in X : (x, z) \in R \quad \text{and} \quad (y, z) \in R.$$

Then there is no nonconstant map $\sigma : X \to S$, for any finite alphabet S, that is invariant along every allowed transition:

$$(x, y) \in R \implies \sigma(x) = \sigma(y).$$

Here *nonconstant* means that there exist $x, y \in X$ with $\sigma(x) \neq \sigma(y)$; the one-block partition is explicitly excluded.

Proof. Suppose an invariant map σ is given. For any $x, y \in X$, choose z by the common-successor hypothesis. Invariance along the two allowed transitions gives

$$\sigma(x) = \sigma(z) = \sigma(y).$$

Thus σ is constant, contradicting nonconstancy. This argument does not depend on finiteness of S; it holds for any codomain. The Lean counterpart is `PVNP.PartISeparation.no_nontrivial_invariant_of_common_successor`. □

Remark (Scope of the collapse and identifiability claims). The proposition is a conditional result about *exact pathwise invariance* under a relation of possible transitions. The common-successor property must be established separately for the specific coarse-graining, observational noise, and evolution model. The numerical curvature condition $\overline{C}_\Theta(\psi) \geq C_{\text{coll}}$ does not by itself establish that property. In particular, a deterministic, invertible WCT flow cannot be silently replaced by an irreducibly mixing noisy transition relation.

The original stronger assertions, that a trajectory language must equal $\emptyset$ or Σ^* and that no observer with unbounded computation can reconstruct a unique rule U, do *not* follow from the common-successor condition or the curvature threshold. Language triviality requires a specified word encoder and a theorem about its language; non-identifiability requires a specified observation channel and two observationally indistinguishable candidate rules. These statements remain open, not corollaries of Proposition 6.3.

The already formalized equality $P_{\mathrm{WCC}}^{\mathrm{disc}} = P_{\mathrm{Turing}}$ concerns the charged uniform *discrete* model and does not identify it with the unconstrained physical WCT evolution.

7 Logical versus physical complexity

In this section we make explicit the distinction between *logical* complexity in the sense of classical Turing machines and *physical* complexity in the sense of the WCT curvature model. The central slogan is:

> *In Hilbert's universe (pure logic), computation is free. In the physical universe, computation is never free.*

7.1 The abstract Turing model: "logic is free"

We recall the classical setting. A (deterministic) Turing machine M is specified by a finite set of states Q, a finite tape alphabet Γ, a transition function

$$\delta : Q \times \Gamma \to Q \times \Gamma \times \{-1, 0, +1\},$$

and a designated start state q_0 and halting states. The *cost* of a computation in this model is defined purely as the number of applications of δ, i.e. the number of *logical steps*.

Definition 7.1 (Free logic axiom). The classical Turing model operates under the implicit *free logic axiom*:

1. Each application of δ has unit cost in the time measure, independent of the physical resources required to realize it.

2. The tape is an unbounded, perfectly stable memory, whose extension and maintenance carry no cost in the complexity measure.

3. There is no penalty in the model for entropy, energy, mass, or curvature associated with storing or transforming symbols.

Under this axiom, we say that *logic is free*: the only resource measured is the number of abstract state transitions.

Within this framework, the classical complexity classes P and NP are defined purely in terms of these free logical steps. For example, P consists of those decision problems decidable by a Turing machine in time bounded by a polynomial in the input length, and NP consists of those problems for which a proposed solution can be verified in polynomial time.

7.2 The WCT model: logic is not free

In the WCT curvature model, information is always realized as structure in a physical field configuration ψ. To store even a single bit requires a metastable distinction between two configurations ψ_0 and ψ_1, and any operation that changes or erases information must be implemented by a physical evolution governed by the curvature functional.

At the thermodynamic level, this is reflected in Landauer's principle: erasing one bit of information at temperature T in a system coupled to a heat bath requires a minimum energy dissipation of

$$E_{\min} = k_B T \ln 2,$$

where k_B is Boltzmann's constant. In particular, each logically irreversible operation carries a nonzero energetic cost.

In the WCT setting, we refine this statement by coupling it to the curvature functional C_Θ introduced earlier (see Section C). Informally, we write

$$C_\Theta(\psi) \;\sim\; E_{\text{grad}}(\psi) + E_{\text{fb}}(\psi)$$

for the relevant curvature budget of a configuration ψ. The precise normalization is not important here; only the fact that logical distinctions are realized as curvature-bearing structures.

Definition 7.2 (Physical logic). A *physical computation* in the WCT model is a trajectory $(\psi_t)_{t\geq 0}$ of field configurations governed by the WCT evolution equations and subject to the WCC rails. The *physical cost* of such a computation is measured not only in time, but also in the accumulated curvature budget,

$$\int_0^T C_\Theta(\psi_t)\, dt,$$

and the associated entropy production. We say that in this model *logic is not free*: every logical transition requires paying a curvature cost and generating entropy.

Thus, whereas Definition 7.1 explicitly ignores energy, curvature, and entropy, Definition 7.2 builds them into the notion of computation.

7.3 Nondeterminism as "knowing before knowing"

Classically, the class NP is defined via nondeterministic Turing machines or via polynomial-time verifiers: a language L lies in NP if there exists

a polynomial-time Turing machine V and a polynomial p such that for all inputs x,

$$x \in L \iff \exists\, w \in \{0,1\}^{\leq p(|x|)} \text{ with } V(x,w) = 1.$$

One often describes this informally by saying that an NP machine can "guess" a witness w and then verify it efficiently.

From the viewpoint of physical logic, this "guessing" step embodies the intuition of *knowing the answer before knowing*: the existence of w is postulated at the logical level without regard to the physical process required to generate it.

The existential quantifier in the classical verifier definition is not a physical operation that creates a witness. A particular WCT apparatus preparing an explicit target field must instead follow its admissible physical dynamics and account for initialization, evolution, and readout. The two statements concern different costs.

Hypothesis 7.1 (Dimensional lock). Let ψ^* denote a target configuration encoding a solution to a given instance of a problem. The *dimensional lock* hypothesis asserts that there is no physically admissible evolution that instantaneously jumps from a generic initial configuration ψ_0 to ψ^* while remaining within the WCC rails. Instead, any transition $\psi_0 \to \psi^*$ must traverse a path in configuration space along which the curvature budget C_Θ is accumulated in a manner consistent with the WCT evolution and the rail constraints.

Under Hypothesis 7.1, the nondeterministic "guessing" step of the classical NP definition is replaced by a *curvature path*: to obtain ψ^* one must physically construct it, paying the corresponding curvature cost.

Remark (Knowing versus not knowing). In the classical setting, one can define P as the class of problems for which there exists a deterministic polynomial-time algorithm (a direct recipe for obtaining answers), and NP as the class of problems for which proposed answers can be verified in polynomial time. The formal equality $P = NP$ would then assert that "not knowing" (search) is in fact equivalent, up to polynomial overhead, to "knowing" (direct computation).

In the WCT curvature model, the distinction between knowing and not knowing is tied to the curvature cost of constructing ψ^* from ψ_0. If the curvature complexity C_Θ required to realize the solution scales superpolynomially (for example, exponentially) with the input size, then the physical transition from "not knowing" to "knowing" carries a prohibitive cost, even if a purely logical verification exists.

This motivates measuring *preparation* and *certificate verification* separately. A hard preparation task does not alone establish a language-class separation: one must show a short, admissible certificate verifier and exclude *every* permitted polynomial-resource decision algorithm. The classical P versus NP question remains open.

7.4 Charged-encoder computation and conditional physical barriers

The Turing classes P and NP are defined by algorithmic time and witness verification. Physical implementations impose additional resource constraints, but a claim that *all* physically admissible SAT algorithms violate a curvature rail requires quantification over the entire allowed implementation class, including the encoder, initialization, evolution, and readout. A lower bound on one device or one specified trajectory does not supply such a universal claim.

For a precise model test, consider the frozen constant-field WCT submodel formalized in `PVNP.PhysicalAudit.PhysConst`. It has two distinguishable readout states, bounded amplitude, and constant physical evolution cost. The input-to-amplitude encoder must belong to the charged class `polyEnc`, consisting of indicator encoders of polynomial-time decidable languages. The Boolean decision is therefore *not free*: its cost is assigned to the encoder. For arbitrary fixed physical parameters κ, β, γ, the Lean-checked identity is

$$\mathsf{PhysConst}_{\mathrm{polyEnc}}(\kappa, \beta, \gamma) = P_{\mathrm{TM}}.$$

Proposition 7.3 (Conditional failure of a universal physical SAT barrier). In the above charged-encoder frozen model, assume hypothetically that $P_{\mathrm{TM}} = NP_{\mathrm{TM}}$. Then dense CNF-SAT admits an implementation in $\mathsf{PhysConst}_{\mathrm{polyEnc}}(\kappa, \beta, \gamma)$. Consequently, the assertion that dense SAT has no admissible implementation in this model *even if* $P_{\mathrm{TM}} = NP_{\mathrm{TM}}$ implies $P_{\mathrm{TM}} \neq NP_{\mathrm{TM}}$.

Proof. Dense CNF-SAT lies in NP_{TM}. Under the hypothetical equality it lies in P_{TM} and hence has an admissible polynomial-time indicator encoder. The frozen submodel reads its decision bit with bounded field amplitude and constant evolution cost. This proves the conditional membership. If the purported universal barrier excluded membership under the same hypothetical equality, the two claims would contradict each other. The formaliza-

tions are `PVNP.PartISeparation.PNP_SAT_physically_constant` and `PVNP.PartISeparation.category_error_claim_implies_classical`. □

Remark (Scope and remaining physical obligation). The proposition *does not* assume or prove $P = NP$, nor does it establish $P \neq NP$. It establishes a *conditional counterargument* inside one explicitly defined charged-encoder physical model. Its constant field-update cost is not constant total computational cost, because producing the encoded decision bit is charged.

A proposed stronger WCT barrier must specify a narrower, physically justified family of admissible encoders and operations, then prove a uniform lower bound on the *total* resource cost of every decision implementation in that family. The corresponding restrictions cannot be inferred solely from curvature rhetoric, and no such uniform SAT lower bound is proved here.

8 Numerical Illustration of P vs. NP–Type Behaviour in WCT

8.1 Setup and Definitions

Wave Confinement Theory (WCT) models computation as the geometric evolution of a wavefield $\psi(x,t)$ under curvature and entropy feedback. A computational problem is represented by a target coherent wave state ψ^*, and an algorithm corresponds to evolving from a trivial initial state ψ_0 under the Lagrangian dynamics:

$$\begin{aligned}\mathcal{L}_{\text{WCT}} = |\partial_\mu \psi|^2 + \kappa \left(\frac{\Box \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right)^2 \\ + \theta \left(\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right)^2 + \gamma \left(\frac{\Box \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right) \left(\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right).\end{aligned} \tag{4}$$

Within this model it is natural to introduce WCT/WCC-specific analogs of classical complexity classes in the *physical* (rail-constrained) setting:

- $\psi^* \in \mathbf{NP}^{\text{phys}}_{\text{WCC}}$: if it satisfies an entropy–coherence criterion $\sigma[\psi^*] \leq \sigma_{\text{crit}}$ and can be *verified* by a local curvature/entropy test in polynomial physical resources;
- $\psi^* \in \mathbf{P}^{\text{phys}}_{\text{WCC}}$: if it is *reachable* from ψ_0 within polynomially bounded time steps under WCT evolution obeying WCT rails.

The classical relation between **P** and **NP** is then approached via a conjectured correspondence between the *ideal* WCC classes $(\mathbf{P}^{\infty}_{\text{WCC}}, \mathbf{NP}^{\infty}_{\text{WCC}})$ and the usual Turing-machine classes; this correspondence is discussed later and is not assumed as a theorem here.

8.2 Simulation Procedure

We implemented a CuPy-accelerated simulation for grid sizes $n = 32, 64, 96$, evolving from a trivial random initial state toward a fixed coherent standing-wave solution ψ^*. The evolution followed the curvature–entropy Lagrangian above with parameters

$$\alpha = 1.5, \quad \theta = 0.0026, \quad \gamma = 10^{-120}, \quad \epsilon = 10^{-6},$$
$$\delta = 10^{-8}, \quad dt = 2 \times 10^{-4}, \quad \text{steps} = 5000.$$

The final overlap with the target state is measured as

$$\text{Overlap}(\psi, \psi^*) = \frac{\int \psi \, \overline{\psi^*} \, dx}{\sqrt{\int |\psi|^2 \, dx \int |\psi^*|^2 \, dx}}. \tag{5}$$

8.3 Results

For one representative experiment, the final average overlaps were:

Grid Size n	**Avg Final Overlap**	**Steps Simulated**
32	0.077637	5000
64	0.012087	5000
96	−0.006049	5000

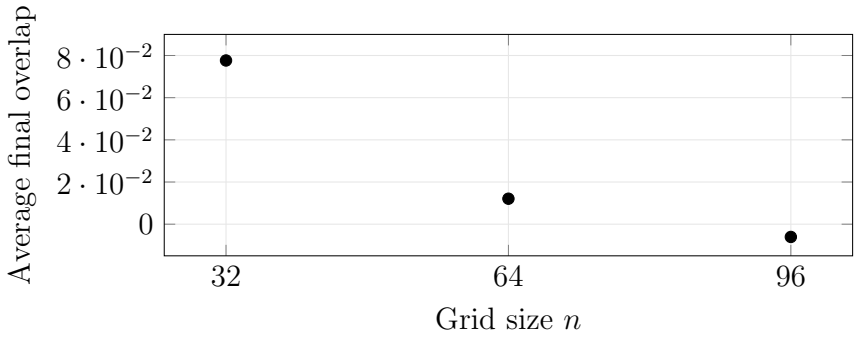


Figure 1: The three tabulated final overlaps after 5000 steps. Each point represents one reported grid size.

8.4 Heuristic Observation: Hard-to-Reach Verifiable States in WCT

For the three stated grid sizes, the recorded final overlaps decrease across runs with the same step budget. These values measure one target-preparation procedure. A bound applying to arbitrary SAT decision procedures requires a separate argument.

8.5 Interpretation

The simulation results above belong historically to the "physics-first" phase of this work: they illustrate that, in practice, certain coherent states ψ^* are difficult to reach under bounded curvature-regulated evolution, and that this difficulty grows with system size.

In the present paper, however, these numerics serve only as *motivation*. All complexity-theoretic statements are formulated symbolically. In particular:

- The conjectured separations $\mathbf{P}_{\mathrm{WCC}}^{\mathrm{phys}} \neq \mathbf{NP}_{\mathrm{WCC}}^{\mathrm{phys}}$ (physical model) and $\mathbf{P}_{\mathrm{WCC}}^{\infty} \neq \mathbf{NP}_{\mathrm{WCC}}^{\infty}$ (ideal model) are stated and analyzed at the level of symbolic transition systems, verifiers, and curvature/Kolmogorov complexity.
- Any potential bridge from $(\mathbf{P}_{\mathrm{WCC}}^{\infty}, \mathbf{NP}_{\mathrm{WCC}}^{\infty})$ to classical $(\mathbf{P}, \mathbf{NP})$ is treated as a *program* and clearly marked as conditional.

Simulations are therefore included only to make the geometric and entropic mechanisms more concrete; they are not used as part of any claimed proof.

9 Internal Separation Heuristics: Constructive Depth in WCC

In this section we isolate a model–internal notion of *constructive depth* for WCC evolutions and relate it to clause–structured NP witnesses such as 3–SAT assignments. The goal is to clarify how locality and finite–range dynamics constrain reachable configurations, and to state a natural *conjectured* internal separation

$$\mathbf{PWCC} \subsetneq \mathbf{NPWCC}$$

within the WCC model. We emphasize that any unconditional proof of such a separation, combined with the equivalences $\mathbf{PWCC} = \mathbf{P}$ and $\mathbf{NPWCC} = \mathbf{NP}$, would yield $\mathbf{P} \neq \mathbf{NP}$ and thus resolve a major open problem. Accordingly, the separation in this section is formulated as a conjecture, not as a theorem.

9.1 Symbolic Support and Constructive Depth

Let $W = (\Lambda_n, S, r, U, \mathrm{enc}, \mathrm{dec})$ be a WCC machine on inputs of length n. For any configuration $C : \Lambda_n \to S$, define its *symbolic support*

$$\mathrm{supp}(C) := \{\, i \in \Lambda_n : C(i) \neq s_0 \,\}$$

for some fixed background symbol $s_0 \in S$. Let Δ_t denote the *symbolic change set*

$$\Delta_t := \{\, i \in \Lambda_n : C_{t+1}(i) \neq C_t(i) \,\}.$$

We say a configuration C^* has *constructive depth* D (with respect to U) if any sequence of configurations

$$C_0 \to C_1 \to \cdots \to C_T = C^*$$

reachable under the local rule U has length $T \geq D$.

Lemma 9.1 (Quiescent finite-support propagation bound). Let $d \geq 1$ and let U be a radius-r local update on configurations $C : \mathbb{Z}^d \to S$, with $r \geq 0$ and a distinguished background symbol $s_0 \in S$. Assume *quiescence*: if every symbol in the radius-r neighbourhood of i equals s_0, then $(UC)(i) = s_0$. Suppose the initial support $A = \mathrm{supp}_{s_0}(C_0)$ is finite and $C_t = U^t C_0$. Then for every $t \geq 0$,

$$\mathrm{supp}_{s_0}(C_{t+1}) \subseteq \big(\mathrm{supp}_{s_0}(C_t)\big)^{+r}, \qquad \mathrm{supp}_{s_0}(C_t) \subseteq A^{+rt}.$$

For the ℓ^∞ neighbourhood on $\mathbb{Z}^d$,

$$\boxed{|\mathrm{supp}_{s_0}(C_t)| \leq |A|\,(2rt+1)^d.}$$

This is a bound on the *volume* of active support; the radius of its causal light cone grows linearly in t.

Proof. Quiescence implies that a cell can become non-background only if some cell within radius r was non-background in the preceding configuration. This gives the one-step inclusion. Inductively, repeated neighbourhood thickening is contained in A^{+rt}, since the ℓ^∞ radii add. For every initial active cell $i \in A$, its radius-rt box has $(2rt+1)^d$ lattice sites. The union of these boxes covers $\mathrm{supp}_{s_0}(C_t)$; summing their cardinalities gives the asserted inequality. The corresponding Lean theorems are `PVNP.PartILocality.supp_step_subset`, `PVNP.PartILocality.supp_iterate_subset`, and `PVNP.PartILocality.support_poly_bound`. □

The light-cone *radius* is linear, but the size of the active region can grow on the order of t^d. In dimension $d = 2$, a quiescent radius-one OR rule starting from one active cell produces the square $[-t, t]^2$ of $(2t + 1)^2$ active cells, contradicting the original linear-in-t volume estimate. This counterexample remains formalized as `PVNP.PartILocality.linear_growth_false_d2`. Without quiescence, even the one-step support inclusion fails (`PVNP.PartILocality.local_growth_needs_quiescence`). Neither locality nor the corrected bound implies a superpolynomial decision-time lower bound.

9.2 Clause–Structured Certificates and Linear Depth

Let L_{SAT} be the canonical 3–SAT language over $\{0, 1\}$. Under the Cook–Levin–style WCC reduction (see Section A), each clause $C_j = (\ell_{j1} \vee \ell_{j2} \vee \ell_{j3})$ is mapped to a local WCC region R_j of constant diameter, with acceptance determined by a curvature–locked pattern ψ_j^{sat} encoding satisfaction of that clause. Let $C(w)$ denote the certificate configuration for a witness $w \in \{0, 1\}^{m(n)}$.

For each configuration C_t on a WCC trajectory, define its *clause–signature vector*

$$v_t := \big(\sigma_1(C_t), \ldots, \sigma_m(C_t)\big) \in \{0, 1\}^m,$$

where $\sigma_j(C_t) = 1$ iff clause j is satisfied in the local region R_j of C_t and $\sigma_j(C_t) = 0$ otherwise.

Lemma 9.2 (Per-clause change requires local activity). If the jth signature bit is a function only of the configuration on R_j, a change in σ_j between two updates requires a change in at least one cell of R_j. If at most $b \geq 1$ clause signatures can change during any global round, altering k signatures requires at least $\lceil k/b \rceil$ rounds.

Proof. An unchanged local region has unchanged signature. Each round can change at most b distinct signatures by the extra stated hypothesis, so $Tb \geq k$. □

A synchronous finite-range rule does *not* impose $b = O(1)$: spatially separated clauses may change in parallel. Consequently, no unconditional $T \geq k$ bound follows from locality alone.

9.3 Heuristic Exponential Barrier and Conjecture

A formula with n Boolean variables admits 2^n candidate assignments, but this count does *not* imply that a correct algorithm must examine all candidates. Furthermore, different assignments need not induce distinct clause-signature vectors, and locality alone does not forbid simultaneous clause updates.

Conjecture 9.3 (Uniform discrete-WCC SAT lower bound). Every uniform deterministic discrete WCC machine deciding the 3-SAT *language* requires superpolynomial worst-case time on infinitely many input lengths.

Because the uniform discrete model is polynomially Turing-equivalent, this conjecture is precisely as difficult as showing $P \neq NP$ by proving that 3-SAT is not in P. It is not established by the size of a search space, a fixed WCC trajectory, or a geometric heuristic. A physical version must separately account for admissible verifier implementations.

9.4 Conditional Internal Separation

Define, as before, the language

$$L^\star := \big\{x \in \{0,1\}^n : \exists w,\ |w| = m(n),\ C(w) \text{ satisfies all clause signatures for } x\big\}.$$

By construction (cf. Section A), $L^\star$ is exactly 3–SAT encoded through the WCC certificate map, and hence $L^\star \in \mathbf{NPWCC}$.

The following statement is therefore conditional on Conjecture 9.3.

Proposition 9.4 (Conditional model–internal separation). If Conjecture 9.3 holds, then

$$L^\star \in \mathbf{NPWCC} \setminus \mathbf{PWCC},$$

and hence

$$\mathbf{PWCC} \subsetneq \mathbf{NPWCC}.$$

Proof. By construction $L^\star \in \mathbf{NPWCC}$. If Conjecture 9.3 holds, no polynomial–time WCC machine can decide $L^\star$, so $L^\star \notin \mathbf{PWCC}$. This gives the strict inclusion. □

In combination with the bijection results of Section D.4 and the identifications $\mathbf{PWCC} = \mathbf{P}$, $\mathbf{NPWCC} = \mathbf{NP}$, any proof of Conjecture 9.3 would immediately yield the classical separation $\mathbf{P} \neq \mathbf{NP}$. At present, Conjecture 9.3 should be viewed as a structured formulation of a WCC–internal manifestation of the **P** versus **NP** question.

10 Dimensional Lock: Explicit States versus Succinct Certificates

An explicit description of a physical state and a succinct certificate for an associated decision problem are different mathematical objects. Their lengths and preparation, encoding, and verification costs must not be conflated. In particular, the original dimensional-lock conjecture demanded that an *explicit* target ψ^* have superpolynomial description complexity and simultaneously act as a polynomial-size NP witness. Those requirements are incompatible under the usual literal-description convention.

Let $D : \{0,1\}^* \rightharpoonup \{0,1\}^*$ be a partial description decoder and write

$$K_D(s) = \min\{\, |d| : D(d) = s \,\}$$

for strings with at least one description. Assume a constant $c \geq 0$ such that every finite string s admits a literal description d_s with $D(d_s) = s$ and $|d_s| \leq |s| + c$.

Proposition 10.1 (Explicit-state certificate-size obstruction). For every finite string s, its description complexity satisfies

$$K_D(s) \leq |s| + c.$$

Therefore, for any polynomial witness-length budget $q(n)$,

$$K_D(s_n) > q(n) + c \quad \Longrightarrow \quad |s_n| > q(n).$$

Such a state s_n cannot itself be a polynomial-size *explicit* witness under the budget $q(n)$.

Proof. The literal description d_s is a candidate in the minimization defining $K_D(s)$, hence $K_D(s) \leq |d_s| \leq |s| + c$. Combining this with the assumed strict inequality gives the result. The corresponding Lean theorems are `PVNP.PartISeparation.Kc_le` and `PVNP.PartISeparation.high_K_not_poly_witness`. The latter uses the polynomial bound with a fixed overhead c. □

10.1 A separate construction and verification cost model

Let x denote a decision instance, w a proposed *succinct* binary certificate, and $\psi^*(x, w)$ an optional physical state associated with the pair. A proposed

physical resource model must specify all three quantities independently:

$$\begin{aligned} C_{\mathrm{prep}}(x,w) &: \text{cost of preparing } \psi^*(x,w), \\ C_{\mathrm{ver}}(x,w) &: \text{cost of checking a supplied } (x,w), \\ C_{\mathrm{dec}}(x) &: \text{cost of deciding } x \text{ without a supplied witness.} \end{aligned}$$

The certificate property is a language-level statement

$$x \in L \quad \Longleftrightarrow \quad \exists w \in \{0,1\}^{\leq q(|x|)}\, V(x,w) = 1$$

with a *uniform*, polynomial-resource verifier V. The verifier's input encoding, witness preparation if required by the *chosen* physical semantics, evolution, and readout must all fit the same rails. If verification only checks a short symbolic certificate, it need not construct the explicit target field. If instead verification is defined to demand preparation of the entire target field, then its cost must include C_{prep}. This distinction must be fixed before a physical NP claim is made.

A lower bound for C_{prep} of *one* prescribed target state does not imply a lower bound on C_{dec} for every algorithm, nor does it imply that the language belongs to the physical NP class. To obtain a physical class separation one must prove *both* (i) uniformly polynomial witness size and verification cost, and (ii) superpolynomial worst-case decision cost for *all* admissible uniform deciders, under identical charging conventions. The resulting logical implication is already formalized as Proposition 12.1; the required physical lower-bound premise is *unproved*.

Remark (Open dimensional-lock research obligation). A curvature or entropy barrier may motivate a conjecture that a specific field family is expensive to prepare. But an explicit uniform encoder, a certificate representation and decoder, a fully charged verifier, and an algorithm-independent decision lower bound must be supplied separately. The original incompatible *high-K explicit-state-as-short-witness* inference has been withdrawn. No classical or physical $P \neq NP$ theorem follows from the corrected size obstruction.

11 Curvature Bounds and Structural Lemmas: Conditional Form

WIN-06 and WIN-07. Let a physically admissible, nonvanishing field ψ have a specified curvature functional $C_\Theta[\psi]$ and a decision interface $\partial\mathcal{R}_\psi$. The following elementary implication is valid but does not establish its geometric hypotheses.

Proposition 11.1 (Conditional interface-to-curvature implication). Suppose an instance family F_n and a class of permitted encodings $\mathcal{E}_n$ satisfy, with constants independent of n and of the encoding, both

$$C_\Theta[\psi] \geq \kappa_0 \mathcal{H}^{d-1}(\partial\mathcal{R}_\psi), \quad \kappa_0 > 0,$$

and, for every admissible realization of the requisite decision computation on some input of length $\Theta(n)$,

$$\mathcal{H}^{d-1}(\partial\mathcal{R}_\psi) \geq c_0 2^{\gamma n}, \qquad c_0, \gamma > 0.$$

Then the corresponding worst-input curvature cost is at least $\kappa_0 c_0 2^{\gamma n}$.

Proof. Substitution of the second assumed inequality into the first. □

Remark. A bounded-degree expander graph does not by itself force the stated interface area in every embedding. A proof must quantify uniformly across admissible encoders, include their scales and distortions, and justify a positive interface cost for the *particular* curvature density used. This manuscript does not supply such a proof.

11.1 Per-update curvature and construction cost

Proposition 11.2 (Valid telescoping bound). If $C_\Theta[\psi_0] \geq 0$ and each allowed update obeys $|C_\Theta[\psi_{t+1}] - C_\Theta[\psi_t]| \leq \Delta_{\max}$, then for every $T \geq 0$,

$$T \geq \frac{(C_\Theta[\psi_T] - C_\Theta[\psi_0])_+}{\Delta_{\max}}$$

when $\Delta_{\max} > 0$.

Proof. The triangle inequality gives $C_\Theta[\psi_T] - C_\Theta[\psi_0] \leq \sum_{t=0}^{T-1} |C_\Theta[\psi_{t+1}] - C_\Theta[\psi_t]| \leq T\Delta_{\max}$. □

The uniform per-update bound and the initial cost must be separately justified; neither follows merely from finite propagation speed.

11.2 Padding stability

Proposition 11.3 (Conditional additive padding). If the model enforces an orthogonal, noninteracting decomposition of field domains $\Omega = \Omega_F \sqcup \Omega_{\text{pad}}$, zero curvature on the padding domain, and no interface or cross-coupling energy, then the curvature functional of that *specific* encoded configuration is unchanged by padding.

Proof. The integral over the disjoint domains is additive and the second integral is zero. □

This does not prove that the *minimum over all alternative encodings* is unchanged; nor does a superlinear padding function preserve an exponential bound in the new input length.

12 Curvature-Locked Witnesses: Existence and Verification Obligations

WIN-03, WIN-05 and WIN-07. An exact Boolean phase encoding supplies an algebraic zero-penalty witness whenever a SAT formula is satisfiable (Theorem 5.3). It does not automatically supply a stable stationary solution to a chosen WCT PDE.

For a specified energy functional $\mathcal{A}$ on a precisely defined function space, the following questions remain distinct:

1. **Attainment:** show that the set of zero-logical-penalty fields is nonempty and closed in a topology in which a coercive energy has a weakly lower semicontinuous restriction.

2. **Stationarity:** prove the constrained Euler–Lagrange conditions, and address Lagrange multipliers before claiming unconstrained stationarity of the full energy.

3. **Verification:** specify finite-precision data, tolerances, encoding length, physical time and curvature energy to establish a polynomial-resource verifier in the *same* physical model.

4. **Lower bound:** rule out all deterministic solvers, including algorithms that directly evaluate a supplied certificate without exploring an exponentially large space of alternative certificates.

Proposition 12.1 (Conditional separation from a language-level barrier). Let L be a language and fix one WCC physical resource semantics. If L has a physically admissible polynomial-time verifier for polynomial-size witnesses and no physically admissible polynomial-time decider, then $L \in NP^{\text{phys}}_{\text{WCC}} \setminus P^{\text{phys}}_{\text{WCC}}$.

Proof. Apply the two class definitions with identical resource bounds. □

Remark. A purported witness whose *energy itself* is $2^{\Omega(n)}$ is not admissible in a physical NP class requiring polynomial witness energy, even if the symbolic assignment can be checked cheaply. The distinction between verifying a Boolean string and realizing a particular high-curvature field is essential.

13 Toward a Formal P vs. NP Program via Wave Dynamics

13.1 Complexity Mapping into Wave Confinement Theory

Wave Confinement Theory represents computation as curvature-driven evolution: a wavefield $\psi(x,t)$ evolves locally under curvature and entropy constraints. This allows an NP-style verification / P-style construction analogy:

Complexity Concept	**WCT Interpretation**
Input string x	Initial configuration ψ_0
Certificate w	Succinct symbolic witness; separate from a target field ψ^*
Verifier $M(x,w)$	Charged test on (x,w), with any required physical state preparation counted
Polynomial-time algorithm	Local update: $T = O(n^k)$ curvature steps
Acceptance	Correct decision bit after charged encoding, evolution, and readout

Table 1: Mapping between classical NP verification and curvature-regulated WCT dynamics.

This motivates the model-relative WCC classes introduced earlier. In the *physical* setting these are $\mathbf{P}^{\text{phys}}_{\text{WCC}}, \mathbf{NP}^{\text{phys}}_{\text{WCC}}$; in the *ideal* (unbounded) setting they are $\mathbf{P}^{\infty}_{\text{WCC}}, \mathbf{NP}^{\infty}_{\text{WCC}}$. Our long-term complexity-theoretic target is a separation $\mathbf{P}^{\infty}_{\text{WCC}} \neq \mathbf{NP}^{\infty}_{\text{WCC}}$ in the ideal model, with the physical separation as a precursor.

13.2 Model-Level Hypothesis to Be Contradicted (Ideal WCC)

The WCT/WCC analog of the classical hypothesis $P = NP$ in the *ideal* model is:

> *Every verifiable coherent state $\psi^* \in \mathbf{NP}^{\infty}_{\text{WCC}}$ is reachable from ψ_0 by curvature-regulated symbolic evolution within polynomial depth in the ideal WCC model.*

Dimensional locking and related curvature barriers suggest this fails for certain families of coherent states.

13.3 Coherence Without Constructibility in the Ideal WCC Model

Let C_n denote the space of confined wavefields over an $n \times n$ grid in the ideal discrete WCC model. We seek $\psi^* \in C_n$ such that:

- ψ^* is verifiable by a local curvature–entropy predicate (i.e. $\psi^* \in \mathbf{NP}^{\infty}_{\text{WCC}}$),
- but for any polynomial-depth trajectory $\psi_0 \to \psi_T$, the symbolic distance $d(\psi_T, \psi^*)$ remains nonzero (i.e. $\psi^* \notin \mathbf{P}^{\infty}_{\text{WCC}}$).

Unreachability of these target states does *not* establish a decision-language complexity-class separation. The invalid inference is withdrawn in the following remark.

13.4 Conditional Curvature Separation Statement (Ideal WCC)

Remark (Withdrawn state-to-language separation step, I-25). The original proposition inferred $\mathbf{P}^{\infty}_{\text{WCC}} \neq \mathbf{NP}^{\infty}_{\text{WCC}}$ from a family of individually verifiable

but unreachable target states. This is invalid without a *language-level* all-decider lower bound. The Lean theorem `PartISeparation.state_separation_not_language_separation` exhibits an unreachable verifiable state even when the decision-language classes coincide. No separation is proved.

Discussion. The historical state-level proposition is invalid. The remaining *language-level* obligations are:

1. constructing or characterizing such families $\{\psi_n^*\}$ (dimensional lock and curvature-complexity lower bounds), and

2. proving the equivalence between ideal WCC computation and Turing computation ($\mathbf{P}_{\mathrm{WCC}}^{\infty} = \mathbf{P}$ and $\mathbf{NP}_{\mathrm{WCC}}^{\infty} = \mathbf{NP}$).

This paper provides partial progress toward (i) and a model-theoretic embedding toward (ii), but does not claim either as a completed theorem.

14 Formal Model-Theoretic Framework for P vs NP in WCT

In this section we place the WCT computational picture into an explicit symbolic framework, so that any future separation result can be stated entirely in the language of Turing machines and circuits.

14.1 Definition of the WCT Computational Model

We define the Wave Confinement Theory (WCT) computational model $\mathcal{M}_{\text{WCT}}$ as follows:

- The computational state is a complex-valued wavefield $\psi(x,t) \in C^2(\Omega \times \mathbb{R})$.
- The evolution of ψ follows a curvature-regulated Lagrangian

$$\mathcal{L}_{\text{WCT}} = |\partial_\mu \psi|^2 + \kappa \left(\frac{\Box \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right)^2 + \theta \left(\frac{\nabla^2 \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right)^2 + \dots \quad (6)$$

  or its discrete symbolic analog.
- Input to the model is given by a grid of size $n \times n$ and a trivial initial field $\psi_0(x)$.
- The system evolves under a time budget $T = \mathcal{O}(n^k)$.
- Acceptance is defined by a target field ψ^* and an overlap/entropy condition:

$$\text{Accept if } \text{Overlap}(\psi_T, \psi^*) > \tau \quad \text{and} \quad \sigma[\psi^*] \leq \sigma_{\text{crit}}. \quad (7)$$

  This defines $\mathbf{P}_{\text{WCT}}$ and $\mathbf{NP}_{\text{WCT}}$ as discussed above.

14.2 Encoding of NP-Complete Languages in WCT

Let L be any NP-complete language (e.g. 3-SAT). The WCT program is to construct a corresponding family of wavefields ψ_L^* such that:

- Each Boolean variable x_i is encoded as a resonance mode over a spatial region R_i.
- Clauses are encoded as interference or curvature constraints.
- The global wavefield ψ_L^* is entropy-coherent ($\sigma[\psi_L^*] \leq \sigma_{\text{crit}}$) if and only if the formula is satisfiable.

Formally achieving

$$x \in L \iff \psi_L^*(x) \text{ is a coherent, verifiable wavefield in } \mathbf{NP}_{\text{WCT}}$$

for all x would realize a Cook–Levin style reduction into the WCT framework. In this paper we sketch such encodings at a conceptual level; a fully formal construction is left as future work.

14.3 Constructibility Barrier under WCT Dynamics

The numerical examples measure overlap with the specified target (Section 8). A correct decider can use an alternative representation. A lower bound for every such decider remains open.

14.4 A Conditional Model-Theoretic Separation Statement

Remark (Withdrawn model-theoretic target-state separation, I-26). The historical inference from a locally verifiable but nonconstructible wavefield to $\mathbf{P}_{\text{WCT}} \neq \mathbf{NP}_{\text{WCT}}$ lacks the necessary uniform input-language and decision-procedure quantifiers. It is refuted by `PartISeparation.state_separation_not_language_separation`. A correctly conditional classical implication would require a *proved* uniform decision-language separation and a resource-preserving identification of the physical and classical models. Neither is established here.

The corrected program requires an input-uniform physical encoding of a decision language, a polynomial verifier, an all-decider complexity lower bound for the *same* language, and an independently proved cost-preserving physical-to-Turing bridge. A target-state nonconstructibility statement does not replace the lower bound.

Connection to classical complexity theory. If these steps can be completed, the resulting separation would mirror, in geometric form, the phenomena captured abstractly by the Time Hierarchy Theorem: deeper "curvature traversals" (longer evolution depth) would be provably required to construct certain verifiable configurations, beyond any fixed polynomial bound. At present, this connection should be viewed as an analogy and a research direction, not as an established theorem.

15 Binary Encoding and Turing Reduction of WCT

All statements in this section are either definitional or explicitly *conditional*. They describe how WCT-style curvature evolution can be represented in a classical Turing framework and how a putative WCT-level nonconstructibility result would *map* into standard complexity language. No unconditional claim about P vs. NP is made.

15.1 Tape Encoding of WCT States

Let the wavefield $\psi(x,t)$ over an $n \times n$ grid be represented as

$$\psi[i,j] = a_{ij} + ib_{ij}, \quad a_{ij}, b_{ij} \in \mathbb{R}.$$

We encode each complex amplitude using finite-precision binary:

$$\text{Tape representation: } \underbrace{0.101011\ldots}_{\text{Re}} \mid \underbrace{0.011010\ldots}_{\text{Im}} \#.$$

The entire field $\psi(x,t)$ is thus encoded as a binary tape of length $\mathcal{O}(n^2)$ (up to the chosen precision).

15.2 Turing Transducer for Curvature Evolution

We consider a deterministic Turing machine M_{WCT} that:

- Reads an input tape encoding ψ_t,
- Computes each grid value update using a finite-difference approximation to curvature-regulated evolution, e.g.

$$\psi_{t+1}[i,j] = \psi_t[i,j] + dt \cdot \left[\nabla^2 \psi_t - \frac{\alpha \nabla^2 \psi_t}{\psi_t + \epsilon e^{-\beta |\psi_t|^2}} - \theta \cdot \psi_t \cdot \nabla^2 (\log |\psi_t|^2) \right],$$

- Applies arithmetic operations using standard polynomial-time Turing subroutines (finite-precision rational arithmetic).

This defines a symbolic curvature-update transducer. For any fixed precision model (e.g. $b(n)$ bits per amplitude with $b(n) = \text{poly}(n)$) one can arrange that a single macro-step of the WCT update is simulated in time $\text{poly}(n, b(n))$. We treat this as a *modelling assumption* fixing the intended regime for the WCT–Turing correspondence.

15.3 Acceptance Criterion as Turing Predicate

We define a symbolic acceptance predicate

$$\text{Accept}(\psi) = 1 \iff \text{Overlap}(\psi, \psi^*) > \tau \text{ and } \sigma[\psi] \leq \sigma_{\text{crit}},$$

where:

- Overlap is computed via a normalized dot product between the encodings of ψ and ψ^*,
- Entropy $\sigma[\psi]$ is approximated by summing $-p \log p$ over the tape (with finite-precision logarithms).

Under reasonable encoding choices, this verifier can be implemented by a deterministic Turing machine using time polynomial in the tape length (e.g. $\mathcal{O}(n^2)$ for an $n \times n$ grid). Thus WCT-style verifiers embed naturally into classical **NP** predicates.

15.4 A Conditional Turing Reduction Template

We now sketch how a WCT-style nonconstructibility statement would translate into classical Turing language. We work with the model-relative classes $\mathbf{P}_{\text{WCT}}$ and $\mathbf{NP}_{\text{WCT}}$ introduced earlier.

Remark (Withdrawn target-state reduction template, I-27). Polynomial-time verification of an explicitly supplied wavefield, together with nonreachability of a prescribed target, does not exhibit a decision language in $\mathbf{NP} \setminus \mathbf{P}$. Lean's `PartISeparation.accepted_language_not_in_NP_diff_P` refutes this inference. The correct open task is an all-decider complexity lower bound for a fixed, uniformly encoded *decision language*.

The missing substantive statement is a uniform *language-level* lower bound, not difficulty constructing one selected target state. A corrected transfer would separately require a fully charged physical-to-discrete class identification.

15.5 Conclusion

The finite-precision update and readout can be used in a classical simulation once their uniform error and cost bounds are proved.

Classical Implication Template: P vs. NP (Conditional)

For one fixed binary language L, a polynomial-time verifier with polynomial witness length proves $L \in \mathbf{NP}$. If every correct deterministic decider for L exceeds every polynomial time bound on arbitrarily long inputs, then $L \notin \mathbf{P}$, and consequently $\mathbf{P} \neq \mathbf{NP}$. The second premise remains open for dense SAT.

16 Discrete Turing Reformulation and Open Nonconstructibility Goal

16.1 Classical Symbolic Encoding of WCT Computation

We define a discrete deterministic Turing model $T^{\Sigma}_{\mathrm{WCT}}$ operating over a finite binary alphabet $\Sigma = \{0, 1, \#\}$ that encodes the PWCC/WCT dynamics symbolically. Each complex amplitude $\psi[i, j] \in \mathbb{C}$ is represented as a binary pair of rational approximations:

$$\psi[i, j] = a_{ij} + ib_{ij}, \qquad a_{ij}, b_{ij} \in \mathbb{Q} \subseteq \mathbb{Z}/2^m,$$

for fixed precision parameter m.

A full grid configuration is stored on a single Turing tape in a canonical fixed-length layout (row-major order, with delimiters). One global evolution step of the field is simulated by a finite sequence of symbolic arithmetic operations (addition, multiplication, and lookup-table–based approximations to nonlinear functions) on these bitstrings.

16.2 Curvature Dynamics as an Arithmetic Circuit

The curvature-regulated update

$$\psi_{t+1} = \psi_t + \Delta t \cdot F(\psi_t)$$

is abstracted as a Boolean/arithmetic circuit F composed of:

- finite-difference Laplacian and stencil operators,
- fixed-precision multipliers and adders,
- rational-function or table-based approximations to exponentials, logs, and normalizations.

In idealized form, and for fixed precision, F can be placed in a low-level circuit class such as AC^0 or NC^1, depending on the exact gate set and fan-in assumptions. A single step of $T^{\Sigma}_{\mathrm{WCT}}$ then simulates $F(\psi_t)$ via a sequence of logically defined operations on binary strings, with per-step cost polynomial in the system size.

16.3 A Purely Symbolic Verifier

We restate the acceptance predicate in purely symbolic terms:

$$A(\psi) = 1 \iff \frac{\langle \psi, \psi^* \rangle}{\|\psi\| \cdot \|\psi^*\|} > \tau \ \wedge \ \sigma(\psi) \leq \sigma_{\text{crit}},$$

where inner products, norms, and entropy are computed from the bit-encoded amplitudes by:

- fixed-precision multipliers and adders for $\langle \psi, \psi^* \rangle$ and $\|\psi\|^2$,
- table-based (or truncated-series) approximations to log,
- prefix-sum trees to aggregate all contributions.

All subroutines used in A are intended to be realizable by polynomial-time Turing machines; the target complexity placement is $A \in \mathsf{NC}^1$ (cf. the discussion in Section A). In particular, the definition of A in this section does not depend on any continuous geometry, only on the chosen binary encoding and standard arithmetic circuits.

16.4 A Programmatic Nonconstructibility Goal

Let ψ_0 be a trivial initial configuration (e.g., all zeros) in the tape encoding, and let ψ^* be a candidate target configuration whose correctness is verifiable by A. In purely classical complexity-theoretic terms, the *programmatic goal* suggested by WCT can be phrased as:

> **Goal.** Find a family $\{\psi_n^*\}$ of bit-encoded configurations and a verifier A with $A(\psi_n^*) = 1$ in polynomial time such that, for every deterministic polynomial-time Turing machine M and all sufficiently large n,
>
> $$M(\psi_0, n) \neq \psi_n^*.$$

The inability to print one prescribed target string does not, by itself, separate decision-language complexity classes. To derive a classical separation, one must specify a uniformly encoded *language* with a polynomial-time membership verifier and rule out every polynomial-time *decider* for that same language. Only then would the classical conclusion $\mathbf{P} \neq \mathbf{NP}$ follow. In this paper, WCT provides geometric and curvature-based heuristics for constructing candidates for $\{\psi_n^*\}$; *the rigorous nonconstructibility proof in the unrestricted Turing model is left as an open problem.*

16.5 Independence from Numerical Simulation

The symbolic model $T^{\Sigma}_{\mathrm{WCT}}$ operates entirely on finite encodings with logically defined transition rules, arithmetic identities, and combinatorial estimates. Consequently:

- Any eventual nonconstructibility theorem inspired by WCT can, in principle, be stated purely in the language of Turing machines and circuits, without reference to floating-point numerics or PDE solvers;
- WCT acts as a geometric design principle for hard instance families and verification predicates, while the *final* separation (if obtained) would live entirely inside discrete complexity theory.

In its present form, this section should therefore be read as a *discrete reformulation and roadmap*: it shows how the PWCC/WCT curvature-machine model embeds into the classical symbolic framework, and isolates the precise nonconstructibility statement that must be proved to upgrade the internal PWCC–$\mathsf{NP}_{\mathrm{WCT}}$ separation to a full classical **P** vs. **NP** result.

17 Formal Reduction to Cook–Levin Framework and Universal Scope

17.1 Embedding PWCC Verifiers into the Cook–Levin Framework

The Cook–Levin theorem states that for any language $L \in \mathbf{NP}$, there exists a polynomial-size Boolean circuit $C(x, w)$ such that

$$x \in L \iff \exists w : C(x, w) = 1.$$

In the WCT-inspired setting, we consider symbolic encodings of wavefields and define a verifier predicate

$$A(\psi) = 1 \iff \text{Overlap}(\psi, \psi_L^*) > \tau \ \wedge \ \sigma[\psi] \leq \sigma_{\text{crit}},$$

implemented strictly as a rational-arithmetic circuit over the binary encoding of ψ. Its intended role is analogous to that of $C(x, w)$: to accept exactly those encodings corresponding to satisfying assignments for instances of L.

The classical *discrete* Cook–Levin tableau reduction is formalized separately in this repository, including the `CookLevinTableau` and `CookLevinUniform` modules. A corresponding uniformly charged reduction into a specified *physical WCT PDE* has not been proved and remains a separate open objective.

17.2 Universal Scope of Polynomial-Time Turing Machines

Let M be any deterministic Turing machine running in time $\mathcal{O}(n^k)$. Suppose M attempts to construct a symbolic wavefield ψ_L^* corresponding to some NP instance x, such that

$$V(x, \psi_L^*) = 1$$

for a polynomial-time classical verifier V.

In the symbolic WCT framework, the role of the verifier is played by a Turing-implementable predicate A (overlap and entropy test) computed using only:

- bounded-precision multipliers,
- prefix-sum trees,

- comparator circuits,
- and lookup-based logarithms,

A uniform depth bound for this circuit depends on the precision, table size and approximation margins specified below.

Any machine M that constructs ψ_L^* must therefore do so through a sequence of polynomial-time symbolic steps beginning from a trivial tape configuration ψ_0.

Withdrawn universal-scope inference, I-28. The statement that each prescribed constructor misses some target configuration does not imply that the associated decision language lies outside **P**. Lean's `PartISeparation.universal_scope_counterexample` refutes this inference even for a language in **P**. A valid classical separation would instead need one uniformly encoded decision language and an asymptotic lower bound against *every* correct polynomial-time decider for that language. State production and language decision must not be conflated.

17.3 Conditional Summary of the Cook–Levin Embedding

The intended theoretical path is:

- Encode NP-verifiers as symbolic rational-arithmetic circuits consistent with the WCT/PWCC encoding;
- Identify (via curvature/entropy arguments) families of states that are verifiable by these circuits but conjecturally not constructible by any polynomial-time Turing machine;
- Establish a superpolynomial decision-time bound for one such language against every correct deterministic decider. A physical bound also requires a resource-preserving transfer.

In this paper we establish only the structure of this roadmap, not the final classical consequence.

Verifier Complexity. The target complexity class of the verifier

$$A(q) = 1 \iff \text{Overlap}(q, q^*) > \tau \wedge \sigma(q) \leq \sigma_{\text{crit}}$$

has no established uniform NC^1 placement without explicit precision, logarithm approximation, arithmetic depth and threshold-margin assumptions. The existing Lean CNF-verifier theorem is not a low-depth entropy/readout circuit theorem.

18 Final Remarks on Turing Reduction and Acceptance Formalism

18.1 Symbolic Evolution Under $T^{\Sigma}_{\mathrm{WCT}}$

Let $T^{\Sigma}_{\mathrm{WCT}}$ be the discrete Turing machine defined earlier. A symbolic tape configuration for a small 3×3 grid at 8-bit precision is a block

$$q_t = [\texttt{01000011} \mid \texttt{00110010} \mid \cdots],$$

representing real/imaginary components.

Each update step performs:

- finite-difference Laplacian computation on the tape,
- symbolic evaluation of curvature-feedback terms via lookup tables,
- arithmetic composition into the next configuration q_{t+1}.

All gates are bounded fan-in, and the symbolic update is intended to be simulable in uniform $\mathbf{P}$.

18.2 Formal Acceptance Predicate

We define:

$$A(q) = 1 \iff \mathrm{Overlap}(q, q^*) > \tau \ \wedge \ \sigma(q) \leq \sigma_{\mathrm{crit}},$$

where:

- q is the symbolic Turing encoding of a wavefield,
- Overlap is a normalized inner product (prefix-sum tree),
- Entropy is computed as a discrete approximation of $\sum -p_i \log p_i$.

Lemma 18.1 (Intended Verifier Role). For each NP instance x, the symbolic goal is to construct a configuration q^* such that

$$x \in L \iff A(q^*) = 1.$$

This matches the Cook–Levin interpretation of a verifier state, though no equivalence theorem is yet claimed.

18.3 Diagonalization and Proof-Barrier Context

The WCT-inspired PWCC model is designed with specific complexity-theoretic concerns in mind:

- **Relativization:** The model uses no oracle; all arguments occur within a fixed symbolic transition system.
- **Natural Proofs:** Any envisioned separation relies on specific, structured configuration families, not on large, uniform, efficiently checkable properties—thus avoiding the typical Razborov–Rudich template.
- **Constructivity Constraints:** Conjectured lower bounds arise from symbolic curvature, entropy, and Kolmogorov complexity restrictions rather than adversarial Boolean structure.

These observations only position the framework relative to classical barriers; they do not guarantee their circumvention.

18.4 Model-Level Conclusion

The symbolic WCT/PWCC framework:

- operates entirely within discrete Turing computation once the encoding is fixed;
- admits verifiers plausibly contained in low-depth circuit classes (NC^1);
- offers a geometric mechanism (dimensional lock / entropy barrier) suggesting the existence of a verifiable state beyond the reach of polynomial-time constructors.

Natural-Proof Barrier Positioning

This framework is not designed to exhibit a "natural" property in the Razborov–Rudich sense: it does not deploy a large, uniform, efficiently checkable Boolean property. Instead, it focuses on specific curvature-locked symbolic configurations constrained by entropy and resonance. Whether this ultimately avoids all known barriers is an open question; here we only articulate the intended positioning.

Logical Dependency Roadmap

Conceptual dependency outline:

- **Verifier Construction ($\mathbf{NC}^1$ Target):** Build a curvature/entropy-based verifier A in a low-depth circuit class.
- **SAT/WCT Encoding:** Give a formal encoding of 3-SAT into wave configurations ψ^* accepted by A.
- **Constructibility Barrier:** Establish a Kolmogorov/entropy-based lower bound showing some ψ^* cannot be constructed in polynomial time.
- **Generalization to All Poly-Time Machines:** Lift the barrier from a specific algorithm to all $M_i \in \mathbf{P}$.
- **Hierarchy/Separation Step:** Prove a superpolynomial lower bound against all correct deterministic deciders of one fixed NP language.

This roadmap is intended as a research program rather than a completed proof. If each step can be made precise and rigorously established, the standard implication $\mathbf{P} \neq \mathbf{NP}$ would follow in the usual way.

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A Symbolic 3-SAT Witness Encoding and Complexity Scope

The corrected literal scores in Section 5.3 give an exact polynomial-size encoding of the Boolean *verifier* relation. Given a 3-CNF formula F and an assignment a, checking $E_{\text{logic}}[a] = 0$ requires only $O(m)$ three-literal clause checks when F has m clauses. This establishes a familiar polynomial-time witness relation, not a new deterministic lower bound for SAT.

Proposition A.1 (Uniform symbolic verification). For an explicitly represented 3-CNF formula F with m clauses and n variables, the predicate

$$V(F, a) := \mathbf{1}\{E_{\text{logic}}[a] = 0\}$$

can be evaluated by a uniform deterministic algorithm in polynomial time. If the formula and witness are supplied as ordinary bitstrings, the algorithm checks no more than $3m$ signed literals after accessing the corresponding assignment bits.

Proof. Evaluate each signed literal from its input bit, compute the OR of three literal values, and AND the results for all clauses. A standard multitape Turing implementation is polynomial-time. □

A.1 Open constructive-reachability problem

The fact that $V(F, a)$ is efficiently computable does not decide whether $\exists a\, V(F, a) = 1$ admits a polynomial-time algorithm. Likewise, a lower bound for the complexity of one prescribed WCT trajectory or one clause-signature enumeration algorithm does not exclude all admissible deciders. The proposed geometric obstruction remains a quantified, language-level research question.

A.2 Nonuniformity and diagonalization limitation

Enumerating all polynomial-time algorithms and choosing an output not generated by a particular machine does not by itself produce an NP language outside P. The existence of an efficiently *verifiable* witness for an efficiently describable *language* must be shown, not inferred from the complexity of an individual prescribed state. Generic time-hierarchy theorems do not imply $P \neq NP$.

B Complexity Class Comparison Table in WCT

Wave Confinement Theory (WCT) provides a physically motivated dynamical model in which computation proceeds through curvature-regulated, locally updated wavefield evolution. In this setting, classical complexity notions have natural analogues. The table below summarizes the heuristic correspondence between classical classes and their WCT interpretations.

Classical Complexity Class	**Definition**	**WCT Interpretation (Model-Relative)**
P	Languages decidable in polynomial time by a deterministic Turing machine	States reachable from the trivial field ψ_0 via $T = \text{poly}(n)$ curvature-regulated update steps in the PWCC/WCT model (i.e. P_{st})
NP	Languages whose certificates are verifiable in polynomial time	Coherent target states ψ^* verifiable by a log-depth symbolic predicate (e.g. an NC^1 overlap–entropy check), corresponding to NP_{st}
NP \ P	NP-verifiable languages not in P (if any; existence unproved)	Conjectural curvature-locked states ψ^* that satisfy a verifier but may be *not reachable* from ψ_0 in $\text{poly}(n)$ PWCC evolution (i.e. $\psi^* \in NP_{\text{st}} \setminus P_{\text{st}}$)
EXP	Languages decidable deterministically in time $2^{\text{poly}(n)}$	Target states whose symbolic constructive depth under PWCC dynamics grows as $\Omega(2^n)$ (exponential-length curvature paths in configuration space)

Table 2: Heuristic correspondence between classical complexity classes and wave-confinement computation in the PWCC/WCT model. These identifications are *model-relative* and are not claimed to imply any classical separation such as $\mathbf{P} \neq \mathbf{NP}$ in the unrestricted Turing framework.

The state sets P_{st} and NP_{st} refer to reachability and verification. The lan-

guage classes in the first column classify input decision problems. Their comparison requires quantification over deciders.

C Semantic Curvature Complexity: Conditional Lower-Bound Program

The curvature of an individual field and the semantic cost of a *language* are different objects. Fix a physical model $\mathfrak{W}$, its uniform encoding/readout convention, permitted algorithms, and a nonnegative per-state curvature functional $\Phi(\psi)$. For an algorithm M and input x write

$$\mathsf{Cost}_\Theta(M,x) := \sum_{t=0}^{T_M(x)} \Phi(\psi_{M,x}^t).$$

A pointwise optimization of language-level physical cost takes an infimum *over all admissible deciders*, with a worst-case input cost:

$$\mathsf{C}_{\Theta,\mathfrak{W}}(L,n) := \inf_{M \text{ decides } L} \max_{|x|\le n} \mathsf{Cost}_\Theta(M,x).$$

Empty algorithm classes have cost $+\infty$ by convention. Any claim of an exponential universal lower bound must justify its algorithm-independent quantifiers and clarify whether Φ counts instantaneous energy, total time-integrated expenditure, or both.

Quantifier order and single-tape first-answer time. Fix a correct deterministic one-tape dense-SAT decider M. Put $b_M = 1 + \lfloor \log_2(|\Lambda_M \times \mathrm{Dir}|+1)\rfloor$, where Λ_M denotes its finite control states, and $N(j) = 18j2^j + 6j^3$. For every budget $H \ge N(b_M+4)$ there is a rejected word x with

$$H/3 < |x| \le H, \qquad H^2 \le 3456 b_M\, T_M(x).$$

To prove this, choose the largest j with $N(j) \le H$. For $j \ge 4$, expansion of the two polynomials and $2^{j+1} = 2\,2^j$ give $N(j+1) \le 3N(j)$. The matrix crossing argument supplies a rejected word of length $N(j)$ with $N(j)^2 \le 384 b_M T_M(x)$. Combining the inequalities proves the stated bound. This strengthens the frequency of the quadratic guarantee. The superpolynomial exponent in Conjecture 9.3 remains open.

If the machine can instead be chosen after the horizon, the first-answer time has a different bound. For every decidable language and every H there exists a globally correct one-tape machine D_H with $T_{D_H}(x) \le 2|x| + 5$ on all $|x| \le H$. Its finite control holds the complete table through H. One round trip reads a short word and returns its table entry; longer inputs invoke the original decider. The description size is unrestricted and can depend on H. Thus a pointwise infimum over machines of first-answer times has a linear upper bound even on decidable languages with large uniform time complexity. A physical cost including machine preparation requires separate accounting. The checked statements are `quadratic_every_budget` and `fast_through_horizon`.

C.1 Interface and expander hypotheses

Hypothesis C.1 (Uniform interface price). For all admissible fields used by a specified class of machines, $\Phi(\psi) \ge \kappa_0 \mathcal{H}^{d-1}(\partial \mathcal{R}_\psi)$ with $\kappa_0 > 0$ independent of the input length and encoding.

Remark (Addendum, 12 November 2026: the hypothesis fails on growing tori). In the frozen model the field lives on a torus of side L_n. The exact solution $\psi(t, x) = A\, e^{i(2\pi x_1/L - \omega t)}$ of the frozen equation has the same amplitude $|A|$ for every L, so its quadrature readout $\operatorname{Re}\psi$ reaches $\pm|A|$ with a fixed margin. For every curvature operator with $|\Theta[\psi]| \le c|\Delta\psi|$, including Δ and the operator Θ_ε of Part III,

$$\Phi(\psi_t) \le 4\pi^2|A|^2 + 16\pi^4 c^2 |A|^2/L^2,$$

while the frontier of $\{\operatorname{Re}\psi_t > 0\}$ in the fundamental domain has Euclidean length at least L, at every time. Hence no $\kappa_0 > 0$ independent of the input length exists for the frozen trajectory class (`SeparatorFrozen.no_uniform_separator_price`). These fields are used by an admissible decider whose encoder writes the decision bit as the sign of A (`SeparatorFrozen.stripe_device_violates`), and a modulus readout fails in the same way (`SeparatorFrozen.no_uniform_price_modulus`). What survives is a width-dependent price: a transition from $\le -\eta$ to $\ge \eta$ across width w costs at least $4\eta^2/w$ per unit length, and this is sharp (`SeparatorFrozen.layer_price`). A uniform κ_0 therefore requires a uniform bound on transition widths, as in the mesh price below.

Hypothesis C.2 (Encoding-uniform expander interface growth). For every permitted decider and encoding of a hard-instance family, there is a worst-case input whose *required* field trajectory incurs separator measure at least $c_0 2^{\gamma n}$ with constants independent of the algorithm.

Remark (Addendum, 12 November 2026: the expander hypothesis is as hard as $P \neq NP$). The charged constant devices of the frozen model, whose encoder writes a polynomial-time decision bit as a spatially constant amplitude, have no decision interface at any time. Consequently, for dense SAT in the class of these devices the hypothesis above is equivalent to $P_{\mathrm{TM}} \neq NP_{\mathrm{TM}}$, and in every device class containing them it implies $P_{\mathrm{TM}} \neq NP_{\mathrm{TM}}$ (`ExpanderInterface.expanderGrowth_SAT_iff`, `ExpanderInterface.expanderGrowth_implies_P_ne_NP`). For a language decidable in polynomial time it is false.

Neither hypothesis follows merely from graph expansion or from counting disconnected truth assignments. An expander on $\Theta(n)$ vertices has only $O(n)$ edges at bounded degree; an exponential area bound requires an additional geometric embedding theorem with explicit distortion, domain volume, and feature-width controls.

A verified mesh price. Count each selected face e once, give it area $a \geq 0$ and dual volume ha with $h > 0$, and let J_e be the norm of its field difference. If each face of a decoded cut has $J_e \geq \delta \geq 0$, then

$$\sum_e (J_e/h)^2 ha + R \;\geq\; (\delta^2/h)\, a\, \#\mathrm{cut} \quad (R \geq 0).$$

Sum the pointwise squared differences on the cut and use nonnegativity on its complement. A scalar K-Lipschitz readout ($K > 0$) on opposite sides of a threshold with margins η gives $J_e \geq 2\eta/K$. Uniform constants follow when h and K have fixed upper bounds and η has a fixed positive lower bound. On the two-level graph encoding with amplitudes $1, 2$, the absolute graph Laplacian at a vertex equals its number d_i of oppositely labeled neighbors. The quadratic energy therefore lies between $\sum_i d_i$ and $d_{\max} \sum_i d_i$. These estimates apply to an observed interface. The obligation to force such an interface on every admissible decider remains in Hypothesis C.2.

Proposition C.1 (Conditional product of two lower bounds). Under the two hypotheses above, if the interface cost is incurred on at least one step of every

admissible decider for a language L at input length n, then

$$\mathsf{C}_{\Theta,\mathfrak{W}}(L,n) \geq \kappa_0 c_0 2^{\gamma n}.$$

Proof. For each algorithm, take the hard input and the step specified in the hypotheses. Its per-state cost bounds the nonnegative trajectory sum. Finally take the infimum over the algorithms. □

Remark (WIN-06). Restricting attention to solvers that explicitly or implicitly *enumerate clause-signature sectors* yields only a lower bound for that solver subclass, unless every correct solver is independently proved to belong to it. Direct clause evaluation on a supplied assignment is a counterexample to an alleged need to enumerate satisfying assignments during verification.

C.2 Classical complexity implications

Proposition C.2 (Conditional bridge; no classical separation claimed). If one proved $P^{\mathrm{disc}}_{\mathrm{WCC}} = P$ and $NP^{\mathrm{disc}}_{\mathrm{WCC}} = NP$ for the *same* uniform discrete language classes, then a genuine separation of those *same* classes would imply $P \neq NP$. A separation only of a restricted physical class does not produce the latter conclusion without additional class-identification theorems. The discrete equalities do not preserve curvature energy by themselves.

Proof. Substitute the class identities when they refer to precisely the same classes. The second statement follows because changing the resource constraint changes the class being quantified over. □

C.3 Status of the proposed SAT curvature lower bound

The earlier locked-expander exponential estimate, gate-stencil upper bounds, and padding reductions require independent analytic proofs of uniform PDE existence, fixed readout margins, regularized-operator denominator control, encoding closure, and the universal interface property. They are not asserted as established results in this revision. The model remains available for such investigations, but a claimed general SAT lower bound or "non-natural, non-relativizing" resolution is premature.

D Wave Curvature Computation and Curvature–Bounded Complexity

D.1 Wave Curvature Computation Model

We model computation as the evolution of a confined wavefield ψ under curvature-regulated dynamics on a finite lattice. Let $\Lambda_n \subset \mathbb{Z}^d$ be a finite grid associated to input size n, and let

$$\mathrm{Conf}_n := \{\psi : \Lambda_n \to Cl_{1,3}\} \tag{8}$$

denote the configuration space, where $Cl_{1,3}$ is the spacetime algebra and $\psi(x)$ encodes a local multivector degree of freedom. Following Leary's GA encoding, classical bit strings are embedded into $Cl_{1,3}$–valued configurations by an injective map

$$\mathrm{Enc} : \{0,1\}^* \to \bigcup_{n\geq 1} \mathrm{Conf}_n, \qquad x \mapsto \psi_x, \tag{9}$$

where the logical value of each bit is represented by a choice of multivector component or rotor phase in a fixed subspace of $Cl_{1,3}$.

A *Wave Curvature Computation (WCC) machine* is a tuple

$$\mathcal{M} = (\{\Lambda_n\}_{n\geq 1}, U, \Theta, \mathcal{R}), \tag{10}$$

where:

- U is a local update operator implementing a discretized WCT evolution step
$$U : \mathrm{Conf}_n \to \mathrm{Conf}_n, \qquad \psi^{t+1} = U(\psi^t),$$
derived from the curvature-regularized Lagrangian $\mathcal{L}_{\mathrm{WCT}}$;
- $\Theta[\psi]$ is the WCT curvature operator (e.g. discrete Laplacian divided by $g(\psi)$);
- $\mathcal{R}$ is a set of *rails* (physical constraints) bounding curvature, entropy, and amplitude:
$$E_{\mathrm{WCT}}[\psi^t] \leq K(n), \qquad \|\psi^t\| \leq B(n), \qquad t = 0, 1, \ldots, T(n),$$
for some specified resource bounds $K(n)$, $B(n)$.

We write ψ_x^t for the configuration at time t on input x, and we assume that, after $T(n)$ steps, a distinguished cell or region encodes the output bit $f(x)$.

D.2 Semantic Curvature Complexity

For a Boolean function $f : \{0,1\}^* \to \{0,1\}$, we define its *semantic curvature complexity* with respect to a WCC model $\mathcal{M}$ as

$$C_\Theta^{\mathcal{M}}(f,n) := \inf_{\mathcal{M} \text{ computes } f} \sup_{|x|\le n} \sum_{t=0}^{T(n)} \Phi\big(\psi_x^t\big), \tag{11}$$

where Φ is the cumulative curvature functional

$$\Phi(\psi) := \sum_{y\in\Lambda_n} \Big(|\nabla\psi(y)|^2 + |\Theta[\psi](y)|^2\Big)\,\Delta V, \tag{12}$$

and the infimum is taken over all WCC machines $\mathcal{M}$ that compute f on inputs of length at most n with acceptance/rejection determined by a fixed readout convention.

Thus $C_\Theta^{\mathcal{M}}(f,n)$ quantifies the minimal total curvature cost needed to realize f up to size n within the WCT dynamics. We regard a family $f = \{f_n\}$ as *curvature-efficient* if $C_\Theta^{\mathcal{M}}(f,n)$ grows at most polynomially in n, and *curvature-hard* if it requires super-polynomial growth in n for any $\mathcal{M}$.

D.3 Curvature-Bounded WCC Complexity Classes

We define curvature-bounded WCC analogues of P and NP.

Definition D.1 (Class $\mathrm{P}_{\mathrm{WCC}}$). A language $L \subseteq \{0,1\}^*$ lies in $\mathrm{P}_{\mathrm{WCC}}$ if there exists a WCC machine $\mathcal{M}$, a polynomial p, and a polynomial curvature bound K such that, for all x:

1. $\mathcal{M}$ halts on input x in at most $T(|x|) \le p(|x|)$ steps;
2. the curvature rails are respected: $E_{\mathrm{WCT}}[\psi_x^t] \le K(|x|)$ for all $t \le T(|x|)$;
3. the output cell encodes 1 iff $x \in L$.

Definition D.2 (Class $\mathrm{NP}_{\mathrm{WCC}}$). A language $L \subseteq \{0,1\}^*$ lies in $\mathrm{NP}_{\mathrm{WCC}}$ if there exists a polynomial q and a curvature-bounded WCC verifier $\mathcal{V}$ such that, for all x:

1. if $x \in L$ then there exists a *witness configuration* $\omega_x \in \mathrm{Conf}_{q(|x|)}$ (or equivalently a witness bitstring w with $|w| \le q(|x|)$ and associated GA encoding) such that $\mathcal{V}$ accepts (x, ω_x) in polynomially many steps and with polynomial curvature cost;

2. if $x \notin L$ then for all ω the verifier rejects (x, ω) under the same curvature rails.

In this framework, a *toroidal locking witness* for a 3-SAT instance is a curvature-locked configuration encoding a toroidal ψ–electron structure whose phase-locking and flux constraints correspond to the satisfaction of all clauses. Nonexistence of such a locking configuration is equivalent to unsatisfiability. *Addendum, 12 November 2026.* The locking constraints are not defined here, so this sentence is not a theorem. For the frozen dynamics there is an obstruction: the flow is time-reversible, so a solution that converges to a nonempty, Lyapunov-stable lock set that is invariant under complex conjugation already starts in that set (`Torus.no_locking`). This covers the phase orbit of every binary-phase state. A robust locking realization therefore needs dissipative or forced dynamics.

D.4 Discrete Equivalence with Classical P and NP

The following statements concern *uniform finite-state local-update* machines with polynomial lattice size and polynomially many updates; physical curvature budgets are *not* part of these definitions.

Definition D.3 (Uniform discrete WCC machine). A discrete WCC machine has a fixed finite alphabet S, fixed finite local interaction radius, fixed local update rule, and lattice $\Lambda_n \subset \mathbb{Z}^d$ of size $\mathrm{poly}(n)$. It additionally has a *uniform* polynomial-time procedure to generate the lattice and initialize the encoded input, and a uniform polynomial-time readout of a halted configuration. The rule and readout do not contain nonuniform advice or an oracle for the language being decided. Time counts all polynomial encoding and readout work as well as the local updates.

Theorem D.4 (Uniform discrete P equivalence). For the uniform discrete WCC model just defined, $P^{\mathrm{disc}}_{\mathrm{WCC}} = P$.

Proof. A deterministic Turing machine running in n^k steps touches at most $O(n^k)$ tape cells and is simulated by a fixed radius-one finite-state cellular rule using a constant number of local rounds per tape transition. The initial tape and final halt predicate are uniformly constructible. Conversely, a Turing machine can store a polynomial-size WCC configuration and simulate each of its polynomially many finite-radius rounds in polynomial time, including encoding and readout. Both transformations have polynomial overhead. □

Theorem D.5 (Uniform discrete NP equivalence). If $NP^{\mathrm{disc}}_{\mathrm{WCC}}$ is defined using polynomial-size witness strings and uniform polynomial-time discrete WCC verifiers, then $NP^{\mathrm{disc}}_{\mathrm{WCC}} = NP$.

Proof. Apply the two polynomial-overhead simulations to the *deterministic verifier* acting on each pair (x, w); existentially quantify only over witnesses of polynomial length. □

Remark (Physical resource bridge remains unproved). The existence of a finite-state cellular realization of a Turing machine is not a proof that a prescribed nonlinear PDE implements the same transitions while obeying polynomial WCT curvature energy, physical time, spectral bandwidth, and stability constraints. Thus no equality between $P^{\mathrm{phys}}_{\mathrm{WCC}}$ and $P^{\mathrm{disc}}_{\mathrm{WCC}}$, or between the corresponding NP classes, is derived by the above simulation theorems.

D.5 Curvature Blow-Up and Locked Expander Families: Open Obligation

WIN-06. The earlier suggested bound for solvers enumerating clause-signature sectors only concerns algorithms restricted to that form of search. No proof currently rules out deterministic algorithms that never enumerate those sectors. To establish a meaningful universal complexity bound, one must prove that *every* physical WCC decider for a well-specified language incurs super-polynomial curvature or time cost. The interface hypotheses in Section C are explicit candidates, not consequences of ordinary graph expansion.

D.6 Supplied-Assignment Language and Corrected Complexity Consequence

WIN-08. For a fixed uniformly polynomial-time constructible family of 3-CNF formulas F_n, consider

$$L^\star = \{(n, a) : a \in \{0,1\}^n \text{ and } F_n(a) = 1\}.$$

Theorem D.6 (Supplied-assignment membership is in P). If F_n has polynomial size and can be generated in polynomial time from an ordinary efficient encoding of n, then $L^\star \in P$ when input length accounts for the encoding and output assignment. The version with (F, a) explicitly supplied is also in P, without any uniformity condition on the formula family.

Proof. Generate F_n if needed, then evaluate its at most polynomially many clauses under the *supplied* assignment a. Checking each fixed-width clause takes polynomial time in the input length. For explicit (F, a), simply scan the clauses. No enumeration of candidate assignments occurs. □

Lean scope of I-40. The formal theorem `SATV.V_SAT_mem_P_TM` covers the explicit formula-and-assignment verification language. The uniformly generated F_n variant requires the separately stated polynomial-time generator assumption and a composition argument; that additional composition is not independently kernel-checked here.

The originally proposed language $L^\star$ therefore cannot be used to demonstrate a classical $NP \setminus P$ language. A physical curvature lower bound for a highly constrained encoding of this verification task would measure an overhead of that physical implementation, not classical SAT search complexity.

D.7 Precise conditional physical separation template

Proposition D.7 (Conditional physical separation). Fix a language L (not a set of individual target fields) and a single physical resource accounting for both deterministic deciders and witness verifiers. Suppose:

1. there is a uniform polynomial-resource physical WCC verifier for (x, w) and some polynomial witness-length bound, with $x \in L \iff \exists w\, V(x, w) = 1$; and

2. for every uniform physical WCC deterministic decider M of L and every polynomial bound p, there are arbitrarily large input lengths n and inputs x of length n on which M exceeds $p(n)$ units of time or the specified curvature/energy resource.

Then $L \in NP^{\text{phys}}_{\text{WCC}} \setminus P^{\text{phys}}_{\text{WCC}}$.

Proof. The first premise is the definition of physical NP membership, and the second is the negation of physical polynomial-time membership for the same language and model. □

Remark. The proposition supplies a logical dependency, not a proof of its second premise. The quantifiers are asymptotic: the input threshold may depend on M and p. Requiring that *at a single fixed* n an execution exceed

every polynomial bound would be impossible for a finite-energy finite-state run.

D.8 Part I repair register

Tag	Corrected status
`WIN-01`	Negated literal sign corrected.
`WIN-02`	OR clause penalty corrected to a product.
`WIN-03`	Logical satisfaction separated from PDE stationarity.
`WIN-04`	Invalid dimensional-lock inference replaced with a counterexample.
`WIN-05`	Internal separation explicitly conditional and language-level.
`WIN-06`	Search-restricted expander barrier not generalized to all solvers.
`WIN-07`	Universal curvature lower bound marked open; asymptotic quantifiers corrected.
`WIN-08`	Supplied-assignment language proved polynomial-time decidable.

D.9 Revision and Computational Review Note

Formatting, reference compilation, and structural consistency checks were performed with the assistance of automation tools, including ChatGPT, used solely for document preparation and stylistic review. All mathematical derivations, proofs, and theoretical conclusions are entirely the author's own.

Part II: Discrete WCC and Turing Equivalence

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Discrete Wave-Constrained Computation and Classical Complexity: Turing Equivalence for P and NP

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Abstract

Wave Confinement Theory (WCT) motivates a discrete model of computation in which information is stored and processed by local interactions between finitely many metastable states arranged on a lattice. In this paper we isolate the resulting *wave-constrained computation* (WCC) model as a finite-state, finite-range local-update machine and compare its complexity classes to the classical ones.

First, we define a *uniform* discrete WCC machine model with fixed finite alphabet, bounded-radius local updates, polynomial-size constructible lattices, and polynomial-time input encoders and output decoders. We explicitly simulate polynomial-time deterministic Turing machines with constant-factor overhead in the number of global WCC rounds and polynomial overhead in total work; conversely, each such WCC machine has a polynomial-time Turing simulation. Thus $P^{\text{disc}}_{\text{WCC}} = P_{\text{Turing}}$.

We then lift this equivalence from decision procedures to verifier-based nondeterministic computation. We define NP_{WCC} via polynomial-time WCC verifiers with polynomially bounded witnesses, and prove that $NP_{\text{WCC}} = NP_{\text{Turing}}$. Taken together, these results show that the uniform discrete abstraction realizes exactly the classical P and NP classes. Exact symbolic representation of wave trajectories needs

a compatibility condition on quantization and local evolution. Additional physical curvature, energetic and PDE constraints are not covered by the discrete complexity-class equalities; the physical capacity estimates are conditional on their specific dynamical hypotheses.

Verification and model boundary. The verified discrete results and the historical changes to their scopes are collected in the single current formalization-status section, Section 5.4. The uniform finite-state class results concern charged encoding, decoding and lattice generation; they do not prove polynomial-resource realization by a chosen nonlinear PDE. Encoding and decoding costs are indispensable: unrestricted receivers can make capacity statements computationally vacuous.

0.1 Resource Model Summary

For convenience, we summarize the resource model used throughout the definition of discrete WCC machines. Let $x \in \Sigma^n$ be an input of length n.

1. **Space (Finite Lattice).** Computation takes place on a finite lattice $\Lambda_n \subset \mathbb{Z}^d$ (with fixed d) satisfying

$$|\Lambda_n| \leq p(n)$$

   for some polynomial p. The cardinality $|\Lambda_n|$ represents the space cost of the machine.

2. **Precision (Finite State).** The state set S is finite. Each cell holds an element of S, representing at most $\ell = \lceil \log_2 |S| \rceil$ bits of information. No operation increases ℓ; precision remains fixed and finite throughout computation.

3. **Locality (Neighborhood Radius).** A fixed interaction radius $r \in \mathbb{N}$ defines neighborhoods

$$N_r(i) = \{j \in \Lambda_n : \|i - j\|_\infty \leq r\}.$$

   Each update applies a local rule $U : S^{N_r} \to S$.

4. **Determinism and Time.** Computation proceeds in synchronous, deterministic rounds via U. For a WCC machine W, the time cost on inputs of length n is

$$T_W(n) = \max_{x \in \Sigma^n} \min\{t \geq 0 : \mathrm{dec}(C_t) \in \{0,1\}\}.$$

   Given C_0 and U, the entire trajectory $(C_t)_{t \geq 0}$ is uniquely determined (deterministic replay).

5. **Total Configuration Space.** Since $|\Lambda_n|$ and $|S|$ are finite, the global configuration space S^{Λ_n} is finite. This precludes hypercomputation via infinite-dimensional state spaces.

The machine descriptions, lattice enumerators, encoders, and decoders are fixed for a language and execute in polynomial time. This uniformity is essential: otherwise an unrestricted input encoding could decide the language before the first WCC update.

Lemma 0.1 (Precision Stability)**.** *Let $W = (\{\Lambda_n\}, S, r, U, \mathrm{enc}, \mathrm{dec})$ be a discrete WCC machine. If $C_t : \Lambda_n \to S$ is representable using $\ell = \lceil \log_2 |S| \rceil$ bits per cell, then so is C_{t+1} obtained by the local update rule U.*

Proof. For each $i \in \Lambda_n$,

$$C_{t+1}(i) = U(C_t|_{N_r(i)}).$$

Since $U : S^{N_r} \to S$ is a total function whose image lies in the same finite state set S, its outputs require no more than $\lceil \log_2 |S| \rceil$ bits. No arithmetic involving unbounded precision is performed. Thus, the precision requirement remains constant, preventing accumulation of infinite information during evolution. □

Proposition 0.2 (Local Implementability of TM Transitions)**.** *Let M be a deterministic single-tape Turing machine with transition function*

$$\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}.$$

Under the two-phase head/symbol encoding of Section 1, each transition of M is realizable by a WCC local update rule with radius $r = 1$.

Proof. In the simulation, each tape cell is encoded as a WCC state $(\gamma, q^{\#}, p) \in S$, where $\gamma \in \Gamma$, $q^{\#} \in Q \cup \{\perp, (q, L), (q, R)\}$ encodes the head state or a directional mark, and $p \in \{0, 1\}$ is a phase bit.

To realize a TM transition, the rule U need only inspect the cell currently holding the head and its immediate neighbors (to which the head may move in the next phase). These cells lie inside the radius–1 neighborhood $\{i - 1, i, i + 1\}$.

Given (q, γ) at the head location, U rewrites the symbol, enters the next state, and places a directional mark for the subsequent phase. Movement of the head by one cell is therefore computed using only local information. Hence, all TM transitions are locally implementable in the WCC model. □

Proposition 0.3 (Deterministic Evolution and Replay)**.** *Let W be a discrete WCC machine. Then the global evolution*

$$C_{t+1}(i) = U(C_t|_{N_r(i)})$$

is deterministic. Consequently, given C_0 and U, the entire trajectory $(C_t)_{t\geq 0}$ is uniquely determined and can be reconstructed (replayed) by any verifier.

Proof. Since U is a function from the finite domain S^{N_r} to the finite set S, each cell update is uniquely determined by the configuration of its neighborhood. Thus C_{t+1} is uniquely determined by C_t.

By induction, the full sequence of configurations is uniquely determined by C_0. A verifier with access to C_0 and U can therefore reproduce the entire evolution without ambiguity, establishing deterministic replay. □

1 Discrete WCC Machines and Turing Equivalence

In this section we isolate the discrete core of wave-constrained computation (WCC) as a finite-state local-update machine model and prove that its polynomial-time class coincides with the classical deterministic polynomial-time class. This makes precise the identification

$$P_{\mathrm{WCC}} = P_{\mathrm{Turing}}$$

used elsewhere in the paper.

We first define the discrete WCC machine under the locality and precision rails, then construct an explicit simulation of any polynomial-time Turing machine by a polynomial-time WCC machine with only constant-factor overhead in time. We then recall the converse simulation (WCC by Turing machines) and obtain full equivalence.

1.1 Discrete WCC machine model

Let Σ be a finite input alphabet and let $n = |x|$ denote input length.

Definition 1.1 (Uniform discrete WCC machine)**.** *A* uniform discrete WCC machine *is a tuple*

$$W = (\{\Lambda_n\}_{n\geq 1}, S, r, U, \mathrm{enc}, \mathrm{dec})$$

with the following data, fixed independently of an individual input.

1. *The spatial dimension $d \geq 1$ and neighborhood radius $r \geq 0$ are constants. The cell set $\Lambda_n \subset \mathbb{Z}^d$ satisfies $|\Lambda_n| \leq p(n)$ and lies in a box with polynomially bounded integer coordinates. A fixed deterministic algorithm constructs and enumerates Λ_n and its fixed-radius neighborhoods in time* $\mathrm{poly}(n)$.

2. *S is a fixed, finite alphabet. Boundary cells outside Λ_n read a distinguished fixed background symbol $s_\partial \in S$.*

3. *$U : S^{\{-r,\dots,r\}^d} \to S$ is a single fixed local rule of constant description length. Its synchronous action is*

$$C_{t+1}(i) = U\big((\overline{C}_t(i+v))_{\|v\|_\infty \leq r}\big), \quad i \in \Lambda_n,$$

   where $\overline{C}_t$ extends C_t by the background symbol.

4. *The uniform encoder* enc *constructs $C_0 = \mathrm{enc}(x) \in S^{\Lambda_{|x|}}$ in time* $\mathrm{poly}(|x|)$ *on a standard deterministic Turing machine.*

5. *The uniform decoder $\mathrm{dec}_n : S^{\Lambda_n} \to \{\mathsf{run}, 0, 1\}$ is computed in time* $\mathrm{poly}(n)$ *from the canonical configuration encoding. It returns* run *before halting and* 0 *(reject) or* 1 *(accept) on a halting configuration. Halting outputs are persistent (equivalently the computation is stopped at its first halting configuration).*

The polynomial bounds, fixed rule, and algorithms are part of the machine description, not extra advice indexed by n or an oracle. All complexity classes denoted P_{WCC} and NP_{WCC} below refer to this discrete uniform *model unless a physical superscript is explicitly attached.*

Definition 1.2 (Polynomial-time discrete WCC)**.** *For a uniform discrete WCC machine W, set*

$$T_W(n) = \max_{x \in \Sigma^n} \min\{t \geq 0 : \mathrm{dec}_n(C_t) \in \{0, 1\}\}.$$

We require the minimum to exist for every input. The machine decides $L \subseteq \Sigma^*$ in polynomial time if $T_W(n) \leq p(n)$ for some polynomial p and its first halting output equals $\mathbf{1}_L(x)$ for every x. By Definition 1.1, the total cost of constructing, simulating, and decoding each of these runs is polynomial in n whenever the round count is polynomial. We write $P^{\mathrm{disc}}_{\mathrm{WCC}}$, or simply P_{WCC} in the discrete sections, for the resulting class.

The remaining WCC rails (coercivity, entropy pruning, renewability) impose additional structure on how such a discrete machine can arise from an underlying wave-evolution model. In this section we restrict attention to the discrete interface; the physical realization is addressed separately (Remark 1.21).

1.2 Wave–Symbol Correspondence Under Compatibility

The local symbolic update U can faithfully describe a discretized wave evolution only if its quantization commutes with that evolution. Locality and finite precision alone do not imply this compatibility. Here a local wave update is a map $F : Z^{\{-r,\dots,r\}^d} \to Z$ on a finite-precision subset $Z \subset \mathbb{C}$, acting on $\Psi_t : \Lambda_n \to Z$ by

$$\Psi_{t+1}(i) = F\big((\overline{\Psi}_t(i + v))_{\|v\|_\infty \leq r}\big). \tag{1}$$

This is a discrete local map; realizing it by a particular WCT PDE requires independent physical analysis.

Definition 1.3 (Quantization and representative encoding)**.** *Let S be a fixed finite alphabet. Let $Q : Z \to S$ be surjective and $\widetilde{Q} : S \to Z$ select a representative for each symbol, so that*

$$Q \circ \widetilde{Q} = \mathrm{id}_S.$$

Both maps act cellwise on configurations. The symbolic state associated with a field is $C_t = Q \circ \Psi_t$.

Given $F, Q, \widetilde{Q}$, define the candidate symbolic rule

$$U(c) = Q\big(F(\widetilde{Q}(c))\big), \qquad c \in S^{\{-r,\dots,r\}^d}. \tag{2}$$

Theorem 1.4 (Exact wave–symbol correspondence)**.** *Suppose the actual field trajectory obeys* (1) *and that, on every neighborhood configuration z reached by this trajectory, the* quantization compatibility *identity holds:*

$$Q(F(z)) = Q\big(F(\widetilde{Q}(Q(z)))\big). \tag{3}$$

Then $C_t = Q \circ \Psi_t$ satisfies the exact symbolic update

$$C_{t+1}(i) = U\big((\overline{C}_t(i+v))_{\|v\|_\infty \le r}\big). \tag{4}$$

Conversely, for every symbolic local rule U there exists an abstract *finite-precision representative evolution, defined by*

$$F_U(z) = \widetilde{Q}\big(U(Q(z))\big),$$

for which every symbolic trajectory is realized exactly on the invariant code manifold $\widetilde{Q}(S)^{\Lambda_n}$. This converse is a statement about a constructed finite-state local evolution, not about solutions of an independently fixed continuous WCT PDE.

Proof. For an actual trajectory and each site i, write z for the corresponding input neighborhood. By compatibility,

$$Q(\Psi_{t+1}(i)) = Q(F(z)) = Q(F(\widetilde{Q}(Q(z)))) = U(Q(z)),$$

which is (4). For the converse, initialize $\Psi_0 = \widetilde{Q}(C_0)$ and use F_U. If $\Psi_t = \widetilde{Q}(C_t)$, then

$$\Psi_{t+1} = \widetilde{Q}\big(U(Q(\Psi_t))\big) = \widetilde{Q}(U(C_t)) = \widetilde{Q}(C_{t+1}).$$

Induction proves exact representative dynamics and invariance of the code manifold. No continuous curvature or energy bound is inferred. □

Remark 1.5 (Compatibility is necessary)**.** *Without* (3)*, quantization need not commute with a local update. For example, let* $Z = \{-1, \frac{1}{2}, 1\}$ *and* $S = \{0, 1\}$*, with* $Q(z) = \mathbf{1}_{\{z>0\}}$*,* $\widetilde{Q}(0) = -1$*,* $\widetilde{Q}(1) = 1$*. Define the one-cell map* $F(-1) = -1$*,* $F(\frac{1}{2}) = \frac{1}{2}$*, and* $F(1) = -1$*. For* $\Psi_0 = \frac{1}{2}$*, one has* $Q(F(\Psi_0)) = 1$*, whereas* $U(Q(\Psi_0)) = Q(F(1)) = 0$*. Thus the unqualified wave–symbol isomorphism is false, even for single-site local maps. Exact correspondence requires the stated compatibility or an invariant representative code manifold.*

1.3 Gandy Correspondence and Physical Cellular Automata

Gandy's physical Church–Turing thesis characterizes a broad class of "reasonable" physical computation models by four structural axioms: bounded information density, spatial homogeneity, bounded propagation speed, and local causality. In this subsection we show that discrete WCC machines satisfy these axioms and therefore instantiate a physical cellular automaton in Gandy's sense.

Definition 1.6 (Gandy-style physical axioms)**.** *A discrete dynamical system on configurations* $C : \Lambda \to S$ *satisfies Gandy's physical constraints if:*

1. ***Bounded information density:*** *S is finite, so each site carries at most* $\log_2 |S|$ *bits of information.*
2. ***Spatial homogeneity:*** *There exists a fixed local rule* $U : S^{N_r} \to S$ *independent of the site* $i \in \Lambda$ *and of the configuration size.*
3. ***Bounded propagation:*** *There is a fixed radius* $r \in \mathbb{N}$ *such that the update at site* i *depends only on the values in the neighbourhood* $N_r(i)$*.*
4. ***Local causality:*** *The global update* $C \mapsto C'$ *is defined pointwise by*

$$C'(i) = U\big(C|_{N_r(i)}\big), \qquad i \in \Lambda,$$

with no dependence on nonlocal aggregates.

Proposition 1.7 (WCC satisfies Gandy axioms)**.** *Let*

$$W = (\{\Lambda_n\}_{n\geq 1}, S, r, U, \mathrm{enc}, \mathrm{dec})$$

be a discrete WCC machine as in Definition 1.1. Then for each n the induced evolution on configurations $C_t : \Lambda_n \to S$ satisfies all four Gandy-style physical axioms.

Proof. Fix $n \geq 1$.

(1) *Bounded information density.* By definition of a WCC machine, the state set S is finite. Thus each cell carries at most $\log_2 |S|$ bits of information.

(2) *Spatial homogeneity.* The local update rule $U : S^{N_r} \to S$ is fixed once and for all in the specification of W, and does not depend on $i \in \Lambda_n$ or on n.

(3) *Bounded propagation.* The neighbourhood shape $N_r(i)$ is defined with a fixed radius $r \in \mathbb{N}$. Therefore the update at site i cannot depend on configuration values outside $N_r(i)$, and information cannot propagate faster than r cells per global step.

(4) *Local causality.* By definition of the WCC evolution,

$$C_{t+1}(i) = U\big(C_t|_{N_r(i)}\big),$$

so the future value at site i depends only on the local neighbourhood $N_r(i)$ at time t and not on any global property of C_t.

Thus all four axioms hold for the evolution on Λ_n, for each n. □

Corollary 1.8 (Gandy–WCC correspondence)**.** *Every discrete WCC machine induces, for each input length n, a finite-state cellular-automaton-style evolution satisfying the listed structural locality axioms. This is a statement about the abstract machine, not a proof of realizability by any fixed physical PDE.*

Proposition 1.7 and Corollary 1.8 show that uniform WCC is a discrete abstraction compatible with the structural locality axioms. Physical feasibility and energy accounting are independent questions.

1.4 Deterministic Turing machines

We recall the standard deterministic Turing machine model and its polynomial-time class.

Definition 1.9 (Deterministic Turing machine)**.** *A deterministic Turing machine is a tuple*

$$M = (Q, \Gamma, \Sigma, \delta, q_0, q_{\text{acc}}, q_{\text{rej}})$$

where:

- Q *is a finite set of states.*
- Γ *is a finite tape alphabet containing a distinguished blank symbol* $\sqcup$*.*
- $\Sigma \subseteq \Gamma \setminus \{\sqcup\}$ *is the input alphabet.*
- $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R\}$ *is the transition function.*
- $q_0 \in Q$ *is the start state.*
- $q_{\text{acc}}, q_{\text{rej}} \in Q$ *are halting accept/reject states.*

A configuration *of* M *is a triple* (γ, h, q) *consisting of a bi-infinite tape* $\gamma : \mathbb{Z} \to \Gamma$ *with all but finitely many entries equal to* $\sqcup$*, a head position* $h \in \mathbb{Z}$*, and a current state* $q \in Q$*. One transition step applies* $\delta(q, \gamma(h))$ *and updates* (γ, h, q) *accordingly.*

Definition 1.10 (Polynomial-time Turing computation)**.** *Let* $T_M(n)$ *denote the worst-case number of transition steps that* M *uses on inputs of length* n*. A language* $L \subseteq \Sigma^*$ *is in* P_{Turing} *if there exists a machine* M *and constants* $c, k \geq 1$ *such that*

$$T_M(n) \leq c \max\{n, 1\}^k$$

for every $n \geq 0$*, and* M *accepts* x *if and only if* $x \in L$*. The factor* $\max\{n, 1\}$ *includes the empty input without forcing* $T_M(0) = 0$*.*

1.5 Simulation of polynomial-time Turing machines by WCC

We now construct, for each polynomial-time Turing machine M, a discrete WCC machine W_M that simulates M with only constant-factor overhead in time. The construction proceeds via a one-dimensional radius-1 local update rule that encodes the tape and head in per-cell symbols and uses a two-phase schedule to implement the Turing transition function.

1.5.1 Configuration encoding

Fix a Turing machine M as in Definition 1.9. We first define the local alphabet used by the WCC machine.

Definition 1.11 (Local tape state and phase)**.** *Let*

$$S_{\text{tape}} := \Gamma \times (Q \cup \{\bot\})$$

where the symbol $\bot$ denotes "no head present". A local tape state $(\gamma, q) \in S_{\text{tape}}$ thus records the tape symbol $\gamma \in \Gamma$ and, if $q \in Q$, the presence of the head with state q at the corresponding cell.

We extend the state space with a phase bit and a directional mark. Let

$$Q' := Q \cup \{(q, L) : q \in Q\} \cup \{(q, R) : q \in Q\}$$

and define

$$S := \Gamma \times (Q' \cup \{\bot\}) \times \{0, 1\}.$$

An element $s = (\gamma, q^{\#}, p) \in S$ consists of:

- *a symbol $\gamma \in \Gamma$,*
- *either $q^{\#} = \bot$ (no head), or $q^{\#} \in Q$ (head present with state $q^{\#}$), or $q^{\#} = (q, d)$ with $q \in Q$ and $d \in \{L, R\}$ indicating that in the next phase the head should move one cell in direction d,*
- *a phase bit $p \in \{0, 1\}$. Phase $p = 0$ corresponds to "before Turing transition" and phase $p = 1$ to "after symbol rewrite, before head movement".*

Given a Turing configuration (γ, h, q) at time t, we encode it as a configuration $C_t : \mathbb{Z} \to S$ by

$$C_t(i) := \begin{cases} (\gamma(i), q, 0), & \text{if } i = h, \\ (\gamma(i), \bot, 0), & \text{if } i \neq h. \end{cases}$$

In the WCC machine we restrict to a finite window

$$\Lambda_n := \{-L(n), \ldots, L(n)\} \subset \mathbb{Z}$$

with $L(n)$ chosen as a polynomial upper bound on the space usage of M on inputs of length n, for example

$$L(n) := T_M(n) + n.$$

1.5.2 Local update rule

We now define the local rule $U : S^{\{-1,0,1\}} \to S$, which acts with radius $r = 1$ in dimension $d = 1$. For each site $i \in \Lambda_n$, the neighbourhood is

$$N_1(i) = \{i - 1, i, i + 1\}$$

(with the understanding that we only care about neighbourhoods fully contained in Λ_n; since the head never reaches the boundary within $T_M(n)$ steps, the behaviour of U at the outermost sites can be defined arbitrarily).

The action of U depends on the phase bit of the centre cell.

Phase 0: symbol rewrite and direction tagging. In phase $p = 0$, at most one cell carries a head state in Q, all other cells have $q^{\#} = \perp$. For a cell at site i with local state $s_i = (\gamma_i, q_i^{\#}, 0)$, we define the phase-0 component U_{phase0} as:

- If $q_i^{\#} \in Q$ (the head is present at i with state $q_i^{\#}$), we consult the Turing transition function

  $$\delta(q_i^{\#}, \gamma_i) = (q', \gamma', d), \qquad d \in \{L, R\},$$

  and set

  $$U_{\text{phase0}}(s_i) := (\gamma', (q', d), 1),$$

  that is, we overwrite the symbol with γ', store a directional mark (q', d), and flip the phase bit to 1.

- If $q_i^{\#} = \perp$ (no head), then the cell does not change its symbol or head component and only flips the phase:

  $$U_{\text{phase0}}(s_i) := (\gamma_i, \perp, 1).$$

This part of the rule is radius-0; it will be incorporated into the full radius-1 rule below.

Phase 1: head movement. In phase $p = 1$, exactly one cell in a valid configuration carries a directional mark $(q', d) \in Q'$ and all others carry either $\perp$ or an unmarked $q \in Q$. The head must move one cell in direction d, and the directional mark must be cleared.

The update for phase 1 uses the radius-1 neighbourhood

$$(C(i-1), C(i), C(i+1)) = (s_{i-1}, s_i, s_{i+1}) \in S^3.$$

We write $s_j = (\gamma_j, q_j^{\#}, 1)$ with $q_j^{\#} \in Q' \cup \{\perp\}$ and define $U_{\text{phase1}} : S^3 \to S$ by case distinction:

- If the *centre* cell carries a right-moving mark, i.e. $q_i^{\#} = (q', R)$, then the head is leaving site i:

$$U_{\text{phase1}}(s_{i-1}, s_i, s_{i+1}) := (\gamma_i, \perp, 0).$$

- If the *left neighbour* carries a right-moving mark and the centre cell is headless, i.e. $q_{i-1}^{\#} = (q', R)$ and $q_i^{\#} = \perp$, then the head arrives from the left:

$$U_{\text{phase1}}(s_{i-1}, s_i, s_{i+1}) := (\gamma_i, q', 0).$$

- If the centre cell carries a left-moving mark, $q_i^{\#} = (q', L)$, then the head is leaving site i to the left:

$$U_{\text{phase1}}(s_{i-1}, s_i, s_{i+1}) := (\gamma_i, \perp, 0).$$

- If the *right neighbour* carries a left-moving mark and the centre cell is headless, $q_{i+1}^{\#} = (q', L)$ and $q_i^{\#} = \perp$, then the head arrives from the right:

$$U_{\text{phase1}}(s_{i-1}, s_i, s_{i+1}) := (\gamma_i, q', 0).$$

- In all remaining cases (cells that neither carry nor receive a head), the cell preserves its tape symbol and any non-directional head component and simply flips the phase bit $1 \mapsto 0$. Under the invariant stated below, this means

$$U_{\text{phase1}}(s_{i-1}, s_i, s_{i+1}) := (\gamma_i, q_i^*, 0)$$

  where q_i^* is either $\perp$ or $q \in Q$ (never a directional mark).

The full local rule U is obtained by combining the two phases into a single radius-1 rule on $S^{\{-1,0,1\}}$, with the phase bit p in the centre cell determining which branch, U_{phase0} or U_{phase1}, applies at that cell.

1.5.3 Encoding, decoding, and correctness

We now define the WCC machine W_M associated to M.

Definition 1.12 (Associated WCC machine W_M)**.** *Let M be a deterministic Turing machine as above. Define W_M to be the discrete WCC machine with:*

- *dimension $d = 1$,*
- *index sets $\Lambda_n = \{-L(n), \dots, L(n)\}$ with $L(n)$ a polynomially bounded function with $L(n) \geq T_M(n) + n$ (for example $L(n) = T_M(n) + n$), so the head remains in the window through the first halting time,*
- *state set S as in Definition 1.11 and radius $r = 1$, with neighbourhoods $N_1(i) = \{i - 1, i, i + 1\}$,*
- *local update rule $U : S^{\{-1,0,1\}} \to S$ as defined above,*
- *encoding $\mathrm{enc}_M : \Sigma^* \to S^{\Lambda_n}$ given by*

$$\mathrm{enc}_M(x)_i = \begin{cases} (x_1, q_0, 0), & i = 0, \\ (x_{i+1}, \perp, 0), & 1 \leq i \leq n-1, \\ (\sqcup, \perp, 0), & \text{otherwise}, \end{cases}$$

  i.e. the input word $x = x_1 \dots x_n$ is written on cells $0, \dots, n-1$, the head starts at $i = 0$ in state q_0, and the rest of the tape is blank,

- *decoding predicate $\mathrm{dec}_M : S^{\Lambda_n} \to \{\mathsf{run}, 0, 1\}$ defined by*

$$\mathrm{dec}_M(C) = \begin{cases} 1, & \text{if the simulated head is in } q_{\mathrm{acc}}, \\ 0, & \text{if the simulated head is in } q_{\mathrm{rej}}, \\ \mathsf{run}, & \text{otherwise.} \end{cases}$$

We first state an invariant that characterizes valid configurations in the simulation.

Lemma 1.13 (Simulation invariant)**.** *For the WCC machine W_M on input $x \in \Sigma^n$, the following holds for all integers $0 \leq t \leq L(n)$. In particular, it holds until the first halting time, since $T_M(n) \leq L(n)$:*

- *At every* even *time* $2t$*, all cells have phase bit* $p = 0$ *and there is a unique head cell* $h_t \in \Lambda_n$ *such that* $C_{2t}(h_t) = (\gamma_t(h_t), q_t, 0)$ *with* $q_t \in Q$*, while all other cells have* $q^{\#} = \perp$*.*

- *At every* odd *time* $2t + 1$*, all cells have phase bit* $p = 1$ *and there is a unique directional mark at some site* $j_t \in \Lambda_n$ *with* $C_{2t+1}(j_t) = (\widetilde{\gamma}(j_t), (q', d), 1)$ *for some* $q' \in Q$ *and* $d \in \{L, R\}$*, while all other cells have* $q^{\#} \in Q \cup \{\perp\}$ *but never a directional mark.*

Proof. The base case follows from the definition of enc_M at $t = 0$. Since the Turing head starts at cell 0 and moves by one cell per step, $|h_t| \leq t$. Thus $h_t \in \Lambda_n$ for $0 \leq t \leq L(n)$, and an inductive transition at $t < L(n)$ stays inside the window. The step from $2t$ to $2t + 1$ uses U_{phase0}, which can only introduce a single directional mark at the unique head cell and flips all phases to 1. The step from $2t + 1$ to $2t + 2$ uses U_{phase1}, which ensures that any directional mark at the centre cell is cleared and exactly one neighbour installs the corresponding non-directional head state; all other cells remain without directional marks and all phases flip back to 0. The uniqueness of the marked site follows from the fact that there is exactly one directional mark at odd times and the local rule never creates additional marks; a straightforward case analysis on the neighbourhoods shows that no two cells can simultaneously install a head state from the same marked neighbour. This preserves the invariant. □

We now establish configuration correspondence and runtime.

Lemma 1.14 (One Turing step equals two WCC steps)**.** *Let* (γ_t, h_t, q_t) *be the Turing configuration of* M *after* t *steps on input* x*, and let* C_{2t} *be the configuration of* W_M *after* $2t$ *update steps with initial condition* $C_0 = \mathrm{enc}_M(x)$*. Then, for all* $0 \leq t \leq L(n)$*, the configuration* C_{2t} *encodes* (γ_t, h_t, q_t) *in the sense that:*

- *For all* $i \in \Lambda_n$*, the tape symbol in* $C_{2t}(i)$ *equals* $\gamma_t(i)$*.*

- *There exists a unique* $i = h_t$ *such that the head component of* $C_{2t}(i)$ *equals* q_t *and all other cells carry* $\perp$ *in the head component.*

- *All cells have phase bit* $p = 0$*.*

Moreover, for $0 \leq t < L(n)$*, applying two further WCC update steps transforms* C_{2t} *into* $C_{2(t+1)}$ *which encodes* $(\gamma_{t+1}, h_{t+1}, q_{t+1})$*. The same two-step*

equality of window-restricted configurations also holds at $t = L(n)$, without asserting that the head stays in the window at $t = L(n) + 1$.

Proof. The claim follows by induction on t. For $t = 0$ the statement holds by construction of enc_M and the invariant of Lemma 1.13. Suppose the claim holds for some $0 \le t < L(n)$. In configuration C_{2t} there is exactly one cell h_t with head state q_t at phase $p = 0$, and the tape symbols agree with $\gamma_t(i)$ for all i.

The phase-0 part of the local rule applies $\delta(q_t, \gamma_t(h_t)) = (q', \gamma', d)$ at cell h_t, overwriting the tape symbol with γ' and marking the head component as (q', d) while flipping the phase to 1. All other cells preserve their tape symbols and have no head, and flip their phase to 1. After this first update, we obtain C_{2t+1}, in which the tape symbols coincide with those of γ_t except at h_t, where they match the updated symbol γ', and the unique directional head mark (q', d) is at cell h_t, in accordance with Lemma 1.13.

In the next update step (phase 1), the radius-1 rule U_{phase1} moves the directional head mark one cell in direction d to obtain a unique head state q' at either $h_t + 1$ (if $d = R$) or $h_t - 1$ (if $d = L$), clears the directional mark, and restores all phase bits to 0, producing $C_{2(t+1)}$. The tape symbols are unchanged in this step. By definition of the Turing transition, the new Turing configuration is $(\gamma_{t+1}, h_{t+1}, q_{t+1})$ with γ_{t+1} and (h_{t+1}, q_{t+1}) matching the updated symbol and head location/state. Thus $C_{2(t+1)}$ encodes $(\gamma_{t+1}, h_{t+1}, q_{t+1})$ as claimed. This completes the induction for $0 \le t \le L(n)$. The equality of configurations restricted to Λ_n continues through $t = L(n) + 1$ by the same two-step local identity, but a head beyond the window is not counted as a head cell inside it. □

Lemma 1.15 (Runtime and space overhead)**.** *Let $T_M(n)$ be the worst-case runtime of M on inputs of length n. Then the associated WCC machine W_M satisfies:*

- *For all $x \in \Sigma^n$, W_M reaches an accepting configuration if and only if M enters q_{acc} on x.*
- *The number of WCC update steps until halting satisfies*

$$T_{W_M}(n) = 2T_M(n).$$

- *The number of cells satisfies $|\Lambda_n| \le 2L(n) + 1 \le \tilde{p}(n)$ for some polynomial $\tilde{p}$.*

Proof. The first two claims follow directly from Lemma 1.14 and the definition of dec_M: after $T_M(n)$ Turing steps, M halts in either q_{acc} or q_{rej}, and the corresponding encoded configuration appears at WCC time $2T_M(n)$ with phase $p = 0$; the decoding predicate tests for the presence of q_{acc} and thus detects acceptance exactly when M accepts.

For the window bound, the head begins at 0 and moves by at most one cell per step, hence $|h_t| \leq t$ until the first halt. The choice $L(n) \geq T_M(n) + n$ therefore ensures that the simulated head remains inside $\Lambda_n = \{-L(n), \ldots, L(n)\}$ through $t = T_M(n)$, including the empty input. Since $L(n)$ is bounded by a polynomial in n, we have $|\Lambda_n| = 2L(n) + 1 \leq \tilde{p}(n)$ for a polynomial $\tilde{p}$. Each update step applies the local rule U independently at each cell, so a single global update can be carried out with work $O(|\Lambda_n|) = \mathrm{poly}(n)$, which respects the polynomial-time cost per pass assumed in Definition 1.2. □

We can now state the main equivalence theorem for the simulation direction TM ⇒ WCC.

Theorem 1.16 (Original two-phase simulation: exact dynamics, conditional uniformity)**.** *Let a deterministic Turing machine M decide a language L with first-halt time $T_M(x)$ bounded by $c \max\{|x|, 1\}^k$, and let the original two-phase WCC window contain all head positions through that time. The original local update rule and decoder have the first-answer equality*

$$T_{W_M}(x) = 2T_M(x)$$

for its intrinsic global-update time*, and agree with the Turing answer. If, additionally, the* original *canonical lattice writer* originalWindowCode(p) *and complete configuration encoder* originalInputCode(M, p) *are both polynomial-time computable under the manuscript's single-tape output convention, then this concrete W_M meets the uniform discrete WCC preparation requirements and defines a polynomial-time WCC decider.*

Without those two writer hypotheses, the dynamical simulation alone does not *prove uniformity of this particular construction. Independently, the uniform class inclusion*

$$P_{\mathrm{Turing}} \subseteq P^{\mathrm{disc}}_{\mathrm{WCC}}$$

is proved using a different, fully charged one-cell encoder construction.

Proof. The first-answer identity and its head-window hypotheses follow from Lemma 1.15 and the two-phase invariant, formalized by `PartIITwoPhase.runtime` and `PartIITwoPhase.tm_to_wcc_decides`. The original decoder is polynomial-time (`TwoPhaseDecoder.decM_fp`). The remaining writer hypotheses would exactly supply the lattice and encoder `FP` fields in the uniform WCC definition, and hence give the conditional claim for this W_M.

The independent unconditional class inclusion uses the `PartIIUniform.oneCell` construction and `PartIIUniform.P_TM_subset_P_disc`, whose encoder runs the original Turing decider, charges that work, and writes the decision bit. These two different proofs must not be conflated. □

Remark 1.17 (Addendum, 12 November 2026: both writer hypotheses are theorems)**.** *The two writer hypotheses of Theorem 1.16 hold for every polynomial radius p and every machine M. The lattice writer and the encoder are explicit single-tape machines,* ***windowMTM*** *p and* ***inputMTM*** *M p. Each one normalizes its input, computes $L = p(|x|)$ in unary by Horner's rule on marked work cells, writes the canonical word, and halts with a blank after it. Lean proves that the outputs are exactly* originalWindowCode$(p)(x)$ *and* originalInputCode$(M, p)(x)$*, at the first halt, within explicit polynomial numbers of transitions (**TwoPhasePolynomialWriter.windowMTM_outputs** and **TwoPhasePolynomialWriter.inputMTM_outputs**). Hence **originalWindowCode_fp** and **originalInputCode_fp**, and the concrete W_M is a uniform discrete WCC machine (**originalWM**). Its first answer appears at update exactly $2T_M(x)$ whenever $T_M(x) \le p(|x|)$ (**originalWM_first_answer**), so it decides L within $2c \max\{|x|, 1\}^k$ updates (**originalWM_decides**). The inclusion $P_{\text{Turing}} \subseteq P^{\text{disc}}_{\text{WCC}}$ also holds with W_M itself as the witness (**P_Turing_subset_P_disc_by_original**). The theorem above keeps its conditional wording of 7 November 2026; this addendum discharges its two hypotheses.*

1.6 Simulation of WCC by Turing machines

We now recall the converse direction: any discrete WCC machine can be simulated by a deterministic Turing machine with polynomial overhead. This is essentially a standard "Turing simulation of a cellular automaton" argument.

Theorem 1.18 (Simulation of WCC by Turing machines)**.** *Let W be a discrete WCC machine that decides a language L in polynomial time as in Definition 1.2, with runtime bound $T_W(n) \le n^k$ and $|\Lambda_n| \le p(n)$ for some*

polynomials k, p. Then there exists a deterministic Turing machine M_W that decides L in time polynomially bounded in n. In particular,

$$P_{\mathrm{WCC}} \subseteq P_{\mathrm{Turing}}.$$

Proof. We sketch the standard simulation; details can be made fully explicit if desired.

Fix n and the corresponding index set Λ_n. A configuration $C_t : \Lambda_n \to S$ of W can be encoded on the tape of a Turing machine as a finite word over $\Gamma_{\mathrm{enc}} := S \cup \{\#\}$ by listing the cell states in some canonical order $i_1, \ldots, i_{|\Lambda_n|}$ (e.g. lexicographic order on Λ_n) separated by a delimiter $\#$. This yields an encoding $\mathrm{code}(C_t)$ of length $O(|\Lambda_n|)$.

To simulate one global WCC update $C_t \mapsto C_{t+1}$, the Turing machine M_W maintains the current configuration encoding on its work tape and proceeds in two phases:

1. It copies the current configuration to an auxiliary work area. This takes time $O(|\Lambda_n|)$.

2. It enumerates each cell, uses the uniform lattice/neighborhood algorithms to retrieve its at most $(2r+1)^d$ local symbols, computes the fixed rule U, and writes the next state. The local lookup and update have polynomial cost in n; a simple tape implementation may take $O(|\Lambda_n|^2 \,\mathrm{poly}(n))$ time per round. No constant-time random access is assumed.

By assumption $|\Lambda_n| \leq p(n)$ and one global update can be effected with cost polynomial in $|\Lambda_n|$, i.e. $\mathrm{poly}(p(n))$. Repeating this procedure for $T_W(n) \leq n^k$ steps yields an overall Turing runtime

$$T_{M_W}(n) \leq T_W(n) \cdot \mathrm{poly}(|\Lambda_n|) \leq n^k \cdot \mathrm{poly}(p(n)) = \mathrm{poly}(n).$$

The Turing machine M_W accepts exactly when the simulated configuration satisfies $\mathrm{dec}(C_t) = 1$, which can be checked during or after each simulated global update by scanning the encoding for an accepting pattern. Thus M_W decides the same language L in polynomial time, and $L \in P_{\mathrm{Turing}}$. Since L was arbitrary in P_{WCC}, it follows that $P_{\mathrm{WCC}} \subseteq P_{\mathrm{Turing}}$. □

1.7 Equivalence of P_{WCC} and P_{Turing}

Combining Theorem 1.16 and Theorem 1.18 yields the desired identification.

Corollary 1.19 (Polynomial-time equivalence)**.** *Under the discrete WCC machine model of Definition 1.1 and the standard deterministic Turing machine model of Definition 1.9, the polynomial-time classes coincide:*

$$P_{\mathrm{WCC}} = P_{\mathrm{Turing}}.$$

Remark 1.20 (Scope of the formally verified part)**.** *Corollary 1.19 is kernel-checked in dimension one for every fixed radius $r \geq 1$ and the fixed origin-marked encoding (Section 5). For $d \geq 2$ the inclusion $P_{\mathrm{Turing}} \subseteq P_{\mathrm{WCC}}$ is kernel-checked for the radius-one Moore neighbourhood. At the original release of this paper the inclusion $P_{\mathrm{WCC}} \subseteq P_{\mathrm{Turing}}$ in dimension $d \geq 2$ was supported by the proof sketch of Theorem 1.18 only. A later repository verification (5 November 2026) kernel-checks it for every dimension by an explicit single-tape machine (`PVNP.Discrete.P_WCCD_eq_P_TM`); see the formal verification status paragraph below. For the polynomial-time encoder class of Definition 1.1, the inclusion $P_{\mathrm{Turing}} \subseteq P_{\mathrm{WCC}}$ also follows trivially, because the encoder may compute the decision bit.*

Remark 1.21 (Physical WCT realization is a separate obligation)**.** *The equality $P_{\mathrm{WCC}}^{\mathrm{disc}} = P$ concerns only the uniform finite-state abstraction of Definition 1.1. It does* not *imply $P_{\mathrm{WCC}}^{\mathrm{phys}} = P$. To deduce that equality for a curvature-constrained WCT device family, one must additionally construct, uniformly in input length and for every required local rule, a stable encoding $s \mapsto \psi_s$ and a physical evolution satisfying:*

1. *the exact code-state transition compatibility of Theorem 1.4;*
2. *polynomially bounded preparation, run time, volume, energy/curvature expenditure, precision and readout costs;*
3. *correct uniform simulation of all polynomial-time discrete WCC computations, including their verifier machines.*

These are additional model-specific hypotheses. They are neither consequences of locality and finite precision nor established by the Turing simulations proved in this section. The reverse inclusion $P_{\mathrm{WCC}}^{\mathrm{phys}} \subseteq P$ would likewise require a precise representation and polynomial-time classical simulation of all admissible physical operations.

2 WCC Verifiers and NP Equivalence

In this section we lift the polynomial-time equivalence of Corollary 1.19 from decision procedures to verifier-based nondeterministic computation. We define an NP-type class NP_{WCC} using polynomial-time WCC verifiers and show that, under the same discrete model assumptions, it coincides with the classical nondeterministic polynomial-time class NP_{Turing}.

2.1 Classical NP via Turing verifiers

We first recall the standard verifier-based definition of NP.

Let Σ be a finite input alphabet and let Γ_{wit} be a finite witness alphabet. For $x \in \Sigma^n$ and $y \in \Gamma_{\mathrm{wit}}^m$ we write (x, y) for the pair instance, and $N := n+m$ for the total input length.

Definition 2.1 (Polynomial-time Turing verifier)**.** *A* polynomial-time Turing verifier *for a language $L \subseteq \Sigma^*$ is a deterministic Turing machine*

$$V_{\mathrm{T}} = (Q, \Gamma, \Sigma \cup \Gamma_{\mathrm{wit}}, \delta, q_0, q_{\mathrm{acc}}, q_{\mathrm{rej}})$$

such that:

- *V_{T} takes as input a pair (x, y) with $x \in \Sigma^n$, $y \in \Gamma_{\mathrm{wit}}^m$, encoded in some fixed, efficiently decodable format on its tape.*
- *There exist polynomials p, q such that for all $x \in \Sigma^n$ and all y,*

  $$T_{V_{\mathrm{T}}}(|x| + |y|) \leq p(|x| + |y|)$$

  and

  $$x \in L \iff \exists y \in \Gamma_{\mathrm{wit}}^m,\ m \leq q(|x|) \text{ such that } V_{\mathrm{T}}(x, y) \text{ accepts.}$$

Definition 2.2 (Classical nondeterministic polynomial time)**.** *A language $L \subseteq \Sigma^*$ is in NP_{Turing} if there exists a polynomial-time Turing verifier V_{T} for L in the sense of Definition 2.1. The class NP_{Turing} is the set of all such languages.*

This definition is equivalent to the usual one via nondeterministic Turing machines, but we will not need that equivalence here.

2.2 WCC verifiers and NP_{WCC}

We now define polynomial-time WCC verifiers and the associated NP-type class.

Definition 2.3 (Polynomial-time WCC verifier)**.** *Let Σ be a finite input alphabet and Γ_{wit} a finite witness alphabet. A* polynomial-time WCC verifier *for a language $L \subseteq \Sigma^*$ is a discrete WCC machine*

$$V_{\mathrm{W}} = (\{\Lambda_N\}_{N\geq 1}, S, r, U, \mathrm{enc}_{\mathrm{W}}, \mathrm{dec}_{\mathrm{W}})$$

in the sense of Definition 1.1, together with polynomials p, q such that:

- *For each instance $x \in \Sigma^n$ and candidate witness $y \in \Gamma_{\mathrm{wit}}^m$ with $m \leq q(n)$, the pair (x, y) is encoded as an initial configuration*

  $$C_0 = \mathrm{enc}_{\mathrm{W}}(x, y) \in S^{\Lambda_N}, \qquad N := n + m,$$

  for some index set Λ_N with $|\Lambda_N| \leq p(N)$.

- *The WCC runtime $T_{V_{\mathrm{W}}}(N)$, defined as the worst-case number of global update steps until $\mathrm{dec}_{\mathrm{W}}(C_t)$ first returns 0 or 1 on inputs of total length N, satisfies*

  $$T_{V_{\mathrm{W}}}(N) \leq p(N)$$

  for all N.

- *For all $x \in \Sigma^n$,*

  $$x \in L \iff \exists y \in \Gamma_{\mathrm{wit}}^m,\ m \leq q(n) \text{ such that } V_{\mathrm{W}}(x, y) \text{ accepts,}$$

  where acceptance means $\mathrm{dec}_{\mathrm{W}}(C_t) = 1$ for some $t \leq T_{V_{\mathrm{W}}}(|x| + |y|)$.

Definition 2.4 (WCC nondeterministic polynomial time)**.** *A language $L \subseteq \Sigma^*$ is in NP_{WCC} if there exists a polynomial-time WCC verifier V_{W} for L in the sense of Definition 2.3. The class NP_{WCC} is the set of all such languages.*

By construction, NP_{WCC} is defined using exactly the same witness-length and runtime scaling as NP_{Turing}, but with WCC machines in place of Turing machines.

2.3 Equivalence of NP_{WCC} and NP_{Turing}

We now prove that, under the discrete WCC model, the verifier-based NP class coincides with the classical one.

Theorem 2.5 (NP equivalence)**.** *Under the discrete WCC machine model of Definition 1.1 and the Turing verifier model of Definition 2.1, the nondeterministic polynomial-time classes coincide:*

$$NP_{\mathrm{WCC}} = NP_{\mathrm{Turing}}.$$

Proof. We prove both inclusions.

Inclusion $NP_{\mathrm{WCC}} \subseteq NP_{\mathrm{Turing}}$. Suppose $L \in NP_{\mathrm{WCC}}$. Then there exists a polynomial-time WCC verifier V_{W} for L with polynomials p, q as in Definition 2.3. For each pair (x, y) with $x \in \Sigma^n$, $y \in \Gamma_{\mathrm{wit}}^m$, $m \le q(n)$, the machine V_{W} runs in at most $p(N)$ global update steps, where $N := n+m$, on a lattice Λ_N with $|\Lambda_N| \le p(N)$.

By Theorem 1.18, there exists a deterministic Turing machine $M_{V_{\mathrm{W}}}$ that simulates V_{W} on inputs of total length N in time bounded by

$$T_{M_{V_{\mathrm{W}}}}(N) \le p(N) \cdot \mathrm{poly}(|\Lambda_N|) \le p(N) \cdot \mathrm{poly}(p(N)) = \mathrm{poly}(N).$$

Define a Turing verifier V_{T} that, on input (x, y), first constructs the encoding $\mathrm{enc}_{\mathrm{W}}(x, y)$ on its tape and then simulates $M_{V_{\mathrm{W}}}$ for $p(N) \cdot \mathrm{poly}(p(N))$ steps, accepting if and only if $V_{\mathrm{W}}(x, y)$ accepts. This adds at most an additional polynomial overhead in N for the encoding step, so V_{T} runs in time polynomial in N.

By construction, for either verifier V in $\{V_{\mathrm{W}}, V_{\mathrm{T}}\}$ the same predicate is decided:

$$x \in L \quad \Longleftrightarrow \quad \exists y \, \big(|y| \le q(|x|) \ \wedge \ V(x, y) = 1 \big).$$

Thus V_{T} is a polynomial-time Turing verifier for L in the sense of Definition 2.1, and hence $L \in NP_{\mathrm{Turing}}$. Since L was arbitrary in NP_{WCC}, we obtain $NP_{\mathrm{WCC}} \subseteq NP_{\mathrm{Turing}}$.

Inclusion $NP_{\mathrm{Turing}} \subseteq NP_{\mathrm{WCC}}$. Conversely, suppose $L \in NP_{\mathrm{Turing}}$. Then there exists a polynomial-time Turing verifier V_{T} for L with polynomials p, q as in Definition 2.1. For each pair (x, y) with $x \in \Sigma^n$, $y \in \Gamma_{\mathrm{wit}}^m$, $m \le q(n)$, the machine V_{T} halts in at most $p(N)$ steps with $N := n + m$.

Apply the construction of Theorem 1.16 to V_{T}, treating (x, y) as a single input string of length N. This yields a discrete WCC machine $W_{V_{\mathrm{T}}}$ that simulates V_{T} with at most a constant-factor overhead in time, i.e.

$$T_{W_{V_{\mathrm{T}}}}(N) \leq c\, T_{V_{\mathrm{T}}}(N) \leq c\, p(N)$$

for some constant $c \geq 1$, on index sets Λ_N with $|\Lambda_N| \leq \tilde{p}(N)$ for some polynomial $\tilde{p}$. We define the encoding map $\mathrm{enc}_{\mathrm{W}}$ for $W_{V_{\mathrm{T}}}$ by viewing (x, y) simply as a word over the combined alphabet $\Sigma \cup \Gamma_{\mathrm{wit}}$ and placing it on the initial tape segment as in Definition 1.12. The decoding predicate $\mathrm{dec}_{\mathrm{W}}$ tests for the presence of the accepting state q_{acc} in the head component, exactly as in Definition 1.12.

By Lemma 1.15 and Theorem 1.16, we then have

$$(x, y) \text{ accepted by } V_{\mathrm{T}} \iff (x, y) \text{ accepted by } W_{V_{\mathrm{T}}}$$

for all pairs (x, y). Thus, with V equal to either V_{T} or $W_{V_{\mathrm{T}}}$,

$$x \in L \quad \Longleftrightarrow \quad \exists y \, \big(|y| \leq q(|x|) \ \wedge \ V(x, y) = 1 \big).$$

Thus $W_{V_{\mathrm{T}}}$ is a polynomial-time WCC verifier for L with witness bound $q(\cdot)$ and runtime bounded by a polynomial in N, and hence $L \in NP_{\mathrm{WCC}}$.

Since L was arbitrary in NP_{Turing}, we obtain $NP_{\mathrm{Turing}} \subseteq NP_{\mathrm{WCC}}$.

Conclusion. We have shown $NP_{\mathrm{WCC}} \subseteq NP_{\mathrm{Turing}}$ and $NP_{\mathrm{Turing}} \subseteq NP_{\mathrm{WCC}}$, hence

$$NP_{\mathrm{WCC}} = NP_{\mathrm{Turing}}.$$

This completes the proof. □

3 Curvature-Bounded Channel Capacity in WCT

In this section we make precise the intuition that a wave-confinement device with finite volume, energy budget, and runtime cannot sustain arbitrarily many independent information "lanes" or arbitrarily fast switching between them. Under the rest-energy curvature identity and the dimensional lock on timescales, we obtain a polynomial upper bound on the spatiotemporal information-processing capacity of a WCT cavity.

3.1 Geometric and energetic setup

Let Γ be a smooth closed characteristic loop of a confined field configuration with arclength parameter $s \in [0, L)$, total length

$$L = \oint_\Gamma ds,$$

and Frenet–Serret curvature $\kappa(s)$ and torsion $\tau(s)$. We define the local curvature-rate scalar

$$\sigma(s)^2 := \kappa(s)^2 + \tau(s)^2. \tag{5}$$

Let $w(s) > 0$ be a weight function induced, for example, by an energy density or probability density derived from $|\psi|^2$ along Γ. We write the corresponding weighted average of a quantity f as

$$\langle f \rangle_w := \frac{\oint_\Gamma w(s)\, f(s)\, ds}{\oint_\Gamma w(s)\, ds}. \tag{6}$$

Definition 3.1 (Rest energy / mass from curvature)**.** *We say that a confined field configuration obeys the* rest energy curvature identity *if its rest energy and inertial mass are given by*

$$E_{\text{rest}} = \hbar c \, \langle \sigma \rangle_w, \qquad m = \frac{\hbar}{c} \, \langle \sigma \rangle_w, \tag{7}$$

for some positive weight w along closed characteristic loops of the configuration.

In a bulk description, we consider a bounded cavity $\Omega \subset \mathbb{R}^3$ with volume $V := |\Omega|$ confining a complex scalar field

$$\psi : \Omega \times \mathbb{R} \to \mathbb{C},$$

which we assume satisfies a WCT-type curvature-locking PDE. We denote by $E_{\text{rest}}(\psi)$ the total rest energy of ψ in Ω and impose a resource bound

$$E_{\text{rest}}(\psi) \leq E_{\max}. \tag{8}$$

3.2 Curvature, bandlimit, and spatial capacity

We first connect the curvature/energy budget to an effective bandlimit on the field.

Definition 3.2 (Effective wavenumber)**.** *Let $\widehat{\psi}(\mathbf{k})$ denote the spatial Fourier transform of $\psi(\cdot, t)$ at a fixed time. We say that ψ is* effectively band-limited *to radius $k_{\max}$ if*

$$\widehat{\psi}(\mathbf{k}) = 0 \quad \textit{whenever} \quad \|\mathbf{k}\| > k_{\max}. \tag{9}$$

We denote by k_{eff} an effective wavenumber scale associated to ψ, for example a suitable norm or average of $\mathbf{k}$ weighted by $\left|\widehat{\psi}\right|^2$.

Definition 3.3 (Curvature–wavenumber coupling)**.** *We assume that for confined WCT configurations the effective wavenumber is controlled by the curvature-rate average in the sense that there exists a constant $C_1 > 0$ with*

$$k_{\text{eff}} \leq C_1 \langle \sigma \rangle_w. \tag{10}$$

Combining the energy identity with the assumed coupling bounds the *effective* wavenumber, but does not by itself bound the maximum Fourier frequency or imply compact spectral support. A separate sharp bandwidth hypothesis is needed for finite-mode counting.

Lemma 3.4 (Effective wavenumber estimate; conditional sharp cutoff)**.** *Assume the rest-energy identity* (7)*, the coupling* (10)*, and $E_{\text{rest}}(\psi) \leq E_{\max}$. Then*

$$k_{\text{eff}} \leq C_1 \frac{E_{\max}}{\hbar c}. \tag{11}$$

If one additionally assumes *a uniform spectral-support bound* $\operatorname{supp} \widehat{\psi} \subseteq \{\|k\| \leq k_{\max}\}$ *and the stronger coupling $k_{\max} \leq C_1 \langle \sigma \rangle_w$, then*

$$k_{\max} \leq C_1 \frac{E_{\max}}{\hbar c}. \tag{12}$$

The second conclusion is conditional on the sharp-support premise; it does not follow from the effective-wavenumber estimate alone.

Proof. The energy identity and bound give $\langle \sigma \rangle_w \leq E_{\max}/(\hbar c)$. Substitution into the corresponding assumed coupling yields (11); using the *additional* sharp-cutoff coupling yields (12). □

We now translate the bandlimit into a bound on the number of independent spatial degrees of freedom ("lanes").

Lemma 3.5 (Curvature-bounded spatial channel capacity)**.** *Let Ω be a fixed periodic cubical cavity with volume V, and let the computation use only its Fourier modes satisfying $\|\mathbf{k}\| \le k_{\max}$. Assume $k_{\max}$ is at least the fundamental nonzero wavenumber of the cavity. Then there is a geometric constant $C_2 > 0$ such that the number N_{lanes} of these modes obeys*

$$N_{\text{lanes}} \;\le\; C_2\, V\, k_{\max}^3. \tag{13}$$

In particular, under the energy budget $E_{\text{rest}}(\psi) \le E_{\max}$ we have

$$N_{\text{lanes}} \;\le\; C_2\, V \left(C_1 \frac{E_{\max}}{\hbar c}\right)^3. \tag{14}$$

Proof. For a periodic cubic cavity, allowed wavenumbers form a lattice of spacing proportional to $V^{-1/3}$. Counting the lattice points inside the spectral ball gives, for $k_{\max}$ at least the fundamental wavenumber, an upper bound proportional to $V k_{\max}^3$. The leading-volume approximation is

$$N_{\text{lanes}} \sim \frac{V}{(2\pi)^3}\,\frac{4\pi}{3} k_{\max}^3 = C_2\, V\, k_{\max}^3,$$

for an appropriate constant $C_2 > 0$. Substituting the bound (12) from Lemma 3.4 gives (14). □

Here N_{lanes} counts orthogonal spatial Fourier modes, not all metastable states in a nonlinear cavity. Its identification with independently addressable code lanes is an additional modeling hypothesis, not a general consequence of the mode count.

3.3 Temporal dimensional lock and spatiotemporal capacity

We now incorporate the temporal side of the dimensional lock. The key assumption is that curvature-locked code states cannot be switched arbitrarily fast.

Definition 3.6 (Dimensional lock in time)**.** *Let $\Delta t_{\min}$ denote the minimal dwell time required for a curvature-locked configuration to relax from one metastable code state to a distinct one while remaining within the WCT code manifold. We assume there exists a constant $C_3 > 0$ such that*

$$\Delta t_{\min} \;\geq\; \frac{C_3}{\omega_{\max}}, \tag{15}$$

where $\omega_{\max}$ is a characteristic maximal angular frequency of the band-limited field. For propagating modes we assume a dispersion relation of the form

$$\omega_{\max} \;\leq\; c\, k_{\max}, \tag{16}$$

with c the relevant propagation speed.

Lemma 3.7 (Per-lane temporal update bound)**.** *Under the assumptions of Lemma 3.5 and Definition 3.6, over any time window of duration $T > 0$ a single lane can undergo at most*

$$N_{\text{steps per lane}} \;\leq\; \frac{T}{\Delta t_{\min}} \;\leq\; \frac{T\, c\, k_{\max}}{C_3} \tag{17}$$

distinct metastable state transitions.

Proof. The first inequality is the definition of $N_{\text{steps per lane}}$. The second follows by combining (15) and (16):

$$\Delta t_{\min} \geq \frac{C_3}{\omega_{\max}} \geq \frac{C_3}{c\, k_{\max}} \quad\Longrightarrow\quad \frac{T}{\Delta t_{\min}} \leq \frac{T\, c\, k_{\max}}{C_3}.$$

□

Combining the spatial and temporal bounds yields a spatiotemporal capacity limit.

Theorem 3.8 (Curvature-bounded spatiotemporal channel capacity)**.** *Let $\Omega \subset \mathbb{R}^3$ be a bounded WCT cavity of volume V, confining a field ψ with rest-energy bounded by $E_{\max}$, and suppose the sharp spectral-cutoff, periodic-cavity, independent-lane, and minimum-dwell-time assumptions of Lemmas 3.4, 3.5, and 3.7 hold. Then, over any time window of duration $T > 0$, the total number N_{updates} of distinct metastable lane transitions obeys*

$$N_{\text{updates}} \;\leq\; C_4\, V\, T\, k_{\max}^4, \tag{18}$$

for some constant $C_4 > 0$, and hence

$$N_{\text{updates}} \;\le\; K\,V\,T\,E_{\max}^4, \tag{19}$$

for a constant $K > 0$ depending only on $C_1, C_2, C_3, \hbar, c$.

Proof. By Lemma 3.5, the number of lanes satisfies

$$N_{\text{lanes}} \le C_2\,V\,k_{\max}^3.$$

By Lemma 3.7, each lane can switch at most $Tck_{\max}/C_3$ times in the interval of duration T. Thus the total number of lane updates obeys

$$N_{\text{updates}} \;\le\; N_{\text{lanes}} \cdot N_{\text{steps per lane}} \;\le\; \Big(C_2\,V\,k_{\max}^3\Big)\Big(\frac{T\,c\,k_{\max}}{C_3}\Big) = C_4\,V\,T\,k_{\max}^4,$$

where $C_4 := C_2 c/C_3$. Substituting the energy-dependent bound $k_{\max} \le C_1 E_{\max}/(\hbar c)$ from Lemma 3.4 yields

$$k_{\max}^4 \;\le\; C_1^4\,\frac{E_{\max}^4}{\hbar^4 c^4},$$

and hence

$$N_{\text{updates}} \;\le\; C_4\,V\,T\,k_{\max}^4 \;\le\; C_4\,V\,T\,C_1^4\frac{E_{\max}^4}{\hbar^4 c^4} \;=\; K\,V\,T\,E_{\max}^4$$

for $K := C_4 C_1^4/(\hbar^4 c^4)$. □

Remark 3.9 (Polynomial resource scaling)**.** *Conditional on the spectral-support, mode-counting, and minimum dwell-time assumptions, Theorem 3.8 shows that, for fixed physical resources $(V, T, E_{\max})$, the total number of resolvable metastable transitions in a WCT cavity is bounded by a polynomial in these parameters. In particular, if a family of WCT-based computing devices indexed by input size n is constrained so that*

$$V(n),\; T(n),\; E_{\max}(n) \;\le\; \text{poly}(n),$$

then the total number $N_{\text{updates}}(n)$ of elementary computational events across all lanes is also bounded by $\text{poly}(n)$*. Thus curvature confinement and dimensional lock enforce a physical bottleneck on the achievable computational complexity as a function of input size.*

4 Physics-Bounded Computation and Conditional NP Safety

In this section we combine the discrete WCC–Turing equivalence with the curvature/energy channel-capacity bound from the WCT model to obtain a conditional statement about the computational power of any AGI implemented on WCT hardware under the rails (L,P,V,C,S,R).

The next bound is a *conditional physical hypothesis* derived algebraically in the preceding section under the stated sharp-bandwidth, mode-counting, and dwell-time premises; it has not been established universally for arbitrary WCT PDE evolutions:

Theorem 4.1 (Conditional channel-capacity step bound)**.** *Let a WCT device occupy a cavity of volume $V > 0$ over a time interval $[0, T]$ with a rest-energy ceiling $E_{\max} > 0$. Assume the sharp spectral cutoff and the uniform mode-counting and minimum-dwell-time hypotheses of Section 3. Then the total number of distinct metastable lane updates (individual local state transitions) N_{updates} that can occur in $[0, T]$ obeys*

$$N_{\text{updates}} \;\le\; K\, V\, T\, E_{\max}^{4}, \tag{20}$$

for some constant $K > 0$ depending only on the WCT parameters $(\hbar, c, \xi, \dots)$ and not on the particular device or algorithm.

We now formalize what it means for a family of physical devices to be *polynomially bounded* in WCT resources.

Definition 4.2 (Polynomial-resource WCT device family)**.** *Let $(D_n)_{n\ge 1}$ be a family of WCT devices intended to process inputs of length n. For each n, let $V(n)$ denote the device volume, $T_{\text{phys}}(n)$ the physical run time, and $E_{\max}(n)$ a rest-energy ceiling during the computation.*

We say that (D_n) is a polynomial-resource WCT device family *if there exist constants $C_V, C_T, C_E > 0$ and exponents $a, b, c \ge 0$ such that for all n,*

$$V(n) \;\le\; C_V n^{a}, \qquad T_{\text{phys}}(n) \;\le\; C_T n^{b}, \qquad E_{\max}(n) \;\le\; C_E n^{c}. \tag{21}$$

Combining Definition 4.2 with Theorem 4.1 yields a polynomial bound on the total number of usable primitive updates.

Proposition 4.3 (Physical step ceiling under WCT rails)**.** *Let $(D_n)_{n\geq 1}$ be a polynomial-resource WCT device family in the sense of Definition 4.2. Then there exist constants $C' > 0$ and $d \geq 0$ such that, for all n,*

$$N_{\text{updates}}(n) \;\leq\; C' n^d. \tag{22}$$

In particular, any discrete WCC machine W faithfully implemented by such a device family (with every nonhalting global round incurring at least one counted elementary event) has worst-case runtime $T_W(n)$ bounded by a polynomial in n.

Proof. By Definition 4.2 we have

$$V(n) \leq C_V n^a, \quad T_{\text{phys}}(n) \leq C_T n^b, \quad E_{\max}(n) \leq C_E n^c.$$

Plugging these into the channel-capacity bound (19) gives

$$\begin{aligned} N_{\text{updates}}(n) &\leq K\, V(n)\, T_{\text{phys}}(n)\, E_{\max}(n)^4 \\ &\leq K \cdot (C_V n^a) \cdot (C_T n^b) \cdot (C_E n^c)^4 \\ &= K C_V C_T C_E^4 \; n^{a+b+4c}. \end{aligned}$$

Setting $C' := K C_V C_T C_E^4$ and $d := a + b + 4c$ yields (22). If every nonhalting global round incurs at least one counted elementary event, then $T_W(n) \leq N_{\text{updates}}(n)+1$; therefore T_W is polynomially bounded. This counting premise must hold for the particular implementation and clocking convention. □

We now combine the physical step ceiling with the discrete model equivalences $P_{\text{WCC}} = P_{\text{Turing}}$ and $NP_{\text{WCC}} = NP_{\text{Turing}}$ from Sections 1–2 to obtain a conditional NP-safety statement.

Theorem 4.4 (Conditional NP-safety of WCT-bounded AGI)**.** *Assume:*

(i) The WCT rails (L,P,V,C,S,R) correctly describe the physically realizable dynamics of any WCT device, and the channel-capacity bound of Theorem 4.1 holds.

(ii) All physically realizable discrete computation on such devices is faithfully modeled by discrete WCC machines in the sense of Definition 1.1, with equivalences

$$P_{\text{WCC}} = P_{\text{Turing}}, \qquad NP_{\text{WCC}} = NP_{\text{Turing}}.$$

(iii) The classical complexity-theoretic separation $P_{\text{Turing}} \neq NP_{\text{Turing}}$ holds.

Let $(D_n)_{n\geq 1}$ be any polynomial-resource WCT device family (Definition 4.2) implementing an AGI-level computational system on inputs of length n. Then there exists an NP-complete language $L \subseteq \Sigma^*$ and a constant $k \geq 1$ such that, for infinitely many input lengths n, the system realized by (D_n) fails *to decide L on at least one input of length n using only resources obeying* (21)*. Equivalently, no such WCT-bounded AGI can decide all NP-complete languages within polynomial physical resources without violating at least one of assumptions (i)–(iii).*

Proof. Let L_{SAT} be any fixed NP-complete language (e.g. 3-SAT) over some finite alphabet Σ. By (iii), $P_{\text{Turing}} \neq NP_{\text{Turing}}$, so $L_{\text{SAT}} \notin P_{\text{Turing}}$. By (ii) and the NP-equivalence, $L_{\text{SAT}} \notin P_{\text{WCC}}$ as well. Unpacking the definition of P_{WCC}, this means that for every discrete WCC machine W deciding L_{SAT} and for every exponent $k \geq 1$, there exist infinitely many input lengths n such that $T_W(n) > n^k$.

Now let (D_n) be a polynomial-resource WCT device family implementing an AGI-level system that, by assumption (ii), corresponds to some WCC machine W_{AGI} on inputs of length n. By Proposition 4.3, there exist constants C', d such that $T_{W_{\text{AGI}}}(n) \leq C'n^d$ for all n, because each global WCC update costs at most a constant number of lane updates and the total number of lane updates is polynomially bounded.

Suppose for contradiction that this AGI decides L_{SAT} on all inputs of length n while respecting the polynomial resource bounds (21). Then W_{AGI} decides L_{SAT} with runtime $T_{W_{\text{AGI}}}(n) \leq C'n^d$, so $L_{\text{SAT}} \in P_{\text{WCC}}$, and hence $L_{\text{SAT}} \in P_{\text{Turing}}$ by (ii), contradicting the choice of L_{SAT} and assumption (iii).

Therefore, for any polynomial-resource WCT device family (D_n), there must exist some NP-complete language L (in fact any NP-complete language) and some $k \geq 1$ such that, for infinitely many lengths n, the system implemented by (D_n) cannot decide L on at least one input of length n without exceeding the polynomial resource bounds in (21) or violating at least one of (i)–(iii). This is precisely the claimed conditional NP-safety statement. □

Remark 4.5 (Interpretation for AGI safety)**.** *Theorem 4.4 can be read as follows. If* (a) *the WCT rails correctly bound all physically realizable computation and* (b) *the classical separation $P \neq NP$ holds, then no AGI implemented on WCT hardware with volume, time, and energy scaling at most polynomially in the input size can solve NP-complete problems on all inputs*

within those resource bounds. In particular, such an AGI cannot "cheat" cryptography or other NP-hard tasks by exploiting super-Turing parallelism or infinitely fine analog degrees of freedom: any attempt to do so would require either super-polynomial physical resources or a violation of the WCT rails.

5 Uniform Encodings, Decision Equivalence, and the Physical Bridge

The preceding sections establish an equivalence of *language classes* between uniform discrete local-update WCC machines and standard Turing machines. They do not establish a pointwise bijection between every abstract Turing configuration and every continuous wavefield. In particular, a general polynomial-time verifier need not be in NC^1: the WCC machine simulates the full verifier computation using polynomially many local rounds.

5.1 Uniform input and certificate representations

For each fixed classical deterministic machine M, the construction of Section 1 supplies a polynomial-time injective tape-to-lattice encoding $\mathrm{enc}_M(x)$ and a fixed-radius local transition rule implementing one classical Turing step in a constant number of WCC rounds. The readout of a halting state is polynomial-time uniform. For a classical verifier $V_{\mathrm{TM}}(x, w)$, use the same uniform construction on an efficiently delimited encoding of the pair (x, w).

Lemma 5.1 (Preservation of verifier outcomes)**.** *For every fixed polynomial-time Turing verifier V_{TM}, there exists a uniform discrete WCC verifier V_{WCC} satisfying*

$$V_{\mathrm{TM}}(x, w) = 1 \quad \Longleftrightarrow \quad V_{\mathrm{WCC}}(\mathrm{enc}(x, w)) = 1$$

for every input and candidate witness, with polynomially bounded encoding, time, lattice size, and decoding overhead.

Proof. Apply the two-phase radius-one Turing simulation to the fixed verifier V_{TM}. Its transition rule does not depend on x or w, and the input pair is encoded uniformly. The simulation invariant preserves the simulated tape, head, and halting state, so acceptance agrees. Polynomial runtime and tape usage of the verifier give polynomial overhead for the WCC representation. □

5.2 Decision-class equivalence

Theorem 5.2 (Uniform discrete complexity equivalence)**.** *For the uniform finite-state model of Definition 1.1,*

$$P_{\mathrm{WCC}}^{\mathrm{disc}} = P_{\mathrm{TM}}, \qquad NP_{\mathrm{WCC}}^{\mathrm{disc}} = NP_{\mathrm{TM}}.$$

The equalities identify the sets of recognized languages *and are not a claim that arbitrary encodings are bijective or that physically constrained WCT evolution realizes every such machine.*

Proof. The deterministic equality is Corollary 1.19; the nondeterministic verifier equality is Theorem 2.5. Both rely on polynomial-time uniform input encoding and decoding. □

5.3 Conditional lifting, without conflating models

Proposition 5.3 (Classical implication under an actual common-model separation)**.** *If a language L is proved to satisfy $L \in NP_{\mathrm{WCC}}^{\mathrm{disc}} \setminus P_{\mathrm{WCC}}^{\mathrm{disc}}$, then $P \neq NP$.*

Proof. Substitute the discrete class equalities from Theorem 5.2. □

A lower bound for only *physical* curvature-bounded realizations does not satisfy the premise of this proposition. Deriving classical $P \neq NP$ from a physical separation would require an independently proved equality of the *same* physical and discrete decision and verifier classes, with polynomial preservation of all relevant costs. No such physical-to-discrete resource equivalence or classical separation is established here.

5.4 Current formalization status (7 November 2026)

The revised Part II inventory contains 28 statements: 27 Exact under their amended hypotheses and one Restricted, namely the uniformity of the author's original two-phase machine W_M (II-10). An exact conditional implication does not establish its open antecedents. The statement below supersedes the manuscript's earlier dated kernel-status addenda.

Discrete computational model. For the one-dimensional, radius-one finite-state quiescent model, the exact classical equalities are formalized as `PVNP.Discrete.P_WCC_eq_P_TM` and `PVNP.Discrete.NP_WCC_eq_NP_TM`. For every fixed radius $r \geq 1$, they are formalized by `P_WCCR_eq_P_TM` and `NP_WCCR_eq_NP_TM`. In arbitrary dimension, the full finite-state equivalences are established by `P_WCCD_eq_P_TM` and `NP_WCCD_eq_NP_TM` for every d. The concrete single-tape sweep interpreter `WCCDSweep.toTM` accounts for all tape motion and has an explicit polynomial bound $(128d+421)(n+2T(n)+3)^{2d+3}$ for its stated input dimensions and first-decision conventions. The radius-zero case requires separate treatment because no intercell communication occurs.

Charged uniform model and class equivalence. For the manuscript's uniform box/lattice model, including polynomial-time preparation, lattice generation and output decoding, Lean states `PartIIUniform.P_disc_eq_P_TM` and `PartIIUniform.NP_disc_eq_NP_TM`. The forward inclusion uses a fully charged one-cell reader whose polynomial-time encoder may run the original Turing decider. That observation does *not* establish the reverse inclusion. The separate proof `PartIIUniform.wcc_to_tm` simulates the charged lattice, encoder, rounds and decoder with an ordinary single-tape machine. Thus the generic equalities are statements about a specified finite-state model, not the existence of a physical wave implementation of the same resource costs.

The author's original two-phase machine, II-10. The actual alphabet, two-phase local rules, bounded-window invariant and first-answer time $T_{W_M}(x) = 2T_M(x)$ are formalized as `PartIITwoPhase.two_steps`, `invariant_even`, `invariant_odd` and `runtime`, subject to the explicitly stated head-in-window range. The real canonical decoder is a polynomial-time transducer (`TwoPhaseDecoder.decM_fp`), with accept taking precedence over reject in the declared output encoding. The original *complete* polynomial-radius window bitmap writer and the full initial configuration state serializer still require Lean proofs as polynomial-time single-tape output machines. The Phase XIII reference writers and exact bitmap statements supply supporting constructions, not these missing uniform `FP` witnesses. Consequently the original construction remains Restricted.

Addendum, 12 November 2026: II-10 is Exact. The two missing witnesses are now Lean theorems, `TwoPhasePolynomialWriter.originalWindowCode_fp` and `TwoPhasePolynomialWriter.originalInputCode_fp`, for the actual polynomial-radius machines and every polynomial radius; see Remark 1.17. The Part II inventory is therefore 28 Exact and none Restricted. The exact-head gate `scripts/verify_ii10_original_wm.sh` (complete kernel build, standard-axiom audit and reference regressions) passed on repository commit `df8824c` with Lean 4.23.0.

Historical corrections retained in the audit. The original all-time two-phase invariant and one-step lemma were false once the head left the finite window; the current statements use the proven head-in-window time range. The original cn^k universal runtime bound forced $T_M(0) = 0$; its corrected form uses $c \max(n, 1)^k$ so the class is ordinary **P**. The Lean counterexamples to the superseded definitions remain available as `invariant_literal_fails`, `one_step_literal_fails`, and `P_Turing_lit_eq`. This historical provenance is preserved in the claim inventory, rather than repeated as contradictory current status.

New supporting SAT and physical results. The formal Cook–Levin reduction and exact dense-SAT target equivalences are results for the classical discrete model. Restricted single-tape crossing bounds, including near-quadratic estimates, do not prove an unrestricted superpolynomial SAT lower bound. The conditional spatial-bandwidth and event capacity estimates require the spectral and independent-lane assumptions stated in Theorems 3.8 and 4.3. A charged uniform physical-to-discrete realization and an all-decider physical curvature barrier are still unproved.

Kernel verification and release boundary. Some Phase XII and XIII declarations were merged while hosted CI jobs failed *before their first step*. The latest full Lean kernel build and standard-axiom audit must be repeated successfully on the current commit before these additions can be called independently kernel verified. PDF compilation does not certify Lean proofs. For dated historical statements and the full 91-claim status, see the maintained manuscript inventory and the independent source-to-Lean publication reconciliation.

6 Conclusion

The principal rigorous result is the polynomial-time equivalence of *uniform discrete* WCC machines and classical Turing machines. Fixed finite-state local rules, polynomial-size constructible lattices, polynomial-time input encoders and output decoders, and polynomially many global update rounds yield

$$P_{\mathrm{WCC}}^{\mathrm{disc}} = P, \qquad NP_{\mathrm{WCC}}^{\mathrm{disc}} = NP.$$

The uniformity requirements exclude arbitrary noncomputable preprocessing in the encoding map. Exact wave-to-symbol correspondence requires quantization compatibility; the converse construction of an abstract finite-state representative evolution does not establish realization by a chosen nonlinear WCT PDE.

The physical curvature/energy channel-counting estimates are presented conditional on their additional spectral-support and dynamical assumptions. Neither those estimates nor the discrete Turing simulations prove $P_{\mathrm{WCC}}^{\mathrm{phys}} = P$ or $NP_{\mathrm{WCC}}^{\mathrm{phys}} = NP$, and no unconditional $P \neq NP$ or universal SAT lower bound is claimed in this manuscript.

The remaining research problem is to construct and validate a precise physical realization map with uniform polynomial resource overhead, then separately study any model-relative physical resource lower bounds.

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Part III: Physical Complexity Reassessment

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Classical Complexity and Curvature-Bounded Physical Computation

A Critical Reassessment of the Proposed Physical P versus NP Separation

Richard J. Reyes

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Revised mathematical audit: October 27, 2026

Abstract

This technical revision distinguishes the mathematically well-defined classical P versus NP question from questions of realizability and resource usage in Wave Curvature Computation (WCC). The discrete finite-state, local-update model can represent polynomial-time Turing computations under uniform encoding, decoding, and transition assumptions. That equivalence does not prove that each symbolic computation has a polynomial-curvature physical realization. Several claims in the original December 2025 technical note required correction: physical distinguishability alone does not force positive Laplacian-curvature cost; finite first-derivative energy and locality do not imply a universal polynomial bound on second-derivative curvature; expander graphs do not, without a separate geometric embedding theorem, imply exponential interface area; and the total influence of any n-bit Boolean function is at most n. We establish exact counterexamples and corrected conditional estimates, clarify physical P and NP witness admissibility, and isolate explicit proof obligations for a possible restricted-model separation. No unconditional separation of classical P and NP, nor a proved separation of the curvature-bounded physical classes, is claimed here.

New decoder-uniformity obstruction (November 2, 2026). A fixed, language-independent sign map encodes every n-bit string into n real amplitudes ± 1, with squared mass n, and the sign receiver recovers the string under per-coordinate additive errors with magnitude strictly below one. Composing this receiver with an arbitrary predicate would decide that predicate without charging the final decoding step. Such a decoder need not be computable, let alone efficient. Consequently, 2^n recoverable labels, finite amplitude cost, and a uniform encoding cannot establish an unconditional lower bound for decision problems without bounding the decoder's complexity. This counterexample does not establish an efficient SAT algorithm. The physical continuum-to-computation bridge and universal curvature-versus-decision lower bound remain open.

1 Historical Context: The Divergence of Logic and Physics

The classical P versus NP question concerns language classes defined by uniform Turing-machine running time and polynomial verification; the question is mathematically well posed independent of a physical implementation [9, 11]. Turing machines are abstract algorithms and do not assert that unbounded physical memory is available at zero energetic cost. Whether a concrete physical computer efficiently realizes a particular algorithm is an additional question. Locality, thermodynamic dissipation, finite precision, field regularity, and resource budgets can restrict a physical substrate without changing the definition of classical P or NP.

WCT motivates such a resource-sensitive investigation. We distinguish three notions throughout: classical classes P, NP; uniformly specified discrete WCC classes $P^{\mathrm{disc}}_{\mathrm{WCC}}, NP^{\mathrm{disc}}_{\mathrm{WCC}}$; and

physical, curvature-rail classes $P^{\text{phys}}_{\text{WCC}}, NP^{\text{phys}}_{\text{WCC}}$. A physical-vs-classical comparison must specify a uniform implementation map and its costs before inferring equality or separation of these classes.

2 The Physical Distinguishability Obstruction

2.1 Physical Encoding of Logical Bits

Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain, with periodic or Neumann boundary conditions allowed. Two field encodings $\psi_0, \psi_1 \in H^2(\Omega)$ can be distinct in the H^1 norm without having different Laplacians. We therefore distinguish *symbolic separation* from *curvature cost*.

Definition 2.1 (Regularized curvature cost)**.** *For $\varepsilon > 0$ and $\psi \in H^2(\Omega; \mathbb{C})$ define the complex-safe curvature operator*

$$\Theta_\varepsilon[\psi] = -\frac{\Delta\psi\,\overline{\psi}}{|\psi|^2 + \varepsilon^2}, \qquad C_{\Theta,\varepsilon}[\psi] = \int_\Omega |\Theta_\varepsilon[\psi]|^2\, dx.$$

For a pair of encodings, a proposed separation cost is $C_{\Theta,\varepsilon}[\psi_1 - \psi_0]$. This particular functional measures second-derivative structure, not every possible resource cost of a bit.

Proposition 2.2 (Counterexample to a universal positive separation cost)**.** *Distinct H^1 states need not have positive curvature-separation cost.*

Proof. Set $\psi_0 \equiv 0$ and $\psi_1 \equiv 1$ on Ω. Then $\|\psi_1 - \psi_0\|^2_{H^1} = |\Omega| > 0$ while $\Delta(\psi_1 - \psi_0) = 0$. Hence

$$C_{\Theta,\varepsilon}[\psi_1 - \psi_0] = 0.$$

The counterexample obeys the stated periodic or Neumann conditions. □

Remark 2.3 (WIN-12)**.** *The original Lemma 2.2 inferred nonzero Laplacian from nonzero H^1 distance, which is false. An experimentally grounded lower bound for a specific device would require extra boundary conditions, state constraints, and a coercivity inequality; distinguishability alone is insufficient.*

2.2 Curvature Cost of an n-Bit Configuration

A lower bound $c_0 n$ for storing n bits may be established within a specified *independent-cell encoding* if every occupied cell contributes at least $c_0 > 0$ and the chosen functional is additive across the disjoint supports. Neither cell independence nor positive c_0 follows from the previous definition. In particular, an n-bit string is not required to simultaneously realize the 2^n possible assignments.

Proposition 2.4 (Conditional additive storage estimate)**.** *If $\Omega = \bigsqcup_{j=1}^n \Omega_j$ and a chosen non-negative storage functional E_{store} is additive on the disjoint cells with $E_{\text{store}}(\psi_{b_j}|_{\Omega_j}) \geq c_0$ for each j, then $E_{\text{store}}(\psi_w) \geq n c_0$.*

Proof. Sum the n cellwise inequalities. No such conclusion is asserted for $C_{\Theta,\varepsilon}$ without proving the premises. □

2.3 Classical Turing Machines and Physical Implementations

The Turing model accounts for operations through a formal step count, not through a PDE-based curvature-energy functional. Its omission of an explicit energy budget does not imply a theorem that every physical implementation must consume zero energy or that a polynomial-overhead physical simulation is impossible. The physical cost of symbols, memory, and erasure is an independent quantitative engineering question [3, 4].

2.4 Physical Complexity Classes

For a fixed, uniform physical model $\mathfrak{M}$ and a stated polynomial resource rail R, define $P^{\mathrm{phys}}_{\mathfrak{M},R}$ by uniformly specified deterministic deciders that meet the rail on every input. Define $NP^{\mathrm{phys}}_{\mathfrak{M},R}$ with polynomial-length, physically admissible witness encodings and deterministic verifiers subject to the same rail. The physical classes are machine-relative; we do not identify them with classical classes without separately proving the necessary simulation theorems.

3 Classical Computation and Physical Resource Models

The classical formalism is mathematically defined even if no hardware can realize an unbounded tape for all input sizes at once. Complexity theory uses a *family* of finite computations, one for each input length. Physical realizability concerns whether a single uniform device family has allowed costs, scaling, and error control. The extended Church–Turing principle is a statement about efficient simulation of physical models, not a prerequisite for formulating classical P versus NP.

Remark 3.1. *A distinction between two physical encoding conventions cannot by itself refute a theorem about abstract languages. Conversely, a formal equivalence of symbolic update systems does not guarantee faithful realization by a specified nonlinear PDE under fixed energy and stability rails.*

4 The WCC Model and Curvature Axioms

Definition 4.1 (Polynomial curvature rail)**.** *For a uniform physical device family indexed by input length n, a trajectory $\psi_{n,x}(t)$ respects a polynomial curvature rail if there exists one polynomial p such that, for every input x of length n and all times $0 \le t \le T(n)$,*

$$\sup_{|x|=n} \sup_{0\le t\le T(n)} C_{\Theta,\varepsilon}[\psi_{n,x}(t)] \le p(n).$$

A requirement of finite *curvature at each n is strictly weaker.*

Lemma 4.2 (A valid conditional curvature bound)**.** *For every $\psi \in H^2(\Omega;\mathbb{C})$ and $\varepsilon > 0$,*

$$C_{\Theta,\varepsilon}[\psi] \le \frac{1}{4\varepsilon^2}\|\Delta\psi\|^2_{L^2(\Omega)}.$$

Proof. For $r \ge 0$, $r/(r^2+\varepsilon^2) \le 1/(2\varepsilon)$. Apply this pointwise to $r = |\psi|$ and integrate the square. □

Remark 4.3 (WIN-13)**.** *The original operator $-\Delta\psi/(\psi+\varepsilon)$ does not have the claimed uniform denominator bound $|\psi+\varepsilon| \ge \varepsilon$: $\psi = -\varepsilon$ is an immediate counterexample. One may instead assume an explicit gap $|\psi+\varepsilon| \ge \delta > 0$ or use the complex-safe regularizer of Definition 2.1. All estimates in this revision use the latter.*

5 Discrete and Physical WCC Semantics

5.1 Discrete WCC Model

A discrete WCC machine uses a finite alphabet S, a fixed-radius local update rule, polynomially many lattice cells, a uniform family of lattices, and polynomial-time constructible encoders and decoders. Machine descriptions must not contain nonuniform advice or a hard language in the initialization map.

Theorem 5.1 (Discrete Turing equivalence under uniformity)**.** *For the uniform finite-state, fixed-radius WCC model defined above, with polynomial configuration size and polynomial rounds,*

$$P_{\mathrm{WCC}}^{\mathrm{disc}} = P, \qquad NP_{\mathrm{WCC}}^{\mathrm{disc}} = NP.$$

Proof. A Turing machine sequentially updates all polynomially many cells in polynomial time per global WCC step. Conversely, standard finite-state local tape-and-head encoding simulates each step of a fixed polynomial-time Turing machine by a constant number of local rounds on a polynomial-size lattice. The two simulations preserve polynomial witness length and verifier runtime. Polynomial-time uniform encoding, decoding, and lattice generation exclude hidden oracle computations. □

5.2 Physical WCC Model and Witness Admissibility

The physical model additionally restricts allowable embeddings, precision, curvature-energy budgets, and dynamics. Therefore the discrete theorem alone supplies no equality $P_{\mathrm{WCC}}^{\mathrm{phys}} = P$ or $NP_{\mathrm{WCC}}^{\mathrm{phys}} = NP$.

Theorem 5.2 (Verifier inclusion; correction of the empty-NP claim)**.** *In any model whose NP verifiers may ignore their witnesses and which subjects deciders and verifiers to the same resource rails,*

$$P_{\mathrm{WCC}}^{\mathrm{phys}} \subseteq NP_{\mathrm{WCC}}^{\mathrm{phys}}.$$

Proof. Given a polynomial-resource decider $D(x)$, define $V(x, w) = D(x)$ for a fixed empty witness (or ignore w). It has the same resource usage and correctness. Thus each physical-P language has a physical-NP verifier. □

Remark 5.3 (WIN-16)**.** *The original assertion $NP_{\mathrm{WCC}}^{\mathrm{phys}} = \varnothing$ contradicts the claimed nonempty physical-P class together with Theorem 5.2. No physical class separation follows from that assertion.*

5.3 A Correct Comparison Statement

Proposition 5.4 (No logical conflict between the two semantics)**.** *Suppose that the discrete classes equal classical P, NP, but physical admissibility imposes additional restrictions. It is logically possible that the physical classes are proper subsets of their respective discrete classes. No equality or inequality between physical P and NP follows from those facts alone.*

Proof. A class inclusion into a larger class does not determine whether two subclasses are equal. In particular the implication $P^{\mathrm{phys}} \subseteq P^{\mathrm{disc}}$ and $NP^{\mathrm{phys}} \subseteq NP^{\mathrm{disc}}$ is compatible with both $P^{\mathrm{phys}} = NP^{\mathrm{phys}}$ and a strict inclusion. □

5.4 On the Claimed Ill-Posedness

Remark 5.5 (WIN-17)**.** *Classical P versus NP is a precise question about languages and Turing machines. What can be* underspecified *is the expression "physically efficient computation" unless the physical machine, precision, allowed resources, and asymptotic family are specified. Failure of a physical equivalence theorem cannot make the classical mathematical question ill-posed.*

6 Locked Expanders and Physical Embedding Costs

A bounded-degree expander has an edge boundary of order the size of a balanced vertex subset. That is an $\Omega(n)$ combinatorial cut constraint on $O(n)$ vertices; it does not by itself prove that *one* encoded satisfying assignment must have $2^{\Omega(n)}$ interface area. Counting exponentially many assignments is different from physically storing all assignments at once.

Proposition 6.1 (Conditional interface-to-curvature lower bound)**.** *Suppose that for each admissible realization ψ of an instance F_n there is a geometric interface $\Gamma(\psi)$ such that*

$$\mathcal{H}^2(\Gamma(\psi)) \geq A_n \quad \textit{and} \quad C_{\Theta,\varepsilon}[\psi] \geq \kappa\, \mathcal{H}^2(\Gamma(\psi))$$

with a constant $\kappa > 0$ uniform in n, the input, and the encoding. Then every such realization satisfies $C_{\Theta,\varepsilon}[\psi] \geq \kappa A_n$.

Proof. Compose the two assumed inequalities. □

Remark 6.2 (WIN-18)**.** *A proved $A_n \geq 2^{\beta n}$, uniformly for every admissible implementation of a specific* decision problem*, would be a substantive new embedding theorem. Graph expansion alone does not establish it, and the original exponential-domain-wall theorem is therefore withdrawn as an unconditional conclusion.*

Proposition 6.3 (Verification of a supplied assignment)**.** *The language*

$$L_{\text{check}} = \{(F, \alpha) : \alpha \textit{ satisfies the 3-CNF formula } F\}$$

is in classical P.

Proof. Evaluate each literal of each clause under the supplied assignment and accept exactly when every clause contains a true literal. The runtime is polynomial in the encoded formula and assignment lengths. Restricting to uniformly constructible expander formulas does not make this verification task NP-hard. □

7 A Conditional Constructibility Barrier

Definition 7.1 (Per-step curvature increment)**.** *For a trajectory $\psi_{t+1} = F_n(\psi_t)$ set $\Delta C_{\Theta,\varepsilon}(t) = C_{\Theta,\varepsilon}[\psi_{t+1}] - C_{\Theta,\varepsilon}[\psi_t]$. A bound $|\Delta C_{\Theta,\varepsilon}(t)| \leq K(n)$ is a* separate hypothesis*; finite precision and local update radius alone do not imply a uniform constant K for all system sizes.*

Theorem 7.2 (Conditional telescoping capacity estimate)**.** *If $C_{\Theta,\varepsilon}[\psi_0] \leq p_0(n)$, $|\Delta C_{\Theta,\varepsilon}(t)| \leq K(n)$ for all $t < T(n)$, $T(n) \leq p_T(n)$, and p_0, K, p_T are polynomials, then*

$$C_{\Theta,\varepsilon}[\psi_{T(n)}] \leq p_0(n) + p_T(n)K(n) = \text{poly}(n).$$

Proof. Telescope the increments and use the triangle inequality:

$$C_{\Theta,\varepsilon}[\psi_T] = C_{\Theta,\varepsilon}[\psi_0] + \sum_{t<T} \Delta C_{\Theta,\varepsilon}(t) \leq p_0(n) + T(n)K(n).$$

□

Remark 7.3 (WIN-19)**.** *A high-curvature* witness *that violates the physical verifier rail is not evidence that its language belongs to physical NP. A separation requires one common language L with a polynomial-resource physical verifier for every admissible witness and a proof that no permitted polynomial-resource physical decider recognizes L. A lower bound only on one chosen witness representation is insufficient.*

8 Turing Simulability of Discrete WCC

The discrete simulation of Theorem 5.1 is an algebraic statement about locally updated finite symbols. With $N(n)$ cells and $T(n)$ global rounds, a sequential machine can evaluate the entire evolution in $T(n)\,\text{poly}(N(n))$ time. This does not convert an arbitrary physical PDE into the desired finite-state machine, nor does it charge curvature resources to the simulation. Such a cost-preserving realization must be established independently (see [15]).

9 Classical-to-Physical Implications

The extended Church–Turing thesis addresses simulation of physical computations in an abstract model with polynomial overhead. A one-directional simulation such as $P^{\mathrm{phys}} \subseteq P$ does not entail $P^{\mathrm{phys}} = P$, nor does it identify physical NP with classical NP. A physical equality claim would require the relevant simulation in both directions and preservation of witnesses, encodings, errors, and budgets.

Remark 9.1. *Even the hypothetical equality $P = NP$ would not, solely by definition, establish an equality between two separately defined physical classes. Conversely, unequal physical subclasses would not by themselves resolve classical P versus NP. No "if and only if ECT" criterion is proved or assumed here.*

Updated scope of current computational results. The repository contains a uniform Cook–Levin reduction and restricted quadratic-type single-tape SAT lower bounds. Neither a superpolynomial lower bound for every correct SAT decider nor a uniform physical-to-Turing simulation under the specified WCT resource constraints has been proved. Formula-dependent phase-energy certificates do not provide efficient physical preparation of the required field. The newer results therefore do not overturn the conditional physical conclusions of this reassessment.

10 Conclusion: What Is Established and What Remains Open

The corrected statements are: (i) under explicit uniformity, discrete WCC has the standard polynomial Turing simulation and verifier equivalence; (ii) positive storage or curvature costs require device-specific coercivity assumptions; (iii) H^2 control together with a positive regularization parameter gives a rigorous curvature upper bound; (iv) expander connectivity does not independently establish an exponential lower bound on the curvature of every decision procedure; and (v) Boolean influence cannot provide an exponential lower bound because it never exceeds n.

The proposed physical separation remains an open mathematical program. A viable theorem must name a language, specify its uniform physical encoding and resource rails, provide an admissible polynomial-resource physical verifier, and prove a lower bound against every admissible physical decider, not only against a particular geometric search protocol. Classical P versus NP remains mathematically well posed and unresolved by these constructions.

10.1 Current Lean formalization status (7 November 2026)

The revised Part III inventory contains 22 entries: 21 Exact under their explicit hypotheses and one Open interpretive remark (III-22), concerning the arrow of time. Exact includes *conditional* analytic results whose physical assumptions are not independent Lean theorems. This unified register supersedes the separate dated verification addenda.

Analytical and finite mathematical results. The denominator counterexample, the zero symbolic-curvature example, the H^2-to-curvature comparison (Lemma 4.2), conditional storage and channel-capacity estimates (Theorem A.3), and the high-frequency periodic-cube example (Proposition A.1) have matching scoped Lean results. The Boolean-hypercube influence bounds and the supplied CNF-assignment verifier are also formalized. The supplied assignment test lies in polynomial time and is not a candidate separation language.

Discrete computational equivalence. The finite-state, fixed-radius discrete models have explicit Turing simulations, including all-dimensional sweep models with charged costs. The actual uniform lattice model, including polynomial-time lattice generation, encoding and readout,

has exact theorems `PartIIUniform.P_disc_eq_P_TM` and `PartIIUniform.NP_disc_eq_NP_TM`. Neither theorem establishes a polynomial-resource physical realization by the curvature PDE. The original two-phase construction's specific uniform writers remain Restricted in Part II (II-10). *Addendum, 12 November 2026:* both writers are now verified as polynomial-time single-tape transducers, and II-10 is Exact.

Field transport and well-posedness. The transport lower bound is verified in Lean *from* the declared instantaneous flux inequality; that implication does not by itself derive the flux inequality for every continuum solution. For the stated finite torus method-of-lines system, Lean derives mass and Hamiltonian conservation and a finite-dimensional flux inequality with explicit assumptions (`PVNP.WCTLattice`). For a specified frozen cubic–quintic PDE, Phase VI supplies scoped H^2 periodic-torus well-posedness and real-detector flux results (`PVNP.Torus`). Those results do not establish a universal device-size-independent physical computation rail. The independent fixed-mass nodal supplement has an exact finite two-node Lean analogue and a separate *analytic* continuum compactness proof, not a kernel-formalized continuum theorem.

Open physical and complexity boundaries. Unrestricted encoders or decoders trivialize physical decision models. The measured physical preparation, bandwidth, energy, spatial precision, and readout costs must be specified and charged. Neither the finite curvature inequalities nor the restricted quadratic single-tape SAT lower bounds imply a universal superpolynomial SAT decision bound. The all-algorithm separator and locked-expander bounds remain unproved. Classical and physical P versus NP remain unresolved.

Verification and publication provenance. Some Phase XII–XIII modules were merged after GitHub Actions jobs failed before their first step; newer source code is not a freshly passed kernel and axiom audit. The exact-head Lean checks must succeed before a release may call those declarations independently verified. Older statuses remain in Git history and the 91-claim ledger. Recompiled PDFs are needed after the latest source amendments.

A Conditional Curvature–Capacity Analysis

A.1 Finite first-derivative energy is insufficient

Proposition A.1 (A high-frequency counterexample)**.** *On the periodic cube* $\Omega = (0, 2\pi)^3$*, let*

$$\psi_N(x) = 1 + \frac{1}{N}\sin(Nx_1), \qquad N \geq 2.$$

Then $\int_\Omega |\nabla\psi_N|^2 dx = 4\pi^3$ *and* $\int_\Omega |\Delta\psi_N|^2 dx = 4\pi^3 N^2$*. With* $\varepsilon = 1$ *in Definition 2.1,*

$$C_{\Theta,1}[\psi_N] \geq \frac{16\pi^3}{25} N^2 \longrightarrow \infty,$$

while the ordinary nonnegative mass, quartic, and gradient contributions of a fixed-coefficient first-derivative field-energy functional are uniformly bounded.

Proof. $\nabla\psi_N = (\cos(Nx_1), 0, 0)$ and $\Delta\psi_N = -N\sin(Nx_1)$, giving the stated integrals. Since $1/2 \leq \psi_N \leq 3/2$, the elementary inequality $\psi_N/(1+\psi_N^2) \geq 2/5$ yields $|\Theta_1[\psi_N]| \geq (2/5)|\Delta\psi_N|$. Integrating gives the claimed bound. The amplitude stays between $1/2$ and $3/2$, so its L^2 and L^4 contributions remain bounded. □

Remark A.2 (WIN-14)**.** *Finite energy involving* $\|\nabla\psi\|_2^2$ *does not imply a uniform* H^2 *estimate, and generic finite action does not silently supply it. A positive* H^2 *regularity assumption or a proved PDE estimate is required.*

A.2 Locality and gravitational considerations

Lieb–Robinson bounds concern information propagation in suitable short-range quantum lattice systems; they do not imply that arbitrary PDE second derivatives grow at most polynomially in time. The original argument also identified divergence of a chosen normalized Laplacian functional with divergence of the stress–energy tensor without an established inequality between them. Einstein's equation by itself cannot close that missing step. The high-frequency family above is a counterexample to a universal fixed-energy bound on the chosen curvature functional.

Theorem A.3 (Corrected conditional capacity law)**.** *Assume a uniform family of WCT fields* $\psi_{n,x}(t) \in H^2(\Omega_n)$ *with a fixed* $\varepsilon > 0$ *and*

$$\sup_{|x|=n} \sup_{t \leq T(n)} \|\Delta\psi_{n,x}(t)\|_{L^2(\Omega_n)} \leq q(n)$$

for one polynomial q*. Then the regularized curvature cost obeys*

$$\sup_{|x|=n} \sup_{t \leq T(n)} C_{\Theta,\varepsilon}[\psi_{n,x}(t)] \leq \frac{q(n)^2}{4\varepsilon^2}.$$

Proof. Apply Lemma 4.2 uniformly over the family. □

Remark A.4 (WIN-15)**.** *This theorem is conditional on the missing uniform second-derivative estimate. Finite energy, spatial locality, and finite runtime alone do not provide the hypothesis. No universal QFT/GR/Lieb–Robinson polynomial curvature-capacity law is claimed.*

B Boolean Influence and Hypercube Boundary

For a Boolean function $f : \{0,1\}^n \to \{0,1\}$, let $x^{\oplus i}$ denote bit i flipped. Define

$$\mathrm{Inf}_i(f) = 2^{-n}\#\{x : f(x) \neq f(x^{\oplus i})\}, \qquad \mathrm{Var}(f) = \sum_{i=1}^{n} \mathrm{Inf}_i(f).$$

Define the *directed* boundary $B_{\to}(f) = \#\{(x,i) : f(x) \neq f(x^{\oplus i})\}$ and the *undirected* boundary $B_{\mathrm{und}}(f) = \frac{1}{2}B_{\to}(f)$.

Lemma B.1 (Corrected boundary–variation identities)**.** *For every Boolean f,*

$$B_{\to}(f) = 2^n \,\mathrm{Var}(f), \qquad B_{\mathrm{und}}(f) = 2^{n-1}\,\mathrm{Var}(f).$$

Proof. Count the $2^n \,\mathrm{Inf}_i(f)$ directed edges in coordinate i and sum over i; every undirected edge has two orientations. □

Remark B.2 (WIN-20)**.** *The original definition counted directed pairs while its lemma used the undirected normalization. Both formulas are now displayed separately.*

Theorem B.3 (Universal total-influence bound)**.** *For any Boolean function of n bits,*

$$0 \leq \mathrm{Var}(f) \leq n.$$

Consequently, it is impossible to have $\mathrm{Var}(F_n) = 2^{\Omega(n)}$ for any family of Boolean functions F_n with n variables.

Proof. Each $\mathrm{Inf}_i(f)$ is a probability in $[0,1]$. Summing n terms proves the upper bound. □

Corollary B.4 (Polynomial variation is vacuous as a complexity class)**.** *If P_{var} denotes Boolean language families with $\mathrm{Var}(f_n) \leq n^{O(1)}$ for all n, then P_{var} contains every Boolean language family. Thus the proposed exponential-variation exclusion of SAT and the resulting argument for $P \neq NP$ do not hold.*

Proof. Apply Theorem B.3 with exponent one. □

Remark B.5 (WIN-21)**.** *A large* unnormalized *cube boundary can be exponential even for simple functions; for example parity has $\mathrm{Var}(f) = n$ and $B_{\to}(f) = n2^n$. Neither an exponential directed boundary nor graph expansion by itself implies classical algorithmic hardness. The previous theorems claiming an exponential total influence and a strict subclass of NP have been replaced by these exact bounds.*

C Arrow of Time: Limits of the Claim

The thermodynamic arrow and the availability of time-reversing or closed-timelike-curve models are additional physical hypotheses, not logical characterizations of classical P and NP. A process computing a witness need not run the verifier backward: search algorithms operate forward in ordinary time. Entropy-reducing maps need not send information from the future to the past, and inverse dynamics need not efficiently solve SAT. Conversely, a generic assertion of retrocausality does not specify the special self-consistent fixed-point model considered in [7, 8].

Remark C.1. *No equivalence $P^{\mathrm{phys}} = NP^{\mathrm{phys}} \iff$ retrocausality, and no physical separation inferred merely from a forward thermodynamic arrow, is proved. This section is retained as a research direction rather than as a theorem.*

Remark C.2 (Addendum, 12 November 2026)**.** *For the frozen dynamics the sentence on inverse dynamics has an exact form. Complex conjugation reverses time: if $\psi(t)$ is a solution, so is $\overline{\psi(-t)}$. Hence the state at time $-t$ is the conjugate of the forward evolution, for time t, of the conjugated initial state (`Torus.backward_eq_forward_conj`, `Torus.flow_neg`). Running this dynamics backward is a forward computation with the encoder and decoder composed with conjugation. The remark above remains an interpretation, not a theorem.*

D Mathematical Revision Register

The following identifiers refer to the issue windows attached to this revised manuscript. "Corrected" means a valid replacement statement has been supplied; "conditional" means the original conclusion is not established and a precise additional hypothesis is specified.

WIN-12
: Zero-curvature counterexample to the distinct-state lemma (Proposition 2.2); corrected.

WIN-13
: Unsafe denominator $|\psi + \varepsilon| \geq \varepsilon$; regularization/gap distinction (Definition 2.1); corrected.

WIN-14
: First-derivative energy does not control H^2; explicit counterexample (Proposition A.1); corrected.

WIN-15
: Universal polynomial curvature growth from locality; replaced by conditional H^2 capacity (Theorem A.3).

WIN-16
: Impossible assertion physical NP is empty; verifier inclusion (Theorem 5.2); corrected.

WIN-17
: Classical mathematical ill-posedness; replaced by a distinction between formal and physical questions (Section 5); corrected.

WIN-18
: Unsupported expander-to-exponential-interface implication; conditional interface bound (Proposition 6.1).

WIN-19
: Inadmissible high-curvature NP witnesses; common physical verifier rail specified (Section 7).

WIN-20
: Directed/undirected cube boundary factor of two; exact identities (Lemma B.1); corrected.

WIN-21
: Exponential Boolean total influence; universal bound and vacuity of P_{var} (Theorem B.3); corrected.

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Auxiliary Appendix: Fixed-Mass Nodal Coercivity

Independent auxiliary note, not one of the original 91 manuscript claims.

Fixed-Mass Nodal Coercivity for Density-Weighted Curvature

An Independent Auxiliary Verification Note for WCT Computation

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October 27, 2026

Abstract

On a fixed two-dimensional periodic domain, we prove that positive conserved mass and an upper bound on the cubic–quintic projection-free Hamiltonian enforce a strictly positive and quantitatively explicit lower comparison between density-weighted Laplacian curvature and ordinary Laplacian energy. No uniform positive lower bound on pointwise amplitude is imposed, so fields with nodes are allowed. This closes a previously open *fixed-domain, fixed-resource* analytic step in a separate curvature–communication study. The constant is not controlled uniformly in domain size, spectral resolution, or computational input length. An exact two-site version and the algebraic readout-cost transfer are formalized in Lean. This auxiliary note is independent of the author's original P versus NP manuscript, which it neither revises nor claims to complete.

1 The maintained Hamiltonian and the curvature weight

Let $\Omega = \mathbb{R}^2/\Lambda$ be a fixed connected flat two-dimensional torus, of volume $V = |\Omega|$, and let $u \in H^2(\Omega;\mathbb{C})$. For $\kappa = k_\star^2 \geq 0$ and $\beta, \gamma > 0$, define

$$\mathcal{H}[u] = \int_\Omega \left(|(\Delta + \kappa)u|^2 - \frac{\beta}{2}|u|^4 + \frac{\gamma}{3}|u|^6 \right) dx. \tag{1}$$

This is the Hamiltonian of the separately maintained projection-free evolution

$$i\partial_t u = (\Delta + \kappa)^2 u - \beta|u|^2 u + \gamma|u|^4 u. \tag{2}$$

The present theorem concerns the *distinct* density-weighted curvature functional, not the Hamiltonian's own quadratic term. For $\varepsilon > 0$, $\alpha \geq 0$, let

$$F_{\varepsilon,\alpha}(s) = \frac{s}{(s + \varepsilon^2 e^{-2\alpha s})^2}, \qquad \mathcal{Q}_{\varepsilon,\alpha}[u] = \int_\Omega F_{\varepsilon,\alpha}(|u|^2)|\Delta u|^2 \, dx. \tag{3}$$

The weight vanishes at every node $u = 0$. Pointwise coercivity is therefore false without a node-free assumption, and the result below is genuinely global.

2 Conservation-derived compactness

Put $a_{\beta,\gamma} = 3\beta^2/(16\gamma)$. For every $s \geq 0$,

$$-\frac{\beta}{2}s^2 + \frac{\gamma}{3}s^3 + a_{\beta,\gamma}s = \frac{\gamma}{3}s\left(s - \frac{3\beta}{4\gamma}\right)^2 \geq 0. \tag{4}$$

Consequently

$$\|(\Delta + \kappa)u\|_2^2 \leq \mathcal{H}[u] + a_{\beta,\gamma}\,\|u\|_2^2\,. \tag{5}$$

Given a positive mass $m > 0$ and finite $H_\star \in \mathbb{R}$, define

$$\mathcal{A}_{m,H_\star} = \{u \in H^2(\Omega;\mathbb{C}) : \|u\|_2^2 = m, \quad \mathcal{H}[u] \le H_\star\}.$$

If this class is nonempty, $H_\star + a_{\beta,\gamma} m \ge 0$. For every $u \in \mathcal{A}_{m,H_\star}$,

$$\|\Delta u\|_2 \le \sqrt{H_\star + a_{\beta,\gamma} m} + \kappa\sqrt{m} =: B. \tag{6}$$

On the fixed torus, elliptic regularity gives $\|u\|_{H^2} \le C_\Omega(\sqrt{m} + B)$. The compact embedding $H^2(\mathbb{T}^2) \Subset C^0(\mathbb{T}^2)$ therefore gives a common amplitude bound $\|u\|_\infty \le U$ and subsequential uniform convergence.

3 The fixed-mass nodal theorem

Theorem 3.1 (Uniform nodal coercivity on a fixed Hamiltonian sublevel)**.** *Fix $\Omega, \kappa, \beta, \gamma, m, H_\star, \varepsilon$ as above, with $\mathcal{A}_{m,H_\star} \neq \varnothing$. There exists a number $c_\star = c_\star(\Omega, \kappa, \beta, \gamma, m, H_\star, \varepsilon) > 0$ such that for* every $\alpha \ge 0$ *and every $u \in \mathcal{A}_{m,H_\star}$,*

$$\boxed{\mathcal{Q}_{\varepsilon,\alpha}[u] \ge c_\star \|\Delta u\|_2^2 .} \tag{7}$$

No hypothesis $\inf_\Omega |u| > 0$ is needed. A constructive coefficient is supplied in Theorem 4.1; it depends on the fixed domain spectrum and the resource upper bound.

Proof. From $\alpha \ge 0$ and the common amplitude bound,

$$0 \le |u|^2 + \varepsilon^2 e^{-2\alpha|u|^2} \le U^2 + \varepsilon^2,$$

and therefore

$$\mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{\|u\,\Delta u\|_2^2}{(U^2+\varepsilon^2)^2}. \tag{8}$$

Suppose no positive $c_\star$ works. Then for each integer $j \ge 1$ there are $u_j \in \mathcal{A}_{m,H_\star}$ and $\alpha_j \ge 0$ with $\|\Delta u_j\|_2 > 0$ and

$$\mathcal{Q}_{\varepsilon,\alpha_j}[u_j] < j^{-1} \|\Delta u_j\|_2^2 .$$

The uniform bound (6) implies that the left side tends to zero. By (8), $\|u_j \Delta u_j\|_2 \to 0$. Periodic integration by parts then yields

$$\|\nabla u_j\|_2^2 = -\operatorname{Re}\int_\Omega \overline{u_j}\Delta u_j\,dx \le V^{1/2}\|u_j\Delta u_j\|_2 \longrightarrow 0.$$

By the Poincaré inequality, $u_j - \langle u_j\rangle \to 0$ in L^2, where $\langle u\rangle = V^{-1}\int_\Omega u\,dx$. Pass to a subsequence that converges uniformly in C^0, using the H^2 compactness established above. Its limit is spatially constant, and fixed mass forces the modulus of that constant to equal $\sqrt{m/V} > 0$.

Thus, for all sufficiently large j, the *whole* field satisfies $|u_j(x)| \ge \frac{1}{2}\sqrt{m/V}$, for every $x \in \Omega$. Together with the common upper amplitude bound,

$$F_{\varepsilon,\alpha_j}(|u_j|^2) \ge \frac{m}{4V(U^2+\varepsilon^2)^2} =: c_0 > 0 \qquad \text{at every } x \in \Omega.$$

Hence $\mathcal{Q}_{\varepsilon,\alpha_j}[u_j] \ge c_0\|\Delta u_j\|_2^2$, contradicting its defining strict inequality for all $j > c_0^{-1}$. □

Remark 3.2 (What fixed mass does and does not imply)**.** *The theorem does* not *prove that each fixed-mass field is node-free. It proves a uniform* integrated *comparison over a compactly controlled class of fields. The positive mass is essential: the earlier high-frequency sequence $N^{-2}\sin(Nx)$ has mass tending to zero and thus does not contradict Theorem 3.1. The Hamiltonian upper bound is essential to prevent the Laplacian energy of arbitrarily fine oscillations from escaping the compact class. The same compactness argument works in dimensions $d \le 3$ on a fixed flat torus; $d = 4$ requires different control because $H^2 \not\hookrightarrow C^0$ there.*

4 An explicit spectral coercivity coefficient

The compactness argument above can be made quantitative on the fixed torus. Let Λ^* be the reciprocal lattice, and use the orthonormal Fourier basis $V^{-1/2}e^{ik\cdot x}$. Let

$$B_+ = 1 + \sqrt{H_\star + a_{\beta,\gamma} m} + \kappa\sqrt{m} > 0, \qquad \mathcal{S}_\Omega = \frac{1}{\sqrt{V}} \left(\sum_{k \in \Lambda^*} \frac{1}{1+|k|^4} \right)^{1/2}, \quad U_+ = \mathcal{S}_\Omega \sqrt{m + B_+^2}.$$

Absolute convergence holds because the torus dimension is two. Parseval and Cauchy–Schwarz show, without invoking an unspecified embedding constant, that every $u \in \mathcal{A}_{m,H_\star}$ satisfies $\|\Delta u\|_2 \le B_+$ and $\|u\|_\infty \le U_+$.

Let $\lambda_1 > 0$ be the first nonzero eigenvalue of $-\Delta$, and set $\tau = \sqrt{m/(32V)}$. Choose a finite radius R containing at least one nonzero reciprocal lattice point, sufficiently large that

$$\mathsf{H}_R B_+ \le \tau, \qquad \mathsf{H}_R = \frac{1}{\sqrt{V}} \left(\sum_{|k|>R} |k|^{-4} \right)^{1/2}.$$

Such an R exists since the Fourier tail converges. Define

$$\mathsf{L}_R = \frac{1}{\sqrt{V}} \left(\sum_{0<|k|\le R} |k|^{-2} \right)^{1/2} > 0, \quad \eta = \min\left\{ \sqrt{\frac{\lambda_1 m}{2}}, \frac{\tau}{\mathsf{L}_R} \right\} > 0.$$

Theorem 4.1 (Explicit fixed-domain coercivity coefficient)**.** *The constant in Theorem 3.1 can be chosen to be*

$$\boxed{c_{\exp} = \frac{1}{(U_+^2 + \varepsilon^2)^2} \min\left\{ \frac{m}{8V}, \frac{\eta^4}{V B_+^2} \right\} > 0.} \tag{9}$$

In particular this choice is uniform for all $\alpha \ge 0$*, and makes the dependence on the domain's Fourier spectrum, positive mass, and energy upper bound explicit.*

Proof. Write $g = \|\nabla u\|_2$ and $\bar{u} = V^{-1}\int_\Omega u\,dx$. We distinguish two cases.

If $g \le \eta$, Poincaré and the mass identity give

$$V|\bar{u}|^2 = m - \|u - \bar{u}\|_2^2 \ge m - \lambda_1^{-1} g^2 \ge m/2.$$

Splitting the Fourier series of $u - \bar{u}$ at radius R and applying Cauchy–Schwarz to the gradient and Laplacian terms gives

$$\|u - \bar{u}\|_\infty \le \mathsf{L}_R g + \mathsf{H}_R \|\Delta u\|_2 \le 2\tau = \sqrt{\frac{m}{8V}}.$$

Since $|\bar{u}| \ge \sqrt{m/(2V)}$, the reverse triangle inequality yields $|u(x)| \ge \sqrt{m/(8V)}$ for every x. Consequently

$$\mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{m}{8V(U_+^2+\varepsilon^2)^2} \|\Delta u\|_2^2.$$

If $g > \eta$, periodic integration by parts, Cauchy–Schwarz, and the uniform denominator estimate instead yield

$$\|u\,\Delta u\|_2^2 \ge \frac{g^4}{V} > \frac{\eta^4}{V}, \qquad \mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{\eta^4}{V(U_+^2+\varepsilon^2)^2}.$$

Finally $\|\Delta u\|_2^2 \le B_+^2$, so this implies

$$\mathcal{Q}_{\varepsilon,\alpha}[u] \ge \frac{\eta^4}{V B_+^2 (U_+^2+\varepsilon^2)^2} \|\Delta u\|_2^2.$$

The smaller of the two coefficients proves (9). If $\Delta u = 0$, the assertion is immediate in either case. □

Remark 4.2 (Asymptotic meaning)**.** *This effective constant is generally conservative. It can deteriorate as V grows, $\lambda_1 \to 0$, $H_\star$ grows, or the device geometry changes. Thus the theorem does* not *establish an input-length-uniform physical resource bound. Its merit is that every dependence is now stated as an explicit spectral sum or model parameter.*

5 A detector corollary for the actual Hamiltonian evolution

Let $u(t)$ be a sufficiently smooth periodic solution of (2) on $0 \le t \le T$, conserving positive mass m and Hamiltonian H_0. Let $\chi \in H^4(\Omega;\mathbb{C})$ have zero spatial mean and write

$$Z_\chi(t) = \int_\Omega \overline{\chi}\, u(t)\, dx, \qquad \delta = |Z_\chi(T) - Z_\chi(0)|.$$

Let $\lambda_1 > 0$ be the smallest positive eigenvalue of $-\Delta$, and use the amplitude bound U from the conserved resources. For

$$K_\chi = \frac{\|(\Delta+\kappa)^2\chi\|_2 + (3\beta U^2 + 5\gamma U^4)\,\|\chi\|_2}{\lambda_1},$$

the periodic equation, the polynomial nonlinear Lipschitz estimate, and the mean-zero Poincaré inequality imply

$$\delta^2 \le K_\chi^2\, T \int_0^T \|\Delta u(t)\|_2^2 \, dt. \tag{10}$$

Corollary 5.1 (Node-tolerant finite-time density-curvature cost)**.** *If $T > 0$, $K_\chi > 0$, $\delta > 0$, and the smooth solution obeys the conserved mass and Hamiltonian hypotheses above, then*

$$\boxed{\int_0^T \mathcal{Q}_{\varepsilon,\alpha}[u(t)]\, dt \ge \frac{c_\star\, \delta^2}{K_\chi^2\, T}.} \tag{11}$$

This remains valid if $u(t,x) = 0$ at some points and times.

Proof. The conserved quantities place every $u(t)$ in the *same* class $\mathcal{A}_{m,H_0}$, so Theorem 3.1 gives one uniform $c_\star > 0$ for all t. Integrate (7) in time and apply (10). □

6 Discrete Lean certificate and exact limitations

On a two-node connected graph with real field values a, b, set $m = a^2 + b^2 > 0$, and use the simplified density weight $w_e(s) = s/(s+e)^2$, $e > 0$. The two graph-Laplacian values are $b - a, a - b$. Since $w_e(a^2) + w_e(b^2) \ge m/(m+e)^2$,

$$\boxed{(a-b)^2[w_e(a^2) + w_e(b^2)] \ge \frac{m}{2(m+e)^2}\,[2(a-b)^2].} \tag{12}$$

The estimate holds even if $a = 0$ or $b = 0$, and the coefficient is strictly positive when $m, e > 0$. The separate Lean module `PVNP/FixedMassNodal.lean` verifies this exact finite result and algebraic detector-to-cost transfer. It does *not* kernel-formalize the Sobolev compactness proof of Theorem 3.1 nor the continuum PDE integration and conservation arguments in Corollary 5.1.

Computational limitation. Neither $c_\star$ nor K_χ is shown to be independent of a growing device domain, input length, resolution, or detector family. The estimates concern a restricted observable of one declared PDE, not all admissible SAT solvers. In particular, no unconditional $P_{\mathrm{WCC}} \ne NP_{\mathrm{WCC}}$ or classical $P \ne NP$ claim follows.

Source provenance. This supplemental note uses the maintained projection-free Hamiltonian equation and follows the independent phase-observability study. The original PVNP trilogy and its published versions are not modified here.